Quantum Mechanics

Lecture Notes

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1. The position of a quantum particle

In your previous unit on quantum physics, you considered a simplified situation where a particle could be found in one of a discrete number of locations — for example, either on the left-hand or right-hand side of a box. Here, we will now consider a more realistic and powerful situation, that considered in the context of **mechanics**, where a particle with one degree of freedom (i.e. constrained to move in one dimension) has a **position** x. Our goal is to develop the quantum formalism to describe the motion of such a particle. This is known as **quantum mechanics**.

We will be able to learn most of the important lessons about the quantum description of the mechanics of a particle by considering only motion in one dimension. This is advantageous, as it is simpler to consider this idealised scenario. At the end of this unit, we will consider the more physically relevant case of motion in **three dimensions**, which describes the universe we live in.

1.1 The position of a quantum particle

Consider the case previously studied, where a particle can be found in one of four possible locations, as summarised in the following figure:

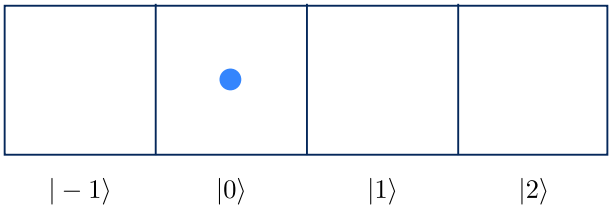


Figure 1.1: Quantum particle that can be in one of four possible locations. Depicted is the quantum particle in the state $|0\rangle$, in the second location. Under each possible location for the particle we have written the corresponding quantum state $|k\rangle$.

In this case, the general quantum state of the particle is a superposition, given by

$$|\psi\rangle = \sum_{k=-1}^{2} \alpha_k |k\rangle, \tag{1.1}$$

where α_k are complex numbers satisfying the normalisation condition $\sum_k |\alpha_k|^2 = 1$, and $|k\rangle$ is the quantum state with the particle at location k. Here, we will now view the labels k not as abstract labels, but really as telling us, in a **coarse-grained** fashion, the position of the particle.

Let us imagine therefore dividing each region into two, to obtain a more **fine-grained** description of the position of the particle. In this case, we would have the following situation:

with a general quantum state of the particle now being a superposition of the form

$$|\psi\rangle = \sum_{k=-3}^{4} \alpha_k |k/2\rangle. \tag{1.2}$$

Notice that our precision has now increased. Whereas before we could only specify the position of the particle in integer steps, we now specify it in twice as many half-integer steps.

We can imagine continuing this process of dividing each region in two **indefinitely** until we have infinite precision, such that each quantum state $|x\rangle$ now represents an actual position for the particle. As a figure, this would now give us a **continuous line**, as depicted below:

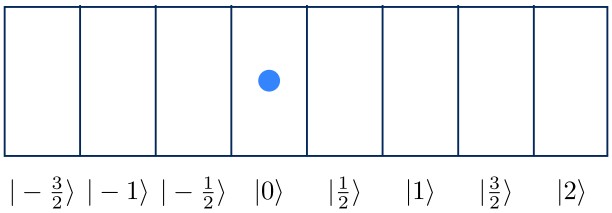


Figure 1.2: Quantum particle that can now be in one of 8 possible locations. Depicted is the quantum particle in the state $|0\rangle$, in the fourth location. As above, under each possible location for the particle we have written the corresponding quantum state $|k\rangle$.

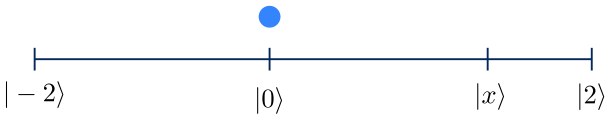


Figure 1.3: A quantum particle that can now be anywhere in the region $-2 \leq x \leq 2$, arrived at by continually dividing the regions from Figure 1.2 until they become infinitely small and continuous. Depicted is a quantum particle at $x = 0$.

Since we have now ended up with a **continuous** position for the particle, the most general quantum superposition will no longer be a summation, but will become an **integral**,

$$|\psi\rangle = \int_{-2}^{2} \psi(x)|x\rangle dx. \tag{1.3}$$

In (1.3), in comparison to (1.2), apart from changing from a sum to an integral, we have also **renamed** the coefficients α_k by $\psi(x)$, and written them now as a **function**. It is useful to realise that α_k could also have been thought of as a (discrete) function of k, although it wasn't particularly useful to write it this way. Now however, it will turn out extremely useful to think of the coefficient $\psi(x)$ as a function. This is the **quantum mechanical wavefunction** of a particle. We will in fact spend the majority of this unit studying the wavefunction, as will become clear below.

In this example so far, we have ended up with a quantum mechanical particle which can be in the region $-2 \leq x \leq 2$. There was nothing special about this region, and in general, we can **extend to arbitrary positions**, so that the particle can be found anywhere. In this case we would have

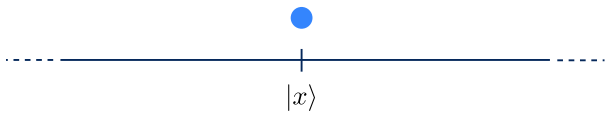


Figure 1.4: A quantum particle that can now be anywhere. Depicted is a quantum particle at an arbitrary position x.

with a general quantum superposition

$$|\psi\rangle = \int_{-\infty}^{\infty} \psi(x)|x\rangle dx. \tag{1.4}$$

This state corresponds to a **superposition** of the particle having a position x, and we will refer to $|x\rangle$ as **position states**. Just as previously, e.g. in (1.1), these states are meant to represent a quantum particle which has — with certainty — position x (we will see below that we have to be slightly careful with this!).

We can use these to introduce the **position operator** $\hat{X}$ of a particle. This will be the operator which has as eigenvalue the position x, with corresponding eigenstate the position state $|x\rangle$. Using the **same rule as before** except replacing a sum by an integral, the position operator is given by

$$\hat{X} = \int_{-\infty}^{\infty} x|x\rangle\langle x|dx. \tag{1.5}$$

1.2 Normalisation of wavefunctions

Recall that we need quantum states to be **normalised**. Mathematically, this corresponded to

$$\||\psi\rangle\| = 1, \tag{1.6}$$

and physically, this ensures that when performing a measurement, the probability of the outcomes always add up to 1, as they must. While we have now moved from discrete superpositions to continuous superpositions of the form (1.4), the quantum state nevertheless must remain normalised, and the normalisation condition **remains the same**. Previously we knew that (1.6) led to the modulus squared of the coefficients summing up to one (i.e. $\||\psi\rangle\|^2 = \langle\psi|\psi\rangle = \sum_k |\alpha_k|^2 = 1$). For the continuous case, the generalisation of this is simply

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1. \tag{1.7}$$

We need to check that this is what we arrive at starting from (1.4). We see that

$$\begin{aligned} \||\psi\rangle\|^2 &= \langle\psi|\psi\rangle, \\ &= \left(\int_{-\infty}^{\infty} \psi^*(x')\langle x'|dx'\right)\left(\int_{-\infty}^{\infty} \psi(x)|x\rangle dx\right), \\ &= \int_{-\infty}^{\infty} \psi^*(x')\left[\int_{-\infty}^{\infty} \psi(x)\langle x'|x\rangle dx\right] dx'. \end{aligned} \tag{1.8}$$

To simplify this expression, we need to recall the definition of the Dirac delta function, which we give below for convenience (along with its key properties, for later reference).

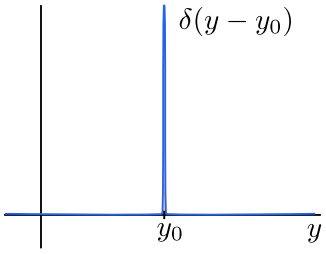
Definition 1.1 — Dirac Delta Function. The Dirac Delta function $\delta(y)$ is defined such that

$$\int_{-\infty}^{\infty} f(y)\delta(y - y_0)dy = f(y_0). \tag{1.9}$$

One representation of the Dirac Delta function is

$$\delta(y) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{iay} da. \tag{1.10}$$

The delta function is an infinitely thin, infinitely tall 'spike', as depicted below:



Two key properties of the delta function are that it is an **even** function,

$$\delta(-y) = \delta(y). \tag{1.11}$$

and that under a **change of scale**

$$\delta(by) = \frac{1}{|b|}\delta(y). \tag{1.12}$$

We see that in the final line of (1.8), if the inner integral (in square brackets) were equal to $\psi(x')$ then we would arrive at the desired expression (1.7). We will achieve this, using the defining equation of the Dirac Delta function (1.9), if

$$\langle x'|x\rangle = \delta(x - x'). \tag{1.13}$$

This is a remarkable equation. What it shows is that **position states have infinite normalisation!**:

$$\||x\rangle\|^2 = \langle x|x\rangle = \delta(0) = \infty. \tag{1.14}$$

However, we know that physically valid quantum states must be normalised. This leads us to an extremely important conclusion:

It is physically impossible in quantum mechanics for a particle to have a definite position.

Nevertheless, as we have seen in (1.4), quantum particles can (in fact must) have a **superposition** of positions, and be in a valid quantum state, so long as the wavefunction satisfies the normalisation condition (1.7). What we see is that, even though the position states $|x\rangle$ are themselves not permissible quantum states, they are **useful**, and provide a convenient **basis** in which to express a quantum state.

1.3 Probabilities to find a particle somewhere

As you learnt in your previous unit on quantum physics, it is an inherently **probabilistic** theory. That is, quantum particles do not have definite properties, but rather we must perform measurements to learn about the properties of a particle, and in general the results will be probabilistic. This is true for the position of a particle, just as with any other property. If we want to ask where a particle is, we have to measure the position operator $\hat{X}$ from (1.5), where x are the position eigenvalues and $|x\rangle$ the associated position eigenstates.

For a state $|\psi\rangle = \int_{-\infty}^{\infty} \psi(x)|x\rangle dx$, what is the probability to find the particle at some location x_0? This question in fact **doesn't make sense** as stated: the position is **continuous**, and therefore there is in fact zero probability to find the particle (precisely) somewhere. As with **any** continuous quantity, we can no longer talk about probabilities, but must rather talk about **probability densities**. That is, we can ask the question: *what is the probability to find the particle in a small region of 'width' dx?*. This will be the probability density[1] $\wp(x_0)$ multiplied by dx, as depicted in the following figure:

where according to quantum mechanics, the probability density is given by (this is the **Born rule**, after Max Born, who first interpreted $|\psi|^2$ as a probability [Born, 1926])

$$\wp(x) = |\psi(x)|^2, \tag{1.15}$$

so that the probability to find the particle in a small region around x_0 is

$$\text{prob}(x_0 \leq x \leq x_0 + dx) = |\psi(x_0)|^2 dx, \tag{1.16}$$

(i.e. nothing but the density, multiplied by the width). Physically, all measurements will have a **finite resolution**, which is here captured by the small interval dx.

[1] Note that this is also referred to as the **probability density function (PDF)**. Here we will simply refer to it as the 'probability density', but the two terms are interchangeable.

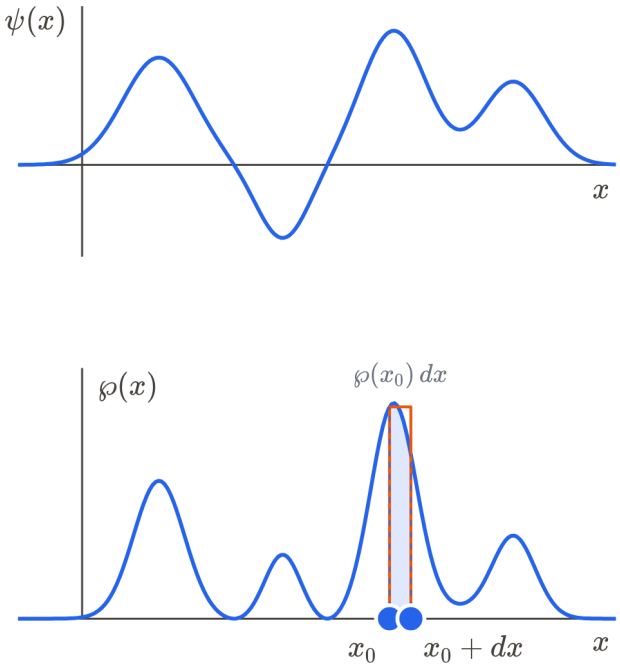


Figure 1.5: **Probability densities in quantum mechanics**. A wavefunction $\psi(x)$ (top) and its probability density $\wp(x) = |\psi(x)|^2$ (bottom). The probability to find a particle in the small region $x_0 \leq x \leq x_0 + dx$ of width dx around x_0 is $\wp(x_0)dx = |\psi(x_0)|^2 dx$. In the above figure, this is the area of the shaded region. A still of the interactive figure in the online notes.

1.4 State after a position measurement

Previously you also considered the **state of a particle directly after a measurement**. The rule that you learnt is that when performing a measurement of some operator, you obtain as the result of the measurement one of its eigenvalues, and the state directly after the measurement will be the corresponding **eigenstate of the operator**.

Here we therefore see that we have a complication: As we saw in (1.14), the eigenstates $|x\rangle$ of the position operator $\hat{X}$ have infinite norm, and are not valid states of a quantum particle! This raises the question of what happens to a particle after we measure its position?

The resolution of this apparent problem is the following:

It is unphysical to be able to perform a perfect measurement of the position of a particle.

That is, just like it was unphysical for a particle to have a definite position, it is also unphysical to perform a perfect measurement, which would determine the position of a particle (to infinite precision!). In reality, the best measurements we can perform will be **coarse-grained** and have a **finite resolution**.

Nevertheless, just as it is useful to consider position states $|x\rangle$ and write quantum states as a superposition of them (as in (1.4)), it is useful to consider the probability density $\wp(x) = |\psi(x)|^2$ that *would* arise from an idealised position measurement. We will consider this **directly** instead of the measurement. We will however **not consider the associated state after the measurement**.

> ***State after a position measurement.*** *While it is possible to describe the state after a coarse-grained measurement, it is beyond the scope of this unit. One reason for this is that there is no **universal** way to do this — it will depend on precisely how the finite precision is taken into account. Roughly speaking however, we expect that right after measuring the position of a particle, its state will now correspond to one **localised** around where it was found. You will often see this referred to as the 'collapse' of the wavefunction.*

One advantage of using directly the probability density $\wp(x) = |\psi(x)|^2$ is that it allows us to answer natural physical questions easily. For example, a natural question we can ask about a quantum particle is: *What is the probability to find it in the region $x_a \leq x \leq x_b$?* (for example, we might be interested in an electron, and the region might be the region close to the proton in a Hydrogen atom). We can answer this question directly: it is just the **total probability** that an idealised position measurement gives a result in this region. The answer is therefore

$$\text{prob}(x_a \leq x \leq x_b) = \int_{x_a}^{x_b} |\psi(x)|^2 dx, \tag{1.17}$$

i.e. we integrate the probability density, just as we would for any question involving probability densities.

1.5 Operators acting on wavefunctions

As this unit progresses, we will see that it is **very often** useful to focus **exclusively** on the wavefunction $\psi(x)$, and leave it **implicit** that the quantum state of the particle is as given by (1.4). The reason for this is that it can often be **overly cumbersome** to work with the full state $|\psi\rangle$, and it is much more convenient to just work with $\psi(x)$ instead.

In fact, in many popular textbooks on quantum mechanics, the presentation often **begins** with the wavefunction $\psi(x)$, before introducing the full quantum state $|\psi\rangle$ later on. One disadvantage of this is that certain aspects of quantum theory are obscured without the link to the quantum state. Here, we have introduced the quantum state first, but would now like to develop the formalism which allows us to focus exclusively on the wavefunction.

In order to do this, there is one key ingredient we need to address, which is how to **describe the action of operators** directly on wavefunctions. We will do this in this section.

Consider the action of the position operator $\hat{X}$ on a quantum state $|\psi\rangle$:

$$\begin{aligned}
\hat{X}|\psi\rangle &= \hat{X}\int_{-\infty}^{\infty} \psi(x)|x\rangle dx, \\
&= \int_{-\infty}^{\infty} \psi(x)\hat{X}|x\rangle dx, \\
&= \int_{-\infty}^{\infty} \psi(x)x|x\rangle dx, \\
&= \int_{-\infty}^{\infty} \psi'(x)|x\rangle dx,
\end{aligned} \tag{1.18}$$

where in the last line we have defined $\psi'(x) = x\psi(x)$.

What this shows us is that the action of the position operator $\hat{X}$ is to transform the wavefunction $\psi(x)$ to the wavefunction $\psi'(x) = x\psi(x)$. We see that this is **still a form of 'operation'** but now at the level of the wavefunction. That is, previously we saw that operators act on ket vectors, and transform them into other ket vectors. Here, in **direct analogy**, we saw that a wavefunction $\psi(x)$ was transformed into another wavefunction $\psi'(x)$, and so we can define an **associated position operator** which we will denote by $\hat{X}_{\text{w}}$ (where the subscript w refers to 'wavefunction') by

$$\hat{X}_{\text{w}}\psi(x) = x\psi(x). \tag{1.19}$$

That is, the action of $\hat{X}_{\text{w}}$ is to multiply (any) wavefunction $\psi(x)$ by the coordinate x everywhere, transforming into a new function $\psi'(x) = x\psi(x)$. What we can observe is that, with this definition, we have the following equality:

$$\hat{X}|\psi\rangle = \int_{-\infty}^{\infty} \left(\hat{X}_{\mathrm{w}}\psi(x)\right)|x\rangle dx. \tag{1.20}$$

In this way, we see that if we are given a wavefunction $\psi(x)$, as a way of describing a quantum state $|\psi\rangle$, then we can **use** $\hat{X}_{\mathrm{w}}$ in order to calculate the action of $\hat{X}$ on $\psi(x)$.

More generally, we can apply the above procedure to **any quantum mechanical operator** $\hat{A}$: There will **always** be an associated operator $\hat{A}_{\mathrm{w}}$ acting on wavefunctions $\psi(x)$, such that

$$\hat{A}|\psi\rangle = \int_{-\infty}^{\infty} \left(\hat{A}_{\mathrm{w}}\psi(x)\right)|x\rangle dx. \tag{1.21}$$

While this is abstract at this stage, it will hopefully become much clearer in the next Chapter, when we begin to investigate the **momentum** of a particle in quantum mechanics. Here we will find a more interesting relationship between the momentum operator $\hat{P}$ and the associated wavefunction momentum operator $\hat{P}_{\mathrm{w}}$.

2. The momentum of a quantum particle

In the previous chapter we began our investigation of the quantum mechanics of a particle moving in one dimension, focusing on the **position** of the particle. Here we will continue our development, by now considering how **momentum** is described quantum mechanically. We will see in this chapter that momentum is related to **wave-like** properties of a particle.

We will introduce the **momentum wavefunction** $\tilde{\psi}(p)$ (in analogy to the spatial wavefunction $\psi(x)$), and give the relationship to the latter, and also introduce the **momentum operator** $\hat{P}$ and its associated operator $\hat{P}_{\text{w}}$ (that acts upon wavefunctions $\psi(x)$).

2.1 The de Broglie relation and states of definite momentum

Our starting point for understanding the momentum of a quantum particle will in fact be the famous **de Broglie** relationship, which states that if a particle has a momentum p, then it has a wavelength of λ equal to

$$\lambda = \frac{h}{p}, \tag{2.1}$$

where h is **Planck's constant** [Planck, 1901]. While this formula is part of '*old quantum theory*' (it was de Broglie's 1924 doctoral thesis [de Broglie, 1925], and electron diffraction confirmed it three years later [Davisson and Germer, 1927]), it turns out that the basic insight is ultimately correct. More concretely, what is correct in quantum mechanics is

> *A quantum particle with definite momentum p has a wavefunction with a definite wavelength $\lambda = h/p$.*

What is a wavefunction with a definite wavelength λ? This could potentially mean a couple of things, for example $\cos(2\pi x/\lambda)$ or $\sin(2\pi x/\lambda)$? These are both examples of plane waves of wavelength λ. Neither of these happens to be the correct answer. What we actually mean is a **complex plane wave**

$$Ae^{2\pi i x/\lambda} \tag{2.2}$$

where A is the amplitude (which we take to be real out of convenience for the moment, and will come back to shortly). This is depicted in the following:

If we substitute the de Broglie relation (2.1) into (2.2), then we see that

$$Ae^{2\pi i x/\lambda} = Ae^{2\pi i p x/h} = Ae^{ipx/\hbar}, \tag{2.3}$$

where

$$\hbar = \frac{h}{2\pi}, \tag{2.4}$$

is the so-called **reduced Planck constant** or simply '**h-bar**' for short.

Putting everything together, in quantum mechanics, **particles that have a definite momentum p have wavefunctions $v_p(x)$ given by**

$$v_p(x) = Ae^{ipx/\hbar}, \tag{2.5}$$

corresponding to **complex plane waves** of wavelength $\lambda = h/p$. We will take this essentially as a **postulate**, and not attempt to derive it.

Using (1.4), i.e. the general relationship between a quantum state and a wavefunction, this means we can **directly write down** quantum states of definite momentum, which we will denote by $|p\rangle$ and refer to

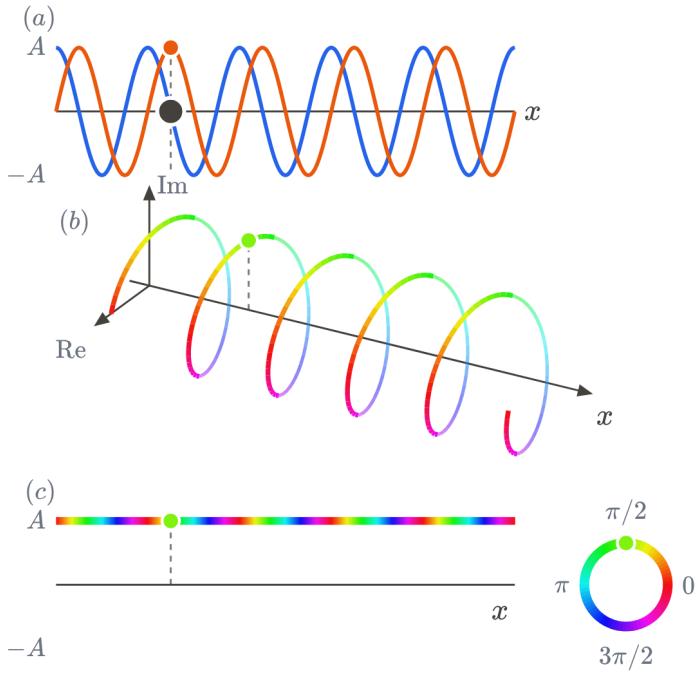


Figure 2.1: **Complex plane waves.** 3 different ways of visually representing a complex plane wave. (a) A plot of the real and imaginary parts of the function. We have $\mathrm{Re}(Ae^{2\pi ix/\lambda}) = A\cos(2\pi x/\lambda)$ and $\mathrm{Im}(Ae^{2\pi ix/\lambda}) = A\sin(2\pi x/\lambda)$. (b) A 3-D plot of this function, with the real and imaginary parts plotted along y and z axes respectively. In this representation, we see that a complex plane wave corresponds to a helix. (c) Polar representation of the function, where the phase is represented by colour. In this representation we have a stripy constant line. A still of the interactive figure in the online notes, in which the helix can be turned and the probe moved.

as 'momentum states' (in analogy to our notation $|x\rangle$ for position states). These are

$$|p\rangle = \int_{-\infty}^{\infty} v_p(x)|x\rangle dx = A\int_{-\infty}^{\infty} e^{ipx/\hbar}|x\rangle dx. \tag{2.6}$$

We still have some ambiguity here: we haven't specified what the 'amplitude' A should be. We can fix this by analogy to position states. Recall that in (1.13) we normalised position states using the Dirac delta function. We will demand that momentum states satisfy the **same** normalisation (and *orthogonality*) condition:

$$\langle p'|p\rangle = \delta(p-p'). \tag{2.7}$$

As an exercise, you will show that this fixes $A = \frac{1}{\sqrt{2\pi\hbar}}$. We thus have

$$|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} e^{ipx/\hbar}|x\rangle dx \tag{2.8}$$

are **quantum states of definite momentum**.

Note that due to (2.7), states of definite momentum have **infinite normalisation** (just like states of definite position). What this means is that they are, once again, **unphysical states**! We thus have another profound conclusion:

It is physically impossible in quantum mechanics for a particle to have a definite momentum.

We can understand this from a second perspective. Our requirement to have a definite wavelength means that a particle has to be spread out **everywhere**. If this were possible, then what would be the probability density $\wp(x)$ to find the particle at some location x_0? This would be

$$\begin{aligned}\wp(x_0) = |v_p(x_0)|^2 &= \left|\frac{1}{\sqrt{2\pi\hbar}}e^{ipx_0/\hbar}\right|^2, \\ &= \frac{1}{\sqrt{2\pi\hbar}}e^{-ipx_0/\hbar}\frac{1}{\sqrt{2\pi\hbar}}e^{ipx_0/\hbar}, \\ &= \frac{1}{2\pi\hbar}.\end{aligned} \tag{2.9}$$

This is a **uniform** probability density — we are equally likely to find the particle anywhere. If we then ask what is the **total** probability to find the particle somewhere, we see that

$$\int_{-\infty}^{\infty}\wp(x)dx = \int_{-\infty}^{\infty}\frac{1}{2\pi\hbar}dx = \infty. \tag{2.10}$$

This is an unphysical result. Furthermore, we can't even 'fix' it by changing our normalisation A, since **any** non-zero choice of normalisation will lead to the same conclusion. We therefore see that it must be physically impossible to have a particle whose wavefunction is a (complex) plane wave, and therefore it is physically impossible to have a definite momentum.

Nevertheless, just as it is extremely **useful** to use position states $|x\rangle$, it is similarly useful to use the momentum states $|p\rangle$, even though they aren't valid quantum states.

2.2 The momentum operator

One use of the states $|p\rangle$ is that they allow us to define the **momentum operator** $\hat{P}$. Recall that, **by definition**, this is the operator for which the momentum states $|p\rangle$ are **eigenstates**, with corresponding momentum eigenvalue p,

$$\hat{P}|p\rangle = p|p\rangle. \tag{2.11}$$

We will return to the momentum operator shortly, and find its associated operator $\hat{P}_{\text{w}}$. Before doing so, we will first introduce the **momentum wavefunction**.

2.3 The momentum wavefunction

Just as we can write any quantum state $|\psi\rangle$ as a **superposition of position states** $|x\rangle$ using the **spatial wavefunction** $\psi(x)$, we can also write it as a **superposition of momentum states** $|p\rangle$,

$$|\psi\rangle = \int_{-\infty}^{\infty} \tilde{\psi}(p)|p\rangle dp, \tag{2.12}$$

where $\tilde{\psi}(p)$ is called the **momentum wavefunction**.

> ***Nomenclature.*** *We will typically refer to $\psi(x)$ as just 'the wavefunction' since it is the wavefunction we will spend the majority of our time studying. When it is important to be clear, we will refer to it more precisely as the 'spatial wavefunction'. Similarly, we will mostly refer to $\tilde{\psi}(p)$ as the 'momentum wavefunction'. If it is clear from context, we will also just refer to this as the wavefunction. $|x\rangle$ and $|p\rangle$ are two different (continuous!) bases, and the wavefunctions are just the coefficients of the quantum state $|\psi\rangle$ in these two different bases.*

Just as we saw in (1.7), the normalisation of the quantum state, $\||\psi\rangle\| = 1$ implies that

$$\int_{-\infty}^{\infty} |\tilde{\psi}(p)|^2 dp = 1. \tag{2.13}$$

We also know that this is related to the normalisation of **momentum measurements**. In particular, if we were to **measure the momentum of a quantum particle** (i.e. measure $\hat{P}$), then the **probability density** to obtain the outcome p_0 is

$$\wp(p_0) = |\tilde{\psi}(p_0)|^2 \tag{2.14}$$

However, just as we saw that idealised position measurements are impossible, **so too are ideal momentum measurements**. These would need to have infinite resolution, which is impossible. In reality, we can also only perform **coarse-grained** momentum measurements, determining the momentum of the particle up to some level of precision. Nevertheless, it is very instructive to consider the momentum probability density $\wp(p) = |\tilde{\psi}(p)|^2$ that would arise from such idealised measurements.

2.4 Relationship between the spatial & momentum wavefunctions

It will prove useful to be able to go **directly** between the spatial wavefunction $\psi(x)$ and the momentum wavefunction $\tilde{\psi}(p)$ associated to the same quantum state $|\psi\rangle$. For example, in many situations we will specify a wavefunction $\psi(x)$, and if we want to consider e.g. measuring the momentum of the particle, we will need to calculate $\tilde{\psi}(p)$ in order to do this. Luckily there is a nice way to go between the two.

To see how, first let us consider a state $|\psi\rangle$ written as a superposition of momentum states, as in (2.12). If we consider the scalar product of $|\psi\rangle$ with a momentum state $|p'\rangle$, we find

$$\begin{aligned} \langle p'|\psi\rangle &= \int_{-\infty}^{\infty} \tilde{\psi}(p)\langle p'|p\rangle dp, \\ &= \int_{-\infty}^{\infty} \tilde{\psi}(p)\delta(p-p')dp, \\ &= \tilde{\psi}(p'), \end{aligned} \tag{2.15}$$

where to obtain the second line we have used the orthonormality condition (2.7). What this shows is a very important relation:

$$\langle p|\psi\rangle = \tilde{\psi}(p). \tag{2.16}$$

Starting from $|\psi\rangle$ written as a superposition of position states as in (1.4), we similarly arrive at

$$\langle x|\psi\rangle = \psi(x). \tag{2.17}$$

We can now use these facts in an interesting way. We start again with a quantum state written in the form (2.12), but we now take the scalar product with $|x\rangle$, leading to

$$\begin{aligned} \langle x|\psi\rangle &= \langle x| \left(\int_{-\infty}^{\infty} \tilde{\psi}(p)|p\rangle dp \right), \\ \psi(x) &= \int_{-\infty}^{\infty} \tilde{\psi}(p)\langle x|p\rangle dp, \\ &= \int_{-\infty}^{\infty} \tilde{\psi}(p) v_p(x) dp, \end{aligned} \tag{2.18}$$

where in the third line we have used (2.17), for the quantum state $|p\rangle$, a state of definite momentum, and used (2.6), to recall that for such states, the wavefunction is $v_p(x)$, as we saw at the beginning of this Chapter. Substituting in (2.5) (with the correct choice for A), we finally obtain

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\psi}(p) e^{ipx/\hbar} dp. \tag{2.19}$$

This gives the spatial wavefunction in terms of the momentum wavefunction. As an exercise, using the fact that $\langle p|x\rangle = (\langle x|p\rangle)^*$, you will show that we can also obtain the other direction,

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi(x) e^{-ipx/\hbar} dx, \tag{2.20}$$

which gives the momentum wavefunction in terms of the spatial wavefunction. It is important to note the symmetry between these two expressions. One key **distinction** is that in the former, the exponential comes with a plus-sign, while in the latter it comes with a minus-sign. Getting this sign wrong is a very easy and common mistake!

You have encountered the above type of equations before — this is nothing but the **Fourier transform** and its inverse! That is, the spatial wavefunction is the Fourier transform of the momentum wavefunction, and conversely the momentum wavefunction is the inverse Fourier transform of the spatial wavefunction! The reason why this might look unfamiliar is that instead of using the usual variable k, we are using the physical variable — the momentum — instead. What is k usually? It is the **wavenumber** of a wave, which we can recall is related to the wavelength via $k = 2\pi/\lambda$. Substituting this into the de Broglie relation (2.1), we see that $2\pi/k = h/p$, which after re-arranging gives $p = \hbar k$! A simple change of variable then takes us back to the 'standard' Fourier transform.

In hindsight, it is clear that this **had** to be the relation between the spatial and momentum wavefunctions. Why? Our starting point of this Chapter was the fact that in quantum mechanics, states of definite momentum correspond to complex plane waves. A Fourier transform is precisely the way of transforming a function $f(x)$, represented in the normal fashion — as a value at each point x — into a superposition of plane waves, where $\tilde{f}(k)$ is the amplitude to have wavenumber k (and hence wavelength $\lambda = 2\pi/k$). The only new ingredient for us is that in quantum mechanics a wavenumber of k corresponds to a momentum of $\hbar k$, and hence when we Fourier transform the wavefunction of a particle, we directly get its momentum wavefunction $\tilde{\psi}(p)$ — the 'probability amplitude' for it to have momentum p.

2.5 Momentum operator on spatial wavefunctions

The final thing we would like to do in this chapter is find the operator $\hat{P}_{\text{w}}$ associated to the momentum operator $\hat{P}$, that acts **directly** on the (spatial) wavefunction $\psi(x)$, as we studied in Operators acting on wavefunctions. What we would like to find is the operator $\hat{P}_{\text{w}}$ such that

$$\hat{P}|\psi\rangle = \int_{-\infty}^{\infty} \left(\hat{P}_{\text{w}}\psi(x)\right)|x\rangle dx. \tag{2.21}$$

In order to find this, we will use the relationship between $\psi(x)$ and $\tilde{\psi}(p)$. Recall from (2.11), that by definition $\hat{P}|p\rangle = p|p\rangle$. If we apply the momentum operator to a state written as a superposition of momentum states as in (2.12), we therefore find

$$\begin{aligned}\hat{P}|\psi\rangle &= \hat{P}\int_{-\infty}^{\infty}\tilde{\psi}(p)|p\rangle dp,\\ &= \int_{-\infty}^{\infty}\tilde{\psi}(p)\hat{P}|p\rangle dp,\\ &= \int_{-\infty}^{\infty}\tilde{\psi}(p)p|p\rangle dp,\\ &= \int_{-\infty}^{\infty}\tilde{\psi}'(p)|p\rangle dp,\end{aligned} \tag{2.22}$$

where in the last line we have defined $\tilde{\psi}'(p) = p\tilde{\psi}(p)$. What this shows is that, just as the position operator multiplies a spatial wavefunction by x, as we saw in (1.18), so too does the momentum operator multiply a momentum wavefunction by p.

The question we then need to answer is: *if a momentum wavefunction changes from $\tilde{\psi}(p)$ to $\tilde{\psi}'(p) = p\tilde{\psi}(p)$, what happens to the associated spatial wavefunction $\psi(x)$?* That is, what is the wavefunction $\psi'(x)$ associated to $\tilde{\psi}'(p)$, and how does it relate to $\psi(x)$?

We can answer this using (2.19):

$$\begin{aligned}\psi'(x) &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\tilde{\psi}'(p)e^{ipx/\hbar}dp,\\ &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}p\tilde{\psi}(p)e^{ipx/\hbar}dp.\end{aligned} \tag{2.23}$$

At this stage we can use a fascinating 'trick'. We can notice that

$$\frac{\partial}{\partial x}e^{ipx/\hbar} = \frac{ip}{\hbar}e^{ipx/\hbar}, \tag{2.24}$$

from which simple rearranging shows that we can write

$$pe^{ipx/\hbar} = -i\hbar\frac{\partial}{\partial x}e^{ipx/\hbar}. \tag{2.25}$$

We can substitute this into (2.23) to obtain

$$\begin{aligned}\psi'(x) &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\tilde{\psi}(p)\left(-i\hbar\frac{\partial}{\partial x}\right)e^{ipx/\hbar}dp,\\ &= -i\hbar\frac{\partial}{\partial x}\left(\frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\tilde{\psi}(p)e^{ipx/\hbar}dp\right),\\ &= -i\hbar\frac{\partial}{\partial x}\psi(x),\end{aligned} \tag{2.26}$$

where in the second line we used the fact that since $\tilde{\psi}(p)$ is independent of x, we can 'pull' the partial derivative through it, and in fact all the way out of the integral altogether, and in the final line we used the fact that the integral that remains is nothing but (2.19). What this shows is that multiplying a momentum wavefunction by p **has the effect of differentiating the spatial wavefunction** (and multiplying it by $-i\hbar$)! This has thus answered our question, and we can see that

$$\hat{P}_{\mathrm{w}}\psi(x) = -i\hbar\frac{\partial}{\partial x}\psi(x). \tag{2.27}$$

That is, the momentum operator $\hat{P}_{\mathrm{w}}$ is the partial derivative multiplied by $-i\hbar$. In what is basically an abuse of notation, we will sometimes write $\hat{P}_{\mathrm{w}} = -i\hbar\frac{\partial}{\partial x}$. This is somewhat dangerous, as that partial derivative needs to act on something! (actually, it will act on **everything** to the right of it). Nevertheless, it is an **operator** in that it acts on a function and returns a function.

Finally, it is instructive to look at what we get if we apply the momentum operator $\hat{P}_{\mathrm{w}}$ onto the wavefunctions $v_p(x) = \frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar}$ (the wavefunctions of definite momentum, associated to the momentum states $|p\rangle$). We see that

$$\begin{aligned}\hat{P}_{\mathrm{w}}v_p(x) &= -i\hbar\frac{\partial}{\partial x}\left(\frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar}\right),\\ &= -\frac{i\hbar}{\sqrt{2\pi\hbar}}\frac{ip}{\hbar}e^{ipx/\hbar},\\ &= p\frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar}.\end{aligned} \tag{2.28}$$

Therefore we have

$$\hat{P}_{\mathrm{w}}v_p(x) = pv_p(x), \tag{2.29}$$

which is precisely an **eigenvalue equation** for wavefunctions of definite momentum. This is a very important form of **consistency**: the wavefunctions $v_p(x)$ which correspond to states of definite momentum are seen to be eigenstates of the operator $\hat{P}_{\mathrm{w}}$ which represents the momentum operator on wavefunctions. It is important to note that we could have in fact used $\hat{P}_{\mathrm{w}}$ to **find** the states $|p\rangle$. In particular, we could have first found the eigenfunctions $v_p(x)$ of $\hat{P}_{\mathrm{w}}$, and then **defined** $|p\rangle = \int_{-\infty}^{\infty} v_p(x)|x\rangle dx$, which would be eigenstates of $\hat{P}$ (a fact you will check as an exercise).

It is also important to realise that we could have in fact started this whole chapter in reverse, and **defined first** the momentum operator $\hat{P}_{\mathrm{w}}$, and used this to arrive at the momentum eigenfunctions $v_p(x)$, rather than postulate them as states of definite momentum as we did. Thanks to de Broglie, we didn't need to do this, as he already had the insight into what form the eigenfunctions should have, i.e. that they corresponded to plane waves.

However, for other physical properties, this won't be the case — we won't know in advance what form wavefunctions should have in order to have a definite value of a given property (e.g. to have a definite energy or angular momentum). Our general procedure in these cases will be precisely to go in the other direction: For a given operator $\hat{A}$, to start by considering the associated operator $\hat{A}_{\mathrm{w}}$ (acting on wavefunctions), find its eigenfunctions, and then use **these** to define quantum states with definite properties, that is, the eigenstates of $\hat{A}$. A key example will be to find states of definite energy (in which case the operator of interest is the **Hamiltonian** $\hat{H}$). We will study this extensively in the following chapters.

3. Incompatible observables & the Heisenberg uncertainty principle

In this Chapter we will learn about one of the basic aspects of quantum mechanics — **incompatible observables**. As we will see in more detail below, quantum mechanics predicts that it is **impossible for a particle to have multiple definite properties at the same time**. For example, a particle cannot have a definite position and a definite momentum simultaneously, or a definite energy and a definite position (as just two examples). In fact, we can say much more than this, and this is really the main lesson: there is a strong **trade-off** between **how well-defined** properties can be. That is, if a particle has a fairly well defined position, this necessitates it having a broad spread of momenta. This is the content of the famous **Heisenberg uncertainty principle** (HUP), which we will introduce here. This is a fundamental fact about nature, and shows that certain observable properties are **incompatible** with each other.

3.1 Gaussian example

We can illustrate the main idea behind the incompatibility of position and momentum by considering a simple example — that of a **Gaussian wavefunction**. Consider the following wavefunction

$$\psi(x) = \sqrt{\frac{1}{a\sqrt{2\pi}}} e^{-x^2/4a^2}, \tag{3.1}$$

with a parameter $a > 0$, which determines the **width** of the wavefunction, and where $\sqrt{\frac{1}{a\sqrt{2\pi}}}$ is a normalisation factor, which ensures that this is a normalised wavefunction (this is left as an exercise). This wavefunction is depicted below:

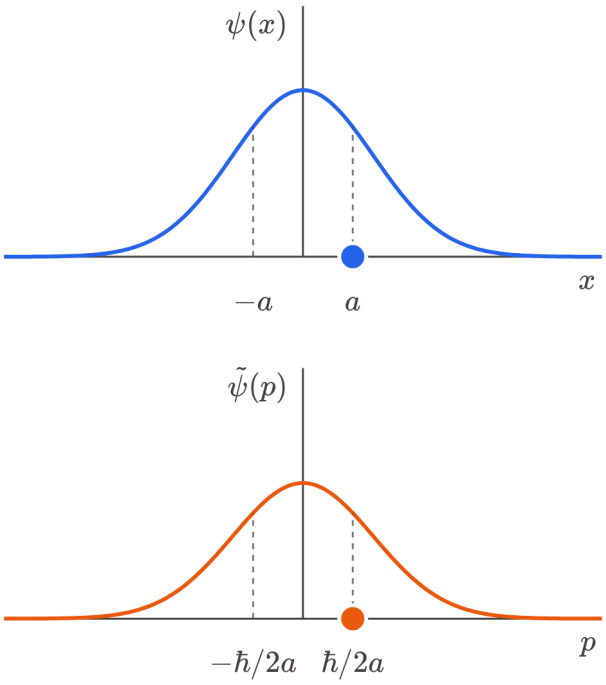


Figure 3.1: A Gaussian wavefunction, as given by (3.1) (top), and its momentum wavefunction (3.3) (bottom). The parameter $a > 0$ determines the 'width' of the wavefunction: $\Delta x = a$, while $\Delta p = \hbar/2a$, so narrowing one broadens the other. A still of the interactive figure in the online notes.

As we can see, it has the Bell-curve shape, and is centred at the origin. Direct calculations show that the expected position and standard deviation are

$$\begin{aligned} \langle x \rangle &= 0, \\ \Delta x &= \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = a. \end{aligned} \tag{3.2}$$

The fact that the standard deviation Δx is equal to a is the reason we say that a determines the width (since it uniquely specifies the standard deviation, which is a very useful and common way of quantifying the width of a probability density function). What does it mean for the particle to have a small width? It means that if we were to measure the position, we would be very likely to find the particle within a small region of space (in this case, close to the origin, since this is the only region where the probability density $\wp(x) = |\psi(x)|^2$ is large). This is where we should think of the quantum particle as being located, roughly speaking. If on the other hand a becomes large, then the particle will be found in a larger region of space when we measure its position.

The question we would now like to ask is what momentum does the particle have, as we consider varying the width of the spatial wavefunction a? We can answer this by finding the momentum wavefunction $\tilde{\psi}(p)$, which determines the probability density $\wp(p) = |\tilde{\psi}(p)|^2$ for the particle to have momentum p.

In an exercise, you will show that

$$\tilde{\psi}(p) = \sqrt{\frac{\sqrt{2}a}{\sqrt{\pi}\hbar}} e^{-a^2 p^2/\hbar^2}. \tag{3.3}$$

(Remarkably) this is **also a Gaussian** (now in the variable p instead of x). However, the expected momentum and standard deviation of momentum are

$$\begin{aligned} \langle p \rangle &= 0, \\ \Delta p &= \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \frac{\hbar}{2a}. \end{aligned} \tag{3.4}$$

What we see is that the spread of momentum is now **inversely proportional** to a. This means that if a becomes smaller — so that the particle has a smaller width in space (more precisely, so that a measurement of the position of the particle is likely to find it closer to the origin), it gains a bigger spread of momentum, meaning that a momentum measurement would find that the particle has large momentum with higher probability. Conversely, if the wavefunction has a wider width (a is larger), the spread of momentum in fact goes down!

3.2 The Heisenberg Uncertainty Principle

We often refer to these widths as **uncertainties**, since they tell us how uncertain our measurements of position and momentum will be. That is, how close or far we are from obtaining a very deterministic outcome when performing a measurement of that property of the particle.

What is remarkable is that what we observed above **isn't just a special property of Gaussian wavefunctions**, but is a general fact about quantum mechanics. In fact,

> *It is impossible in quantum mechanics for a particle to simultaneously have a small uncertainty Δx in position and a small uncertainty Δp in momentum.*

The famous **Heisenberg Uncertainty Principle** makes this statement **quantitative**: it says that the **product** of the uncertainties can never be smaller than a fundamental constant:

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \tag{3.5}$$

> ! *Heisenberg's original 1927 paper [Heisenberg, 1927] argued for the principle through thought experiments; the precise inequality (3.5), with the $\hbar/2$, was proved the same year by Kennard [Kennard, 1927]. We will not prove the Heisenberg Uncertainty Principle in this unit, but merely state it as a fact of quantum mechanics. An interested student will be able to find a proof in any textbook on quantum mechanics.*

This shows that if we have a particle in a particular quantum state $|\psi\rangle$ with corresponding spatial wavefunction $\psi(x)$, and we calculate the uncertainty in the position of the particle Δx, then even before calculating the momentum wavefunction, we already know that $\Delta p \geq \hbar/2\Delta x$. Hence if Δx is small, then this implies that Δp is necessarily going to be large.

Returning to our Gaussian wavefunction example from above, we saw in (3.2) and (3.4) that $\Delta x = a$ and $\Delta p = \hbar/2a$, and so in this case

$$\Delta x \Delta p = a \times \frac{\hbar}{2a} = \frac{\hbar}{2}, \tag{3.6}$$

that is, independent of a, a Gaussian wavefunction **saturates** the HUP: its product of uncertainties in position and momentum is in fact as small as allowed by quantum mechanics! We won't prove this, but it can be shown that **Gaussian wavefunctions are the only wavefunctions which saturate the HUP**. As we will see in numerous examples, $\Delta x \Delta p > \hbar/2$ for any other non-Gaussian wavefunction. For this reason, you will see the Gaussian wavefunction referred to as the **minimum uncertainty wavefunction** in some textbooks. We will also see later on in this unit, that the Gaussian wavefunction is very important, as it corresponds to the ground state (the energy eigenstate of lowest energy) of a harmonic oscillator — i.e. the quantum description of a mass on a spring, or a pendulum, or, in fact, the electromagnetic field!

3.3 Incompatible Observables

Because of the fact that position and momentum satisfy the HUP, we say that they are **incompatible observables**: these are two properties of a quantum particle that are somehow in *conflict* or *tension* with each other — such that having one property well defined precludes the other property being well defined too. This is a completely novel aspect of quantum mechanics compared to classical physics, where there is no such difficulty. For example, in classical mechanics, a particle has both a definite position and a definite momentum at all times.

Crucially, it is not just position and momentum which are incompatible. In fact **most observable properties are incompatible in quantum mechanics!** That is, apart from some special cases, if we consider two distinct physical properties, it is very likely that they are incompatible, and that a particle cannot have definite values for both properties simultaneously.

The question we will answer in the next section is **how to determine whether two observables are incompatible or not?** We will see that this is determined by a purely **mathematical** property of the **operators** associated to each observable property, known as **commutation**:

> *Two physical observables are compatible if (and only if) their associated operators **commute**.*

We will explain what this means in the next section.

3.4 Commutation

Recall that operators are things which 'act' on a quantum state. It is possible to act with **multiple** operators in a **sequence**. We can therefore ask a natural question: whether the **order** matters or not. That is, if we act with one operator and then a second, is this the same as first acting with the second operator, followed by the first? The answer in general is **no**. This is precisely what is captured by **commutation**: two operators commute if the two orders lead to the **same action**. On the other hand, two operators for which the two orders lead to different overall actions are said to **not commute**.

In order to capture this mathematically, we introduce the so-called **commutator** of two operators. We denote the commutator of (two arbitrary operators) $\hat{A}$ and $\hat{B}$ by $[\hat{A}, \hat{B}]$, and it is defined by

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{3.7}$$

That is, the commutator is the **difference** between the two orders. Having introduced the commutator, we can re-state in a more succinct manner what it means for two operators to commute:

> *$\hat{A}$ and $\hat{B}$ commute if (and only if) $[\hat{A}, \hat{B}] = 0$.*

This is because the commutator will vanish if $\hat{A}\hat{B} = \hat{B}\hat{A}$, i.e. precisely when the two different orders coincide.

> ⓘ *We note that there is something that is a little strange here, and can be misleading. While commutation is about the **order** of operators, this doesn't imply that incompatibility of observables is **also** about order. This is a common misconception. The link is in fact a very mathematical one: commutation is a succinct way of determining whether two operators have common **eigenstates** (i.e. whether the eigenstates of one operator happen to coincide with the eigenstates of another operator or not). It is this property which is crucial for compatibility, and it happens that this will be the case only when operators commute.*

In the context of **quantum mechanics**, we are usually interested not in the operators $\hat{A}$ and $\hat{B}$ directly, but rather their associated operators $\hat{A}_{\mathrm{w}}$ and $\hat{B}_{\mathrm{w}}$ (that act on wavefunctions $\psi(x)$ rather than on quantum states $|\psi\rangle$). We will use the same notation to refer to the associated commutator in this context:

$$[\hat{A}_{\mathrm{w}}, \hat{B}_{\mathrm{w}}]\psi(x) = \hat{A}_{\mathrm{w}}\left(\hat{B}_{\mathrm{w}}\psi(x)\right) - \hat{B}_{\mathrm{w}}\left(\hat{A}_{\mathrm{w}}\psi(x)\right), \tag{3.8}$$

where we have additionally included brackets, to make it clear exactly what it means in this case. Note that, in this case, we really **need** the wavefunction in order for this to make sense properly, even though the wavefunction is **completely arbitrary**.

Crucially, the commutation of any two operators $\hat{A}$ and $\hat{B}$ can be determined from $\hat{A}_{\mathrm{w}}$ and $\hat{B}_{\mathrm{w}}$. To explain what this means, we will use the example of $\hat{X}$ and $\hat{P}$. As an exercise, you will show that

$$\begin{aligned}[\hat{X}_{\mathrm{w}}, \hat{P}_{\mathrm{w}}]\psi(x) &= x\left(-i\hbar\frac{\partial}{\partial x}\psi(x)\right) + i\hbar\frac{\partial}{\partial x}\left(x\psi(x)\right) \\ &= i\hbar\psi(x).\end{aligned} \tag{3.9}$$

From this we can 'read off' the commutator of $\hat{X}$ and $\hat{P}$:

$$[\hat{X}, \hat{P}] = i\hbar. \tag{3.10}$$

This is often referred to as the **canonical commutation relation** since it is so important and central.

To arrive at (3.10), we simply replaced everywhere $\hat{X}_{\mathrm{w}}$ and $\hat{P}_{\mathrm{w}}$ by $\hat{X}$ and $\hat{P}$, and removed the wavefunction. This is generally true. Namely,

$$[\hat{A}_{\mathrm{w}}, \hat{B}_{\mathrm{w}}]\psi(x) = \hat{C}_{\mathrm{w}}\psi(x) \implies [\hat{A}, \hat{B}] = \hat{C}. \tag{3.11}$$

That is, we calculate the commutator of $\hat{A}_{\mathrm{w}}$ and $\hat{B}_{\mathrm{w}}$ on a wavefunction $\psi(x)$, and call this $\hat{C}_{\mathrm{w}}\psi(x)$ (in general, it will be some operator acting on the wavefunction). Then, from this, using (1.21) we can find the corresponding operator $\hat{C}$ associated to $\hat{C}_{\mathrm{w}}$.

Coming back to the big picture: we see in (3.10) that $\hat{X}$ and $\hat{P}$ do not commute, and hence (as we already knew in this case!) they are **incompatible observables**. If we now want to consider other pairs of properties, and want to know if they are incompatible or not, we follow the same procedure: we calculate the commutator of their associated operators (most likely the operators acting on their wavefunctions) and whether or not this vanishes answers our physical question.

4. Energy and the Hamiltonian

In this Chapter we will consider now the final important physical property of a particle we will study in this unit — the energy. Not only is this one of the most important physical properties in nature, due to its **conservation law**, it is central to the **time evolution** in quantum mechanics, as we will begin investigating in the next Chapter.

We will reintroduce the **Hamiltonian operator** (the total-energy operator), in the context of mechanics, its associated **eigenvalue equation**, and discuss the important notion of **bound** and **non-bound** states, and their distinct description within quantum mechanics.

4.1 The Hamiltonian

What different types of energy can a particle have? In the context of the mechanics of a single particle, there are two primary contributions to the energy: the **kinetic energy** and the **potential energy**. The total energy of a particle is just the sum of these two. We call the total energy the **Hamiltonian** (after Hamilton), which will be

$$H = K + V, \tag{4.1}$$

where K is the kinetic energy, and V is the potential energy.

> ***Other contributions to energy.*** *There are, of course, other contributions to the energy in general. For instance, thermal energy, rotational kinetic energy, etc. Here, as in classical mechanics, we focus on very simple situations, where we can safely **neglect** all of these other contributions, and focus exclusively on the kinetic and potential energies.*

While up until now you may have found it more customary to write the kinetic energy as $K = \frac{1}{2}Mv^2$, where M is the mass of the particle and v is the velocity, since $p = Mv$, we can alternatively write the kinetic energy as

$$K = \frac{p^2}{2M}, \tag{4.2}$$

which turns out to be the 'correct' way to express the kinetic energy in QM.

The potential energy is associated to the **forces** acting on the particle, through $F = -\frac{\partial V}{\partial x}$. This highlights that the potential energy of a particle is usually a **function of its position** $V(x)$. For example, this might be the gravitational potential energy of a particle (associated to the force of gravity), or the energy stored in a spring for a particle undergoing harmonic motion (associated to the restorative force $F(x) = -kx$, with k the spring constant).

In quantum mechanics, we associate an **operator** with the total energy, the **Hamiltonian operator** which is given by

$$\begin{aligned} \hat{H} &= \hat{K} + \hat{V}, \\ &= \frac{\hat{P}^2}{2M} + \hat{V}. \end{aligned} \tag{4.3}$$

As previously, it will be very useful to introduce the associated operator $\hat{H}_{\mathrm{w}}$ that **acts on wavefunctions** $\psi(x)$. It is given by

$$\hat{H}_{\mathrm{w}} = \frac{\hat{P}^2_{\mathrm{w}}}{2M} + \hat{V}_{\mathrm{w}}. \tag{4.4}$$

To understand the kinetic energy operator, we can see that

$$\begin{aligned}\frac{\hat{P}_{\mathrm{w}}^2}{2M}\psi(x) &= \frac{1}{2M}\hat{P}_{\mathrm{w}}\left(\hat{P}_{\mathrm{w}}\psi(x)\right), \\ &= \frac{1}{2M}\left(-i\hbar\frac{\partial}{\partial x}\right)\left(-i\hbar\frac{\partial}{\partial x}\psi(x)\right), \\ &= -\frac{\hbar^2}{2M}\frac{\partial^2}{\partial x^2}\psi(x),\end{aligned} \tag{4.5}$$

where in the second line we used the explicit form of the momentum operator $\hat{P}_{\mathrm{w}}$ from (2.27). This shows that

$$\hat{K}_{\mathrm{w}} = \frac{\hat{P}_{\mathrm{w}}^2}{2M} = -\frac{\hbar^2}{2M}\frac{\partial^2}{\partial x^2}, \tag{4.6}$$

is the kinetic energy operator (acting on wavefunctions). For the potential energy, the situation is simpler. We have

$$\hat{V}_{\mathrm{w}}\psi(x) = V(x)\psi(x), \tag{4.7}$$

that is, the potential energy operator multiplies the wavefunction at each position x by the value of the potential energy at that position, namely $V(x)$, just as we saw in (1.19) that the position operator multiplies the wavefunction by the position everywhere. Thus, we can write

$$\hat{V}_{\mathrm{w}} = V(x). \tag{4.8}$$

Putting everything together, we can now write down the Hamiltonian operator acting on wavefunctions:

$$\hat{H}_{\mathrm{w}} = -\frac{\hbar^2}{2M}\frac{\partial^2}{\partial x^2} + V(x). \tag{4.9}$$

As you will show in an exercise, the kinetic energy and potential energy **do not commute**, $[\hat{K}, \hat{V}] \neq 0$ in general. This leads to a profound aspect of quantum mechanics:

The kinetic energy and potential energy of a quantum particle are incompatible properties.

That is, a particle cannot have both a well defined kinetic energy and a well defined potential energy at the same time! One intuitive way to see why this **had** to be the case is because the kinetic energy depends only upon the **momentum** of the particle, while the potential energy depends only upon the **position**, and we already know that these two properties are incompatible. Many of the intricacies of how quantum particles behave derive, ultimately, from the fact that these two contributions to energy are incompatible with each other.

4.2 Energy eigenstates

A quantum state where the particle has definite energy corresponds to an **eigenstate of the Hamiltonian** $\hat{H}$. That is, a state $|E\rangle$ of energy E that satisfies the eigenvalue equation

$$\hat{H}|E\rangle = E|E\rangle. \tag{4.10}$$

As with momentum eigenstates, we can write these eigenstates in terms of their associated wavefunctions,

$$|E\rangle = \int_{-\infty}^{\infty} u_E(x)|x\rangle dx, \tag{4.11}$$

where $u_E(x)$ are wavefunctions of the particle corresponding to states of definite energy. We call these the **energy eigenfunctions**. These wavefunctions will satisfy the associated eigenvalue equation

$$\hat{H}_{\mathrm{w}} u_E(x) = E u_E(x). \tag{4.12}$$

For reasons that will become clear in the following chapter, this equation is often referred to as the **Time-independent Schrödinger Equation** or TISE for short. This will in fact be our **central** equation for studying the energy of a particle in quantum mechanics, essentially because of the fact that the potential energy is a function of position.

4.3 Bound and non-bound states

We end this chapter with an important discussion about **qualitatively distinct** situations that can arise, and how this affects the allowed energies a particle can have, and the properties of their eigenstates.

4.3.1 Bound states

In situations where a particle does not have enough energy to leave a finite region of space we say that it is in a **bound state**. This is a very generic situation, which occurs across all of physics, ranging from planets in bounded orbits around stars, all the way down to electrons bound to a nucleus. What unites all such situations is that the object will be **stuck in the corresponding potential well** (whether that be the gravitational potential well of the star, or the electrostatic potential well of the nucleus). This is most readily depicted graphically:

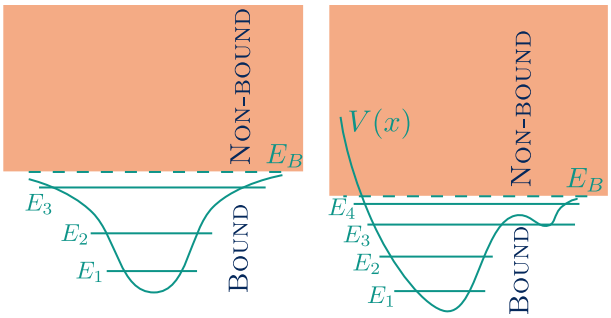


Figure 4.1: Two examples of potential wells. On the left, the well is symmetric. On the right, the well is asymmetric. If the energy of the particle is $E < E_B$, then the particle will be 'trapped' inside the well, which we call a bound state. In this case, in QM the particle will have discrete energy levels. If $E > E_B$ the particle is not bound to the well, but will escape to infinity (on the left in either direction; on the right, only towards the right). In QM, in this regime there is no discretisation of energies; energy is continuous but eigenstates become unphysical.

For any potential well $V(x)$ there will be a value E_B, such that if the energy of the particle is $E < E_B$, then it will not be able to escape from the well, and is hence trapped inside it / bound to it.

> ⓘ *It could be that $E_B = \infty$, i.e. that the particle is **definitely** trapped in the well. We will encounter such situations in this unit, for example a harmonic oscillator, where the particle cannot escape the spring (assuming it doesn't snap!). In general, we expect that E_B is finite, and we should always ask ourselves as a first question, whether particles are always bound, given a specific potential $V(x)$.*

In QM, when a particle is in a bound state, we find that **it can only have certain special and discrete energies** E_k. These are the famous **energy levels** (e.g. of atoms). More specifically and mathematically, we will see that in this case there are **only** physical solutions to the TISE (4.12) for specific values E_k. The corresponding eigenstates $|E_k\rangle$ are **normalised** and **orthogonal** to each other,

$$\langle E_k | E_\ell \rangle = \begin{cases} 1 & \text{if } k = \ell \\ 0 & \text{if } k \neq \ell \end{cases} = \delta_{k,\ell}. \tag{4.13}$$

4.3.2 Non-bound states

The alternative situation is where a particle is **not confined** to one region of space, either because **it has too much energy to be bound in the well** ($E > E_B$), or because **there are no forces**, i.e. $V(x) = 0$ and there is no well to be trapped in. Classically, this corresponds to **non-closed orbits**, such as asteroids passing a star, or an electron that isn't bound to a nucleus. In these cases, in principle, the object will eventually **escape to infinity** (travel as far as possible in some direction).

In QM in this case we find that there is **no discretisation** (no energy levels), but rather the particle can have **any energy** $E > E_B$ (just like it can have any momentum or position in general), and energy remains **continuous** as it is classically.

However, in this case, just like for position and momentum states, the energy eigenstates are **unnormalisable** and therefore **unphysical**. We take them to satisfy the analogous orthonormality condition as for position (1.13) and momentum (2.7), namely

$$\langle E' | E \rangle = \delta(E - E'), \tag{4.14}$$

where $\delta(E - E')$ is the Dirac delta function. Just as in those cases, even though the states $|E\rangle$ themselves are unphysical, they are a very useful **basis** of states, since **superpositions** can be normalised states.

We end by noting that in general, a quantum particle could be in a superposition of being **bound** and being **non-bound**. That is, it could have a superposition of energies, including energy levels below E_B and energies above E_B. This causes no problem. It will nevertheless be **easier** to study the two regimes **separately** at first, before combining them after, if required by the physical situation under consideration.

> ! *For example, it is possible, by carefully tuning a laser pulse, to create a superposition of having an atom ionised (exciting an electron, so that it has enough energy to escape the nucleus) and not (the electron still in a bound state).*

5. Time evolution

We are now ready to embark on one of the central physical questions that we are interested in: *how do quantum particles evolve in time?*

Up until now, we have just been discussing quantum states, and their **static** properties, without making any reference to time, or how the state changes in time. That is, when we have considered a particle with a state $|\psi\rangle$, we have really meant the state at some reference time, e.g. $t = 0$. If we prepare a particle in some state, we don't expect it to **remain in that state**. For example, if the particle has momentum, we know it is moving, and so know its state should change, reflecting this non-zero momentum.

In this chapter, after reintroducing the **equation of motion** of QM, we will see how writing a quantum state as a superposition of energy eigenstates allows us to **solve it directly** and **write down the state at time** t.

5.1 The Schrödinger Equation

In general, if we prepare a quantum particle in the state $|\psi_{\text{in}}\rangle$ at time $t = 0$, we are interested in determining its state at an arbitrary later time t, which we will denote by $|\psi(t)\rangle$.

The **equation of motion** in quantum physics is called the Schrödinger Equation (SE) [Schrödinger, 1926a,b], and is given by

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle. \tag{5.1}$$

If $|\psi(t)\rangle$ satisfies (5.1) with the **initial condition** $|\psi(0)\rangle = |\psi_{\text{in}}\rangle$, then this will be what the quantum state evolves into at time t.

What is the SE in terms of the wavefunction $\psi(x,t)$*?* We can find this by expressing $|\psi(t)\rangle$ as in (1.4), and taking the scalar product with $\langle x'|$. As an exercise, you will show that this leads to

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = \hat{H}_{\text{w}}\psi(x,t). \tag{5.2}$$

This is also referred to simply as the **Schrödinger Equation**, as it is the equation of motion, now for the wavefunction, rather than the quantum state.

5.2 Evolution of bound states

You saw how to solve the SE last year for a d-dimensional quantum system. When considering a quantum particle which is in a **bound state** — by which we mean **can be written as a superposition of energy eigenstates, all with energy** $E < E_B$ (i.e. trapped within a potential well), then this same treatment carries over directly.

In particular, let us assume that our initial state $|\psi_{\text{in}}\rangle$ (at $t = 0$) can be written as

$$|\psi_{\text{in}}\rangle = \sum_{k=1}^{d} \alpha_k |E_k\rangle, \tag{5.3}$$

where we consider an arbitrary Hamiltonian $\hat{H}$, with a potential $\hat{V}$ such that there are d bound energy eigenstates — that satisfy $\hat{H}|E_k\rangle = E_k|E_k\rangle$ with $E_k < E_B$, and $\sum_k |\alpha_k|^2 = 1$ so that $|\psi_{\text{in}}\rangle$ is a normalised quantum state.

Then, according to what you learnt last year, the quantum state at time t will be

$$|\psi(t)\rangle = \sum_{k=1}^{d} \alpha_k e^{-iE_k t/\hbar} |E_k\rangle, \tag{5.4}$$

which we can directly see satisfies the initial condition $|\psi(0)\rangle = |\psi_{\text{in}}\rangle$. We leave it as an exercise to confirm that (5.4) solves the SE (5.1).

Using (2.17), we can also see how the wavefunction evolves. In particular, the particle will have a wavefunction $\psi(x,t)$ at time t. The initial condition is that the wavefunction is $\psi_{\text{in}}(x) = \langle x|\psi_{\text{in}}\rangle$. Taking the scalar product with $\langle x|$ on both sides of (5.4), the wavefunction at time t is given by

$$\psi(x,t) = \sum_{k=1}^{d} \alpha_k e^{-iE_k t/\hbar} u_{E_k}(x), \tag{5.5}$$

where $u_{E_k}(x)$ are the energy eigenfunctions associated to the energy eigenstates $|E_k\rangle$ through (4.11), which satisfy the TISE (4.12).

5.3 Evolution of non-bound states

We now turn our attention to the evolution of **non-bound** states. By this, we mean that the particle **definitely has too much energy to be confined in the potential well**. More concretely, it means that we consider a situation where the initial state can be written as

$$|\psi_{\text{in}}\rangle = \int_{E_B}^{\infty} \phi_{\text{in}}(E)|E\rangle dE, \tag{5.6}$$

where now, since energies are **continuous** for non-bound states, we have to integrate rather than sum to form a superposition, and where we have changed to $\phi_{\text{in}}(E)$ (from α_k), which is now a **function of energy**, just as the position and momentum wavefunctions are functions, respectively, of position and momentum. The lower limit on the integral of E_B is because only in this **energy range** is the particle not bound to the well.

Our goal now is to find the state $|\psi(t)\rangle$ which satisfies the SE (5.1) and has (5.6) as the initial condition. Let us **assume** that we can write $|\psi(t)\rangle$ in the same form as (5.6), that is, in the form

$$|\psi(t)\rangle = \int_{E_B}^{\infty} \phi(E,t)|E\rangle dE, \tag{5.7}$$

for some function $\phi(E,t)$ that we want to determine. Substituting this into the SE (5.1), we find

$$i\hbar \frac{d}{dt} \int_{E_B}^{\infty} \phi(E,t)|E\rangle dE = \hat{H} \int_{E_B}^{\infty} \phi(E,t)|E\rangle dE. \tag{5.8}$$

We can take $\frac{d}{dt}$ inside the integral on the left-hand side, as long as we write it as $\frac{\partial}{\partial t}$ (since inside the integral, we have **both** the variables t and E), and similarly we can take $\hat{H}$ inside the integral on the right-hand side, and use the fact that $|E\rangle$ is an energy eigenstate, satisfying (4.10), to arrive at

$$i\hbar \int_{E_B}^{\infty} \frac{\partial}{\partial t} \phi(E,t)|E\rangle dE = \int_{E_B}^{\infty} \phi(E,t)E|E\rangle dE. \tag{5.9}$$

Finally, if we take the scalar product with $\langle E'|$ on both sides, using (4.14) and (1.9), we obtain

$$i\hbar \frac{\partial}{\partial t} \phi(E',t) = E'\phi(E',t). \tag{5.10}$$

This is however **the same differential equation you solved previously**. The solution is

$$\phi(E,t) = \phi_{\text{in}}(E) e^{-iEt/\hbar}. \tag{5.11}$$

Substituting this back into (5.7), the final solution is

$$|\psi(t)\rangle = \int_{E_B}^{\infty} \phi_{\text{in}}(E) e^{-iEt/\hbar} |E\rangle dE, \tag{5.12}$$

which we can readily confirm satisfies the initial condition $|\psi(0)\rangle = |\psi_{\text{in}}\rangle$, as required. In terms of wavefunctions, we have

$$\psi(x,t) = \int_{E_B}^{\infty} \phi_{\text{in}}(E) e^{-iEt/\hbar} u_E(x) dE. \tag{5.13}$$

5.4 Stationary states

Let us return for the moment to bound states, and consider a particularly simple initial condition: that the particle has a definite energy E_ℓ. That is, the initial state is $|\psi_{\text{in}}\rangle = |E_\ell\rangle$, and so $\alpha_\ell = 1$ and $\alpha_k = 0$ for $k \neq \ell$. According to (5.4) the quantum state at time t will be

$$|\psi(t)\rangle = e^{-iE_\ell t/\hbar} |E_\ell\rangle. \tag{5.14}$$

That is, **in time, the state just acquires a *phase*** $e^{-iE_\ell t/\hbar}$. What we would now like to ask is *what properties of the particle change?*

First of all, we can see that the **energy hasn't changed**. In particular, the particle started off with a definite energy E_ℓ. While it has acquired a phase, **this doesn't change the energy of the particle**. One way to see this is to consider **measuring the energy**. Recall that if we had a general superposition of energy levels, $\sum_k \beta_k |E_k\rangle$, then the probability of the result E_k would be

$$\text{prob}(E_k) = |\beta_k|^2. \tag{5.15}$$

Here we have $\beta_\ell = e^{-iE_\ell t/\hbar}$ (and $\beta_k = 0$ for $k \neq \ell$). So

$$\text{prob}(E_\ell) = |e^{-iE_\ell t/\hbar}|^2 = 1. \tag{5.16}$$

This is in fact what we know really **must** have been the case, since **energy is a conserved quantity** which **should not change in time**. If quantum mechanics had predicted that the energy of a particle would change in time, then this would truly be a radical departure from classical physics! Luckily this isn't the case.

What about the position of the particle? Does this change in time? From (5.5), we see that here

$$\psi(x,t) = e^{-iE_\ell t/\hbar} u_{E_\ell}(x), \tag{5.17}$$

where $u_{E_\ell}(x)$ is the wavefunction associated to the quantum state $|E_\ell\rangle$. If we measure the position of the particle at time t, the probability density to find it at x is

$$\begin{aligned} \wp(x,t) &= |\psi(x,t)|^2, \\ &= |e^{-iE_\ell t/\hbar} u_{E_\ell}(x)|^2, \\ &= |u_{E_\ell}(x)|^2, \end{aligned} \tag{5.18}$$

which is **independent of time**. Therefore, no matter at what time we choose to measure the position of the particle, we obtain the same probabilities to find it in any given region. We thus see that **the particle is not moving**.

We reach a similar conclusion about the momentum, which we leave as an exercise. Namely, we find that the probability density for the particle to have momentum p at time t is also time-independent.

In fact, **none of the physical properties of the particle change**! For this reason, we call energy eigenstates **stationary states**. The simple evolution of acquiring a phase doesn't change any property of the state.

This follows from a more fundamental property of quantum mechanics:

The states $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ lead to identical physical predictions in all situations.

In our example, the only difference in the quantum state of a particle with a definite energy E_ℓ between $t = 0$ and time t is precisely a phase factor $e^{-iE_\ell t/\hbar}$. While this phase **changes in time**, it is a phase nevertheless, and hence the physical predictions are identical at all times.

5.5 Evolution of superpositions

We now want to return to the 'big picture' of time evolution in quantum mechanics, and see how we can **understand** (5.5) and (5.13). The key is to realise that these are **special cases** of a more general lesson about quantum evolution, known as the **superposition principle**.

To understand what this is, we need to consider **two separate initial conditions**, $\psi_{\text{in}}(x)$ and $\psi'_{\text{in}}(x)$ (specified in terms of wavefunctions, rather than quantum states, although the two are equivalent). Let's assume that in each case, we have found the corresponding time evolution $\psi(x,t)$ and $\psi'(x,t)$ (as given through (5.5) or (5.13)).

Now, the question is, *what would happen if we considered a third initial condition, which is a **superposition** of the above two initial conditions?* That is, how does the wavefunction

$$\psi''_{\text{in}}(x) = \beta\psi_{\text{in}}(x) + \gamma\psi'_{\text{in}}(x), \tag{5.19}$$

evolve in time, where β and γ are arbitrary complex numbers, such that $\psi''_{\text{in}}(x)$ is a normalised wavefunction.

The crucial insight is that **we can write down the answer immediately** because of the **superposition principle**, which states that

The evolution of a superposition is just the superposition of evolutions.

That is, since $\psi''_{\text{in}}(x)$ is a superposition of $\psi_{\text{in}}(x)$ and $\psi'_{\text{in}}(x)$, and since we already know that $\psi_{\text{in}}(x)$ evolves into $\psi(x,t)$ and $\psi'_{\text{in}}(x)$ evolves into $\psi'(x,t)$, we are guaranteed that $\psi''_{\text{in}}(x)$ evolves into the superposition of $\psi(x,t)$ and $\psi'(x,t)$. Namely, into

$$\psi''(x,t) = \beta\psi(x,t) + \gamma\psi'(x,t). \tag{5.20}$$

This is very special, and **remarkably powerful**. You will show that it is true as an exercise.

We can now return to (5.5) and use the superposition principle to understand (and therefore hopefully remember!) it. We wrote the initial wavefunction as a **superposition** of energy eigenfunctions. Each energy eigenfunction evolves in a **simple** fashion, as we saw in Stationary states: it only acquires a phase factor $e^{-iE_k t/\hbar}$ in time. (5.5) is then nothing but an application of the superposition principle: the evolution of the superposition is just the superposition of the **simple** evolutions! It is worthwhile taking time to internalise this, as it is one of the most important lessons about quantum mechanics.

The same story is also true for (5.13), the only difference now is the form of the superposition (going from a discrete sum, to a continuous integral).

We thus arrive at a rather important realisation: quantum mechanical evolution is actually very simple: energy eigenstates evolve in very simple ways. We will nevertheless see that quantum mechanical evolution **appears complex**. Where does this complexity come from? It comes from **interference**, and the fact that in a superposition, we take **complex** multiples of energy eigenstates.

This is in fact similar to what we see in many situations, for example when considering **waves on a string**, or **coupled oscillators**. The motion is complex, but there are **modes** which evolve in simple ways, and the complex motion comes merely from interference between modes. The situation is similar in quantum mechanics, the only real distinction being the introduction of complex numbers, allowing for more interesting forms of interference.

6. Free particle

In this chapter we will investigate the simplest possible scenario one can consider in the context of mechanics — a **free particle** — one with no forces acting upon it whatsoever. We will see that, given everything we have learnt up until this point, we can **directly** solve for the evolution of a free particle, since we can immediately find the energy eigenstates (the quantum states of a particle with a definite energy) in this simple setting.

We will see that quantum evolution is more intricate than classical evolution, ultimately due to the HUP. In particular, in a classical setting, if a particle has an initial position x_0 and an initial momentum p_0, then from the equation of motion — Newton's second law — the position of the particle at time t will be

$$x(t) = x_0 + \frac{p_0}{M}t, \tag{6.1}$$

(and the momentum is constant, $p(t) = p_0$).

In quantum mechanics, as we already saw, a particle must necessarily have a **superposition of momenta**. This means that it will be travelling with a superposition of speeds, and therefore after some time, it will be in a superposition of having different displacements from its initial position (i.e. of having travelled different distances). As we will see, this generically leads to **spreading out** of the wavefunction, so that at later times, a position measurement will generally have a **larger standard deviation**.

6.1 States of definite energy

Our starting point, as will always be the case, is to ask the question *what are the energy eigenstates of a free particle?*, that is, what are the **states of definite energy?** While in general this is a subtle question, when there are no forces acting on the particle, the situation is much simpler. This is because in this case, no forces translates into a vanishing potential energy $V(x) = 0$, and hence the **Hamiltonian** becomes simply

$$\hat{H} = \frac{\hat{P}^2}{2M}, \tag{6.2}$$

i.e. the **total energy** of the particle is just the **kinetic energy**. *When will a particle have a definite kinetic energy?* This is easy to answer: it will have a definite kinetic energy when it has a **definite momentum**. This therefore suggests that for a free particle

> *For a free particle, momentum eigenstates are energy eigenstates.*

We can directly verify that this is the case, and find the corresponding energy eigenvalue:

$$\begin{aligned} \hat{H}|p\rangle &= \frac{\hat{P}^2}{2M}|p\rangle, \\ &= \frac{p^2}{2M}|p\rangle, \end{aligned} \tag{6.3}$$

where to arrive at the second line we used the fact that $\hat{P}^2|p\rangle = \hat{P}\left(\hat{P}|p\rangle\right) = p\hat{P}|p\rangle = p^2|p\rangle$, i.e. $|p\rangle$ is an eigenstate of $\hat{P}^2$ with eigenvalue p^2.

What (6.3) shows is that $|p\rangle$ is an eigenstate of $\hat{H}$ (i.e. an energy eigenstate) with energy $E = p^2/2M$. This is **exactly as it should be**, since if a particle has a definite momentum p, then it has a (definite) kinetic energy $K = p^2/2M$.

There are **two important points to note** at this stage. First, there are in fact **two** states each with the same energy: the state $|p\rangle$ and $|-p\rangle$ both have energy $E = p^2/2M$. This is intuitive, since the kinetic energy doesn't care about **the velocity** but only the **speed** of the particle, and is hence insensitive to the

sign of the momentum. This means that **there are two energy eigenstates for every energy eigenvalue**. This is known as **degeneracy** and we will study this further once we move into 3D at the end of this unit. Because of this degeneracy in energy, it will be essential to keep track of the momentum, and hence we will work with momentum eigenstates (and momentum wavefunctions), rather than energy eigenstates.

Second, since momentum is **continuous** we see that for a free particle, **energy is always continuous**. This is consistent with what we saw in the previous chapter: there we learnt that the energy will be continuous when the particle isn't **bound** to a potential well. Since there is no potential well for a free particle, the particle is most certainly not bound, and hence we have a continuous range of energies, as expected.

6.2 Time evolution of a free particle I

Given the above, it is now straightforward to write down the time evolution of a free particle in quantum mechanics. As in the previous chapter, we will begin by expressing the initial state $|\psi_{\text{in}}\rangle$ in terms of the energy eigenstates for a free particle, i.e. given the discussion above, of momentum eigenstates. We can realise that **this is exactly as we did previously in Chapter 2**, in terms of the momentum wavefunction. In particular, what we called $\phi_{\text{in}}(E)$ in the previous chapter, we will call here $\tilde{\psi}_{\text{in}}(p)$, since (5.6) becomes

$$|\psi_{\text{in}}\rangle = \int_{-\infty}^{\infty} \tilde{\psi}_{\text{in}}(p)|p\rangle dp, \tag{6.4}$$

where we add the subscript in to the momentum wavefunction to denote the fact that this is the initially specified momentum wavefunction. We can now use the **superposition principle** (i.e. (5.12)) to **write down** the state $|\psi(t)\rangle$ at time t. In particular, since $|p\rangle$ is an energy eigenstate with energy $E = p^2/2M$, it will evolve into $e^{-iEt/\hbar}|p\rangle = e^{-ip^2t/2M\hbar}|p\rangle$, at time t, and hence the superposition (6.4) will evolve into

$$|\psi(t)\rangle = \int_{-\infty}^{\infty} \tilde{\psi}_{\text{in}}(p)e^{-ip^2t/2M\hbar}|p\rangle dp, \tag{6.5}$$

which is readily seen to satisfy the initial condition $|\psi(0)\rangle = |\psi_{\text{in}}\rangle$.

It is insightful to express this in a slightly different way, solely in terms of **momentum wavefunctions**. What (6.5) shows is that if we know that the momentum wavefunction at $t = 0$ is

$$\tilde{\psi}(p, 0) = \tilde{\psi}_{\text{in}}(p), \tag{6.6}$$

then it will evolve into the wavefunction $\tilde{\psi}(p, t)$ given by

$$\tilde{\psi}(p, t) = e^{-ip^2t/2M\hbar}\tilde{\psi}_{\text{in}}(p). \tag{6.7}$$

This equation restates what we already knew, but makes it much clearer. In particular, it says that for a free particle, if we specify the state in terms of the momentum wavefunction, then the evolution is **remarkably simple**: the wavefunction just acquires a phase $e^{-ip^2t/2M\hbar}$ in time! This is because momentum eigenstates are states of definite energy — and are thus **stationary states**.

6.3 Time evolution of a free particle II

Equations (6.5) and (6.7) in principle give a **complete solution** to the evolution of a free quantum particle. Hence, in principle, we could end the chapter here (after an example maybe). However, this isn't a particularly useful place to stop. The reason for this is that usually we are interested in the **spatial wavefunction** and **its** evolution in time. What we would really like to be able to do therefore is to specify an initial spatial wavefunction $\psi_{\text{in}}(x)$, and find the wavefunction $\psi(x, t)$ that it evolves into. We will do this in this section.

We already have all the ingredients we need, we just need to put them all together. First, if the initial condition is specified by a spatial wavefunction $\psi_{\rm in}(x)$, then in order to use (6.7) we need to recall how to express $\tilde{\psi}_{\rm in}(p)$ in terms of it. This is through the (inverse) Fourier transform, (2.20). That is,

$$\tilde{\psi}_{\rm in}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi_{\rm in}(x) e^{-ipx/\hbar} dx. \tag{6.8}$$

Substituting this into (6.7), we have thus half-solved our problem:

$$\tilde{\psi}(p,t) = e^{-ip^2 t/2M\hbar} \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi_{\rm in}(x) e^{-ipx/\hbar} dx, \tag{6.9}$$

since this now gives a momentum wavefunction at time t in terms of a spatial initial wavefunction. We can finish off our goal, by performing a **second** Fourier transform (now using (2.19)), transforming the momentum wavefunction $\tilde{\psi}(p,t)$ into a spatial wavefunction $\psi(x,t)$. The one **subtlety** that we need to take care with is that we **already used x as our (dummy) integration variable**, but we now want to use x instead as the position variable in $\psi(x,t)$. We therefore **need to change our integration variable to x' for consistency**. Altogether, after some simple re-arranging and simplifying, we obtain

$$\psi(x,t) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi_{\rm in}(x') e^{ip(x-x')/\hbar} e^{-ip^2 t/2M\hbar} dx' dp. \tag{6.10}$$

This is a **formidable** looking expression, involving a double integration! From this perspective, quantum evolution looks highly complicated. However, this is **completely equivalent** to (6.7). This highlights an important lesson: just because something appears complicated, doesn't mean it doesn't have a simpler form. Equation (6.10) is **useful** as it shows how a spatial wavefunction at time t depends upon the spatial wavefunction at time $t = 0$, however if we **only** had this we wouldn't really understand anything. In contrast, we do know how to understand this equation: it is the combination of the relationship between spatial and momentum wavefunctions, combined with the **simple** evolution of momentum wavefunctions.

In general it isn't possible to **simplify** or **evaluate** analytically the right-hand side of (6.10). That is, if you want to consider a generic initial wavefunction $\psi_{\rm in}(x)$, and ask how it evolves in time, then it isn't possible to write down a closed form expression (one not involving integral(s)) for the wavefunction at time t. *Does this mean that (6.10) is useless?* The answer is absolutely not. Even though we can't get a closed form expression, it is very easy to numerically integrate the right-hand side, and thus study the evolution numerically.

> ***Numerical integration.*** *It is worth noting a couple of points about the numerical integration. First, you will need to discretise x, and evaluate the double integral at a finite set of points x_i. The resolution you choose will depend upon the problem, and the amount of time you are willing to spend. Second, a little thought needs to be given to the limits of integration. If the initial wavefunction is localised in space, then the integration over x' can be restricted to this region. Furthermore, we don't need to integrate over all momentum in general, as the particle will also only have momentum in a given range (roughly of the size of $\hbar$/(size of region in x) due to the HUP).*

6.4 Gaussian wavefunction

There is one special initial wavefunction whose evolution can be solved analytically. This is the **Gaussian wavefunction** which we will now study.

Consider that the wavefunction of a particle is initially given by

$$\psi_{\rm in}(x) = \left(\frac{1}{a\sqrt{2\pi}}\right)^{1/2} e^{-x^2/4a^2} e^{ip_0 x/\hbar}, \tag{6.11}$$

where $a > 0$ and $p_0 \geq 0$ are constants. This is closely related to the wavefunction we encountered in (3.1), except for the additional **phase factor** $e^{ip_0 x/\hbar}$. As you will show in an exercise, this has the effect

of **increasing the average momentum of the particle by** p_0. Hence, we are now considering a **moving** particle instead of a stationary one, which initially has a spread $\Delta x = a$.

Substituting this into (6.10), after a lengthy and fiddly calculation, it is possible to solve analytically for $\psi(x,t)$. We find

$$\psi(x,t) = \left(\frac{1}{\alpha(t)\sqrt{2\pi}}\right)^{1/2} e^{-(x-p_0 t/M)^2/4\alpha(t)^2} e^{i\phi(x,t)}, \tag{6.12}$$

where $e^{i\phi(x,t)}$ is a **time-dependent phase factor** (which we won't actually write down, as its form isn't too important), and where

$$\alpha(t) = a\sqrt{1 + \frac{\hbar^2 t^2}{4M^2 a^4}}. \tag{6.13}$$

Here we go through the 'lengthy and fiddly calculation' behind $\psi(x,t)$ one step at a time. To keep the algebra short we write $\beta = a^2\left(1 + \frac{i\hbar t}{2Ma^2}\right)$.

$$\tilde{\psi}_{\text{in}}(p) = \sqrt{\frac{\sqrt{2}\,a}{\sqrt{\pi}\,\hbar}}\, e^{-a^2(p-p_0)^2/\hbar^2}$$

This is the momentum wavefunction of the initial state. The phase factor $e^{ip_0 x/\hbar}$ shifts the Gaussian of (3.3) by p_0 (an exercise in this chapter).

$$\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\psi}_{\text{in}}(p)\, e^{ipx/\hbar}\, e^{-ip^2 t/2M\hbar}\, dp$$

Each momentum state has energy $E_p = p^2/2M$ and evolves with $e^{-iE_p t/\hbar}$; transforming back to x gives the wavefunction at time t (this is (6.10) with the x' integral already done).

$$-\frac{a^2(p-p_0)^2}{\hbar^2} + \frac{ipx}{\hbar} - \frac{ip^2 t}{2M\hbar} = -\frac{\beta}{\hbar^2}p^2 + \frac{2a^2 p_0 + i\hbar x}{\hbar^2}p - \frac{a^2 p_0^2}{\hbar^2}$$

We collect the exponent as a quadratic in p, so that the p^2 terms combine into $-\beta p^2/\hbar^2$.

$$= -\frac{\beta}{\hbar^2}(p - p_c)^2 + E_c, \qquad p_c = \frac{2a^2 p_0 + i\hbar x}{2\beta}, \quad E_c = \frac{(2a^2 p_0 + i\hbar x)^2}{4\beta\hbar^2} - \frac{a^2 p_0^2}{\hbar^2}$$

Complete the square in p.

$$\int_{-\infty}^{\infty} e^{-\beta(p-p_c)^2/\hbar^2}\, dp = \sqrt{\frac{\pi\hbar^2}{\beta}}$$

A Gaussian integral with a complex coefficient (allowed since $\operatorname{Re}\beta > 0$); the complex shift p_c does not matter.

$$\psi(x,t) = \sqrt{\frac{\sqrt{2}\,a}{\sqrt{\pi}\,\hbar}}\, \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{\pi\hbar^2}{\beta}}\, e^{E_c}$$

Put the pieces together. Everything about the shape is in E_c and β.

$$|\psi(x,t)| = \left(\frac{1}{\alpha(t)\sqrt{2\pi}}\right)^{1/2} e^{-(x-p_0 t/M)^2/4\alpha(t)^2}, \qquad \alpha(t) = a\sqrt{1+\frac{\hbar^2 t^2}{4M^2a^4}}$$

Take the modulus: $|\beta| = a\,\alpha(t)$ and $\mathrm{Re}\,E_c = -(x-p_0t/M)^2/4\alpha(t)^2$ (a few lines of algebra with $\beta = a^2 + i\hbar t/2M$). What is left of $e^{E_c}/\sqrt{\beta}$ is a pure phase, $e^{i\phi(x,t)}$: this is equation (6.12).

This wavefunction is remarkably similar in form to (6.11), except for a couple of key differences:

1. The Gaussian becomes centred at $p_0 t/M$ at time t. That is, the Gaussian is moving with **constant velocity** $v = p_0/M$. This is similar to a **classical free particle** which moves with constant velocity!
2. The **width** of the Gaussian changes in time. Recall that the standard deviation of the position of the particle was initially given by $\Delta x = a$. Given the form of the wavefunction (6.12), we can read off that
$$\Delta x(t) = \alpha(t) = a\sqrt{1+\frac{\hbar^2 t^2}{4M^2a^4}}. \tag{6.14}$$
This **grows in time**, from the initial value $\Delta x(0) = a$. It is important to note that $\alpha(t)$ is **independent of** p_0, i.e. independent of how fast the particle is moving. This is thus a **universal** feature of the dynamics of the wavefunction. We will return to this below.
3. Finally, there is a phase factor $e^{i\phi(x,t)}$. This term is complicated, but it accounts for **correlations** which build up, between **where the particle is** and **what momentum it has**.

It is insightful to plot the wavefunction as it evolves in time. We do this below, both for the wavefunction $\psi(x,t)$ and the associated probability density $\wp(x,t) = |\psi(x,t)|^2$, both as an animation and in stills:

6.5 Spreading of wavefunctions

We will end this chapter by discussing in more detail the spreading of wavefunctions of free particles in time. This is a universal feature (not just a feature of the Gaussian example above), and arises due to the HUP, hence is a genuinely quantum phenomenon.

Recall from the HUP, that any initial wavefunction we consider $\psi_{\text{in}}(x)$, will **necessarily have a superposition of momenta**. In particular, it will have an associated momentum wavefunction $\tilde{\psi}_{\text{in}}(p)$, which determines which momenta the particle has. Now, if a particle had a definite momentum, then it would move with a definite velocity. However, it has a superposition of momenta, and hence it will move with a superposition of velocities.

If we consider a fixed time, we therefore see that the particle will have a superposition of **displacements**. This is precisely what **leads to the spreading of wavefunctions**. We can see two interesting hallmarks of this if we look carefully. First, from the HUP, we know that in order for a particle to have an initial wavefunction which is **more localised in space** (i.e. with a smaller Δx), then it will necessarily need to have a larger spread of momentum Δp. From the above discussion, we should therefore expect it to **spread faster** in time. This is precisely what we see analytically for the Gaussian in (6.14): if we make a smaller, then $\alpha(t)$ increases.

Second, this also explains why the phase factor $e^{i\phi(x,t)}$ occurs (and has a complicated form!). As we have explained above, the spreading is due to the superposition of momenta. It therefore follows that if the particle is further away (at the 'front' of the wavefunction) at later times, we would expect it to have higher momentum, and if it is closer to the initial location (at the 'back' of the wavefunction), we would expect it to have lower momentum. That is, **correlations build up in time between where the particle is and what momentum it has**. This is precisely what is captured by the phase factor, and why we should **always** expect to have some phase factor in time.

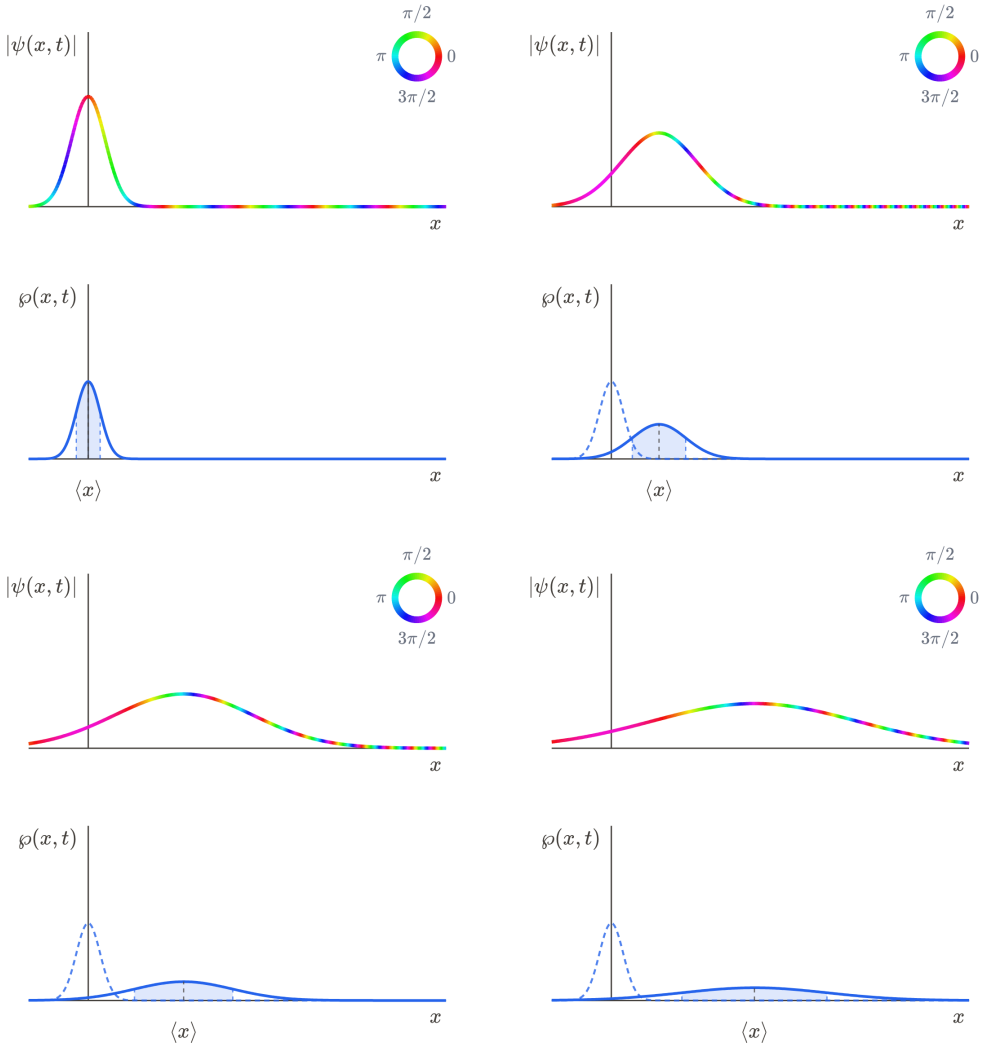


Figure 6.1: Time evolution of a Gaussian wavefunction, as given in (6.12). The parameter $a > 0$ determines the initial 'width' of the wavefunction. Here we represent the complex number $\psi(x,t)$ at each location x and time t in polar form, in terms of the amplitude $|\psi(x,t)|$ (the height), and phase, represented by the colour (with the colour-wheel legend), and below it the probability density $\wp(x,t)$ with the spread $\Delta x(t)$ marked. The particle has a non-zero average momentum $p_0 > 0$, and moves to the right at constant speed, spreading out as it moves. Four stills, at $t = 0$, 2τ, 4τ and 6τ with $\tau = 2Ma^2/\hbar$, of the interactive figure in the online notes.

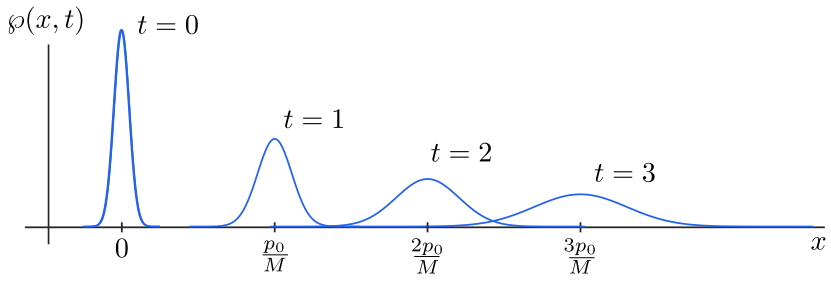


Figure 6.2: Time evolution of probability density $\wp(x,t)$ for a Gaussian wavefunction as given in (6.12).

7. The infinite square well

In the previous chapter we considered the simplest possible scenario, that of a **free particle** where there are no forces acting whatsoever. Here we will begin our exploration of how quantum particles behave when there are forces acting. This is a much more natural and common situation to consider, and is the generic one we encounter in physics.

As already discussed in Chapter 4, there are two **distinct** situations which we can consider. The first is where the forces **confine the particle to a finite region of space**. In this case, we often say that it is **trapped inside a potential well**. As we already stated, in quantum mechanics, it is in this situation where we end up with the famous **quantisation of energy**, the most notable example being the energy levels of atoms, which arise due to the fact that electrons are bound within the potential well of the nucleus (via the electromagnetic attraction).

In this chapter we will study the **simplest** example of a potential well, and see explicitly **how** the energy levels arise. This model isn't very **physical**, but it is mathematically simple, and serves as an ideal starting point from which to study potential wells in quantum mechanics.

The second situation we can encounter — as we did in the previous chapter — is when the forces do not confine a particle to a finite region of space. We will return to more interesting examples — this time involving forces — in Chapter 9.

After introducing the infinite square well, we will show how to find the energy eigenfunctions. We will see that we need a new fact about energy eigenfunctions, relating to their **continuity**. Remarkably, it is in fact from this that the quantisation of energies will arise, as we will see in detail below.

We will then consider a simple example of the evolution in time of a particle inside the infinite square well, seeing that it is very different from what we saw in the previous chapter, for a free particle.

7.1 The infinite square well potential

We are going to consider a situation where a particle is **perfectly confined inside a box** of width L. We will assume that there are walls at $x = 0$ and $x = L$, and want to model these as being **perfectly elastic** so that the particle definitely bounces off of these walls. We can model this using the following potential energy function,

$$V(x) = \begin{cases} 0 & \text{if } 0 \leq x \leq L, \\ \infty & \text{if } x < 0 \text{ or } x > L. \end{cases} \tag{7.1}$$

This potential well is depicted in Figure 7.1.

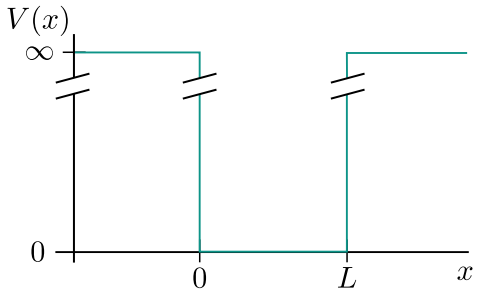


Figure 7.1: **Infinite square well potential.** The potential energy of the infinite square well. The potential vanishes for $0 \leq x \leq L$, and is infinite otherwise. This well is an idealisation for a situation where a particle is trapped inside a 'box', i.e. between two perfectly elastic and impenetrable walls.

Why do we take the potential energy to be infinite outside of the box? Well, this means that the particle would have to have infinite energy to be outside. Thus, under the **physical assumption** that the particle

can only have a **finite energy** then we see that this is a way of modelling a perfectly confined particle. This potential energy is itself unphysical, however we can take it as a **good approximation** to a **deep but finite well**. That is, one where the energy outside the well is V_0, which we take to be as large as we like.

7.2 Energy eigenstates of the infinite square well

The first question we would like to ask about the infinite square well is *what are the states of definite energy?* We are interested in this question, as always, since it is the first step towards understanding the dynamics of a particle confined inside our box.

We are thus interested in finding the energy eigenstates of the Hamiltonian, i.e. of finding solutions to (4.10). Since we have a **potential energy which depends upon position**, we will find the energy eigenstates by solving the TISE (4.12) instead. As we learnt in Chapter 4, this is a completely equivalent way of finding the energy levels and energy eigenstates, and is the natural way to do so in this situation.

Because of the fact that the potential energy is specified **in pieces**, it means that the Hamiltonian operator $\hat{H}_{\mathrm{w}} = \hat{K}_{\mathrm{w}} + \hat{V}_{\mathrm{w}}$ will also be specified in pieces. What this means in practice is that **we will need to consider the different regions separately**, and then understand how to **put them together afterwards**. That is, we will need to consider the TISE in the region $x < 0$, the region $0 \leq x \leq L$, and the region $x > L$ separately. We will refer to these, respectively, as **regions I, II and III**. If we denote a generic energy eigenfunction by $u_E(x)$, then we will specify this **in pieces** accordingly, via

$$u_E(x) = \begin{cases} u_E^{\mathrm{I}}(x) & \text{if } x < 0, \\ u_E^{\mathrm{II}}(x) & \text{if } 0 \leq x \leq L, \\ u_E^{\mathrm{III}}(x) & \text{if } x > L. \end{cases} \tag{7.2}$$

7.2.1 Energy eigenfunctions in Regions I and III

Region I, when $x < 0$, is the region **outside and to the left** of the box. In this region, $V(x) = \infty$ and hence also $\hat{H}_{\mathrm{w}} = \infty$. This is a **strange situation**! We however already implicitly stated the solution. The entire reason for considering the infinite square well is to have the particle **confined within the region $0 \leq x \leq L$, inside the box**. A different way of phrasing this mathematically is that **the wavefunctions must necessarily always vanish outside of the well**. Thus, we have

$$u_E^{\mathrm{I}}(x) = 0. \tag{7.3}$$

> ***Mathematical derivation.*** *You may feel slightly uncomfortable about this, as we haven't shown it mathematically, but just **stated** it as the answer. Furthermore, it isn't clear how we can verify it to satisfy the TISE $\hat{H}_{\mathrm{w}} u_E^{\mathrm{I}}(x) = E u_E^{\mathrm{I}}(x)$, since in region I $\hat{H}_{\mathrm{w}} = \infty$, and so we have an indeterminate expression on the left-hand side $\infty \cdot 0$. If we however consider that the potential energy is **finite** outside the well, and equal to V_0, we can then confirm that we trivially satisfy $0 = 0$ in the TISE. This is ultimately the root of the mathematical justification for (7.3). We will however here accept it on intuitive grounds and not go into the mathematical details further.*

Region III, when $x > L$, is **identical** to region I, in that the potential energy is also infinite here. For the same reasons, we must also take any energy eigenfunction to vanish in this region, and so have

$$u_E^{\mathrm{III}}(x) = 0. \tag{7.4}$$

7.2.2 Energy eigenfunctions in Region II

Region II is our primary region of interest, as this corresponds to $0 \leq x \leq L$, inside the box. In this region $V(x) = 0$, and hence the Hamiltonian is simply

$$\hat{H}_{\mathrm{w}} = \hat{K}_{\mathrm{w}} = \frac{\hat{P}_{\mathrm{w}}^2}{2M} = -\frac{\hbar^2}{2M}\frac{\partial^2}{\partial x^2}. \tag{7.5}$$

That is, inside the well, **the potential energy vanishes** and hence the particle is **essentially free** with no forces acting. We are thus in a very similar situation to Chapter 6. We can use this to realise that we already know what the energy eigenfunctions must be in region II: **they must coincide with momentum eigenfunctions** — i.e. they must be **complex plane waves**. We can check this directly. In particular, we can see that e^{ikx} is an eigenfunction of $\hat{H}_{\mathrm{w}}$, for any value k,

$$\begin{aligned}\hat{H}_{\mathrm{w}}e^{ikx} &= -\frac{\hbar^2}{2M}\frac{\partial^2}{\partial x^2}e^{ikx},\\ &= \frac{\hbar^2k^2}{2M}e^{ikx},\end{aligned} \tag{7.6}$$

and the corresponding eigenvalue is $E = \frac{\hbar^2k^2}{2M}$.

However, as we also saw in Chapter 6, we have **degeneracy**, and the complex plane wave with $k' = -k$ is **an eigenfunction with the same energy eigenvalue**

$$E' = \frac{\hbar^2(k')^2}{2M} = \frac{\hbar^2(-k)^2}{2M} = \frac{\hbar^2k^2}{2M} = E. \tag{7.7}$$

Since e^{ikx} and e^{-ikx} are both eigenfunctions with the **same** eigenvalue, as you will confirm in an exercise, **any superposition** of these two functions is **also** an eigenfunction with the same eigenvalue $E = \frac{\hbar^2k^2}{2M}$. At this stage, we don't yet know **what is the correct superposition** to consider, so we will keep things general, and say that the energy eigenfunction will be of the form

$$u_E^{\mathrm{II}}(x) = Ae^{ikx} + Be^{-ikx}, \tag{7.8}$$

where A and B are **arbitrary (complex) constants** which we don't yet know, and $E = \frac{\hbar^2k^2}{2M}$.

7.2.3 Continuity of energy eigenfunctions & energy quantisation

Where have we made it to? Well, summarising the above, we have analysed the TISE in each of the three regions separately, and so far can conclude that the energy eigenfunctions must have the form

$$u_E(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & \text{if } 0 \le x \le L,\\ 0 & \text{if } x < 0 \text{ or } x > L.\end{cases} \tag{7.9}$$

In order to make progress, we now need to use a new fact about energy eigenfunctions that we haven't needed to introduce until this point:

> *Energy eigenfunctions must be **continuous**.*

Recall that a **continuous function** is one which doesn't have any **jumps**. That is, there cannot be two neighbouring points which differ by a finite amount. An example of a **discontinuous** function is shown in Figure 7.2.

We need to use this fact **at the boundaries between the three regions** at the walls of the box, at $x = 0$ and $x = L$. In particular, because of the fact that the eigenfunctions vanish for $x < 0$, continuity means that the wavefunction **must also vanish at $x = 0$, at the wall.** Substituting $x = 0$ into (7.9), we can therefore conclude that

$$A + B = 0. \tag{7.10}$$

That is, whereas before we **didn't know which superposition to take**, continuity at $x = 0$ shows us that we can only take superpositions with $B = -A$, and hence in region II $u_E(x) = A(e^{ikx} - e^{-ikx})$. However, we can use **Euler's formula** to express this as $u_E(x) = 2iA\sin kx$. We then see nicely, since $\sin 0 = 0$, that the eigenfunction vanishes at the left-hand wall, as we require.

Turning now to the boundary between regions II and III, the right-hand wall, in order for the eigenfunction to vanish at $x = L$, we must take

$$2iA\sin kL = 0. \tag{7.11}$$

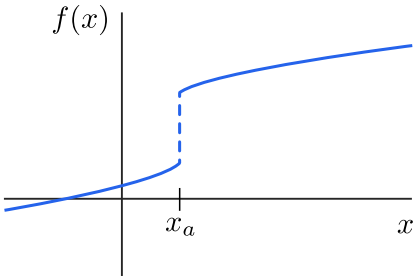


Figure 7.2: **Example of a discontinuous function.** The function which has a discontinuity (i.e. a jump), at $x = x_a$. Energy eigenfunctions must be continuous functions, and cannot have jumps like in this example.

One possible solution appears to be $A = 0$, however a little thought shows that this can't be correct. If $A = 0$, then $u_E(x) = 0$, and hence we have a **trivial mathematical solution**, but not an interesting physical one. In particular, we need energy eigenfunctions — as valid quantum wavefunctions — to be **normalised**. A wavefunction that vanishes everywhere certainly isn't normalised. Hence this isn't what we are looking for.

The only other possibility is that $\sin kL = 0$. While this definitely **isn't true in general**, **there are special values of k for which it will be true**! In particular, we know that $\sin\theta = 0$ if $\theta = 0$, π, 2π, 3π, ..., that is, if $\theta = n\pi$ for n an integer. Thus, we must choose k such that $kL = n\pi$, i.e.

$$k = \frac{n\pi}{L}. \tag{7.12}$$

For these special values of k, we will have $\sin kL = 0$, so the eigenfunction vanishes at the right-hand wall.

Before going on, we can realise quickly that n should be a **positive integer**. First, if $n = 0$, then $k = 0$ and so we have $\sin kx = \sin 0 = 0$. This is just the same as taking $A = 0$, which we already saw was mathematically correct, but not physically meaningful. We therefore are not interested in $n = 0$. For $n = -m$ (with m a positive integer), we can use the fact that sine is an **odd** function, $\sin(-\theta) = -\sin\theta$. This means that $\sin n\pi x/L = -\sin m\pi x/L$ and we have the **same** eigenfunction **up to a minus sign**. Since -1 is just a **phase**, and since we saw previously in Stationary states that an overall phase doesn't change any of the physical properties of a quantum state, n and $-n$ in fact **lead to the same energy eigenfunction**, hence we only need to keep one value of n. It is customary to keep the positive one.

Thus, putting everything together, the requirement of **continuity** has led us to the conclusion that k must be a **positive multiple of π/L**. Since the **energy eigenvalues** were nothing but $E = \hbar^2k^2/2M$, we therefore see that for a particle confined to an infinite square well, there are **discrete energy levels**

$$E_n = \frac{\hbar^2\pi^2n^2}{2ML^2}, \quad n = 1, 2, \ldots \tag{7.13}$$

These are the **only energies that the particle can have with certainty**. It is simply impossible for the particle to have any other energy! The energy levels are depicted below in Figure 7.3.

For the corresponding **energy eigenfunctions** $u_{E_n}(x)$, there is **one final detail to take care of**. We still appear to have one parameter left — A. This is however fixed by **normalisation**: we require $u_{E_n}(x)$ to be a normalised wavefunction. As an exercise, you will show that the normalisation constant is $A = \frac{1}{2i}\sqrt{\frac{2}{L}}$ (independent of n). The **energy eigenfunctions** are thus given by

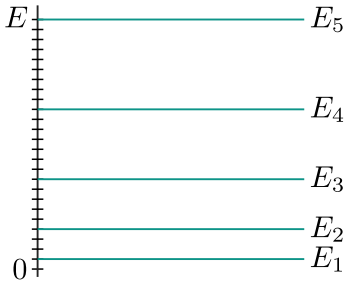


Figure 7.3: **Energy levels of the infinite square well.** The allowed energies of the infinite square well are discrete. The allowed energies are denoted by E_n, and they increase quadratically, proportional to n^2.

$$u_n(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} & \text{if } 0 \le x \le L, \\ 0 & \text{if } x < 0 \text{ or } x > L. \end{cases} \tag{7.14}$$

Note that here we have **changed notation** from $u_{E_n}(x)$ to $u_n(x)$. There are two reasons for doing this. The first is for brevity. The second is that it is much more **common** to label the energy levels by n, and leave it **implicit** that the energy is E_n as given by (7.13). In fact n is often referred to as a **quantum number**, and energy levels and energy eigenfunctions are labelled solely by the quantum number, in this case E_n and $u_n(x)$.

We depict below in Figure 7.4 the first 6 energy eigenfunctions and in Figure 7.5 the associated probability density functions.

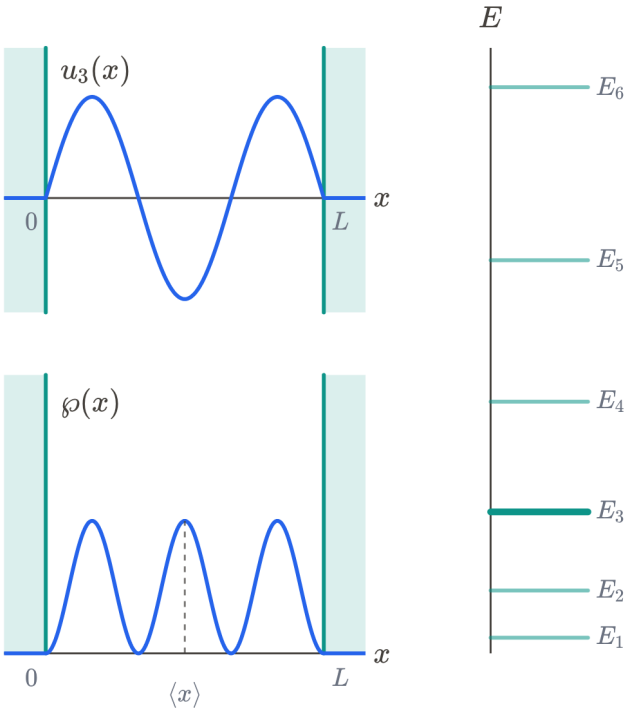


Figure 7.4: **Eigenfunctions of the infinite square well.** The energy eigenfunctions $u_n(x)$ of the infinite square well for $n = 1, \ldots, 6$, their probability densities, and the energy ladder $E_n = n^2 E_1$. Superposing two levels shows a state that is *not* stationary: its probability density sloshes from side to side, as discussed below. A still of the interactive figure in the online notes, showing $n = 3$.

There is clearly some structure here, and also some surprises. The first surprising feature is that the

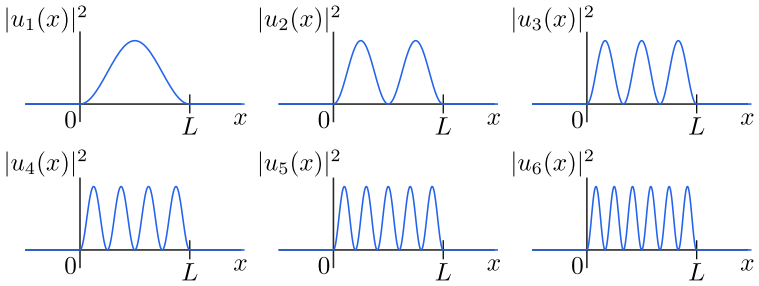


Figure 7.5: **Probability densities of the infinite square well.** The probability densities $\wp(x)$ for the first six energy eigenstates of the infinite square well.

minimum energy the particle can have is strictly bigger than the minimum of potential energy. That is, the lowest energy state — referred to as the **ground state** $|E_1\rangle$ — has energy $E_1 = \frac{\hbar^2\pi^2}{2ML^2} > 0 = V(x)$ inside the well. This is in stark contrast to classical physics, where a particle can be **at rest** at the **minimum of a potential well**, and hence have energy **equal** to the minimum potential energy. In quantum mechanics, in contrast, a particle has **more** energy than this. This additional energy is called the **zero-point energy**. It arises due to the HUP, and the fact that the kinetic and potential energy of a particle are incompatible properties. In particular, because the particle is **confined within the well**, it necessarily has a spread of momenta, and hence a spread of kinetic energy. So the confinement of the particle in fact demands that it will have kinetic energy! The ground state is in fact a **balance** between minimising the potential energy, and minimising the kinetic energy.

Perhaps the second most surprising feature is that when a particle has a definite energy, there are certain regions within the well where **it is very unlikely to find the particle**. For example, in the so-called **first excited state** $|E_2\rangle$ (with associated energy eigenfunction $u_2(x)$) it is very unlikely to find the particle close to the middle of the well!

7.3 Evolution of a particle inside an infinite square well

Having obtained the energy eigenstates and energy eigenvalues, we now again have the ingredients we need in order to study the **evolution** of a particle inside the well. We will once again use the **superposition principle** to achieve this.

Consider a particle prepared initially at $t = 0$ with the wavefunction

$$\psi_{\text{in}}(x) = \sum_{n=1}^{\infty} \alpha_n u_n(x). \tag{7.15}$$

We will start off assuming that **we know the amplitudes** α_n, and come back to the question of how we determine these shortly below. (7.15) specifies the initial wavefunction as a **superposition** of energy eigenfunctions.

As we saw in the previous two chapters, given in this form, we can immediately write down the wavefunction $\psi(x,t)$ that it will evolve into: this is once again an application of the superposition principle: *the evolution of the superposition (of energy eigenfunctions) is just the superposition of the evolutions (of individual energy eigenfunctions)*. Each energy eigenfunction evolves in the usual simple way: by acquiring a phase $e^{-iE_nt/\hbar}$ in time. Thus the wavefunction at time t will be

$$\psi(x,t) = \sum_{n=1}^{\infty} \alpha_n e^{-iE_nt/\hbar} u_n(x), \tag{7.16}$$

precisely as already given in (5.5) (except we updated our notation for energy eigenfunctions from $u_{E_n}(x)$ to $u_n(x)$, as explained above, and now we have infinitely many energy levels, rather than d).

Much like in the previous chapter, there is still a loose end that we should address, namely **how to determine α_n if we are only given $\psi_{\text{in}}(x)$?**

In order to answer this, it is useful to go back to the full quantum state $|\psi_{\text{in}}\rangle$. Written in terms of this, (7.15) is

$$|\psi_{\text{in}}\rangle = \sum_{n=1}^{\infty} \alpha_n |E_n\rangle, \tag{7.17}$$

and we recover (7.15) from this, by taking the scalar product on both sides with a position state $\langle x|$, and recalling that $\langle x|\psi_{\text{in}}\rangle = \psi_{\text{in}}(x)$ and $\langle x|E_n\rangle = u_n(x)$.

If now however we take the scalar product on both sides with an energy eigenstate $\langle E_m|$, recalling from (4.13) that the energy levels are **orthogonal quantum states**, then we see immediately that

$$\begin{aligned} \langle E_m|\psi_{\text{in}}\rangle &= \sum_{n=1}^{\infty} \alpha_n \langle E_m|E_n\rangle, \\ &= \sum_{n=1}^{\infty} \alpha_n \delta_{m,n}, \\ &= \alpha_m. \end{aligned} \tag{7.18}$$

On the other hand, we can use (1.4) to write

$$|\psi_{\text{in}}\rangle = \int_{-\infty}^{\infty} \psi_{\text{in}}(x)|x\rangle dx, \tag{7.19}$$

in terms of its wavefunction. Taking the scalar product on both sides of (7.19) with $\langle E_n|$, we find

$$\begin{aligned} \langle E_n|\psi_{\text{in}}\rangle &= \int_{-\infty}^{\infty} \psi_{\text{in}}(x)\langle E_n|x\rangle dx, \\ &= \int_{-\infty}^{\infty} \psi_{\text{in}}(x) u_n^*(x) dx, \end{aligned} \tag{7.20}$$

where in the second line we used the two facts that $\langle x|E_n\rangle = u_n(x)$ by definition of the energy eigenfunctions, and that $\langle E_n|x\rangle = (\langle x|E_n\rangle)^*$. Combining (7.18) and (7.20), we finally arrive at

$$\alpha_n = \int_{-\infty}^{\infty} \psi_{\text{in}}(x) u_n^*(x) dx. \tag{7.21}$$

> ***Mathematical interpretation.*** *There is a nice mathematical way to understand (7.21): we can understand it as the **scalar product** between the initial wavefunction $\psi_{\text{in}}(x)$, and the energy eigenfunction $u_n(x)$, when we treat the functions as vectors.*

We can use (7.21) to write $\psi_{\text{in}}(x)$ in the form assumed in (7.15). That is, if we aren't given the amplitudes α_n initially, we can use (7.21) to calculate them.

Finally, just as we did for a free particle in (6.10), we can substitute (7.21) into (7.16), to obtain an expression for the wavefunction at time t directly in terms of the initial wavefunction:

$$\psi(x,t) = \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} \psi_{\text{in}}(x') u_n^*(x') u_n(x) e^{-iE_n t/\hbar} dx'. \tag{7.22}$$

This is once again a **formidable expression**, but one which is in fact very similar to (6.10). The difference is that whereas energy was **continuous** for a free particle, it is now **discrete** in the infinite square well, and we see this replaces one of the integrals with a summation. We also can realise that the factor $\frac{1}{2\pi\hbar} e^{ip(x-x')/\hbar}$ is really $v_p^*(x') v_p(x)$, where $v_p(x)$ is the momentum eigenfunction (2.5), which are the

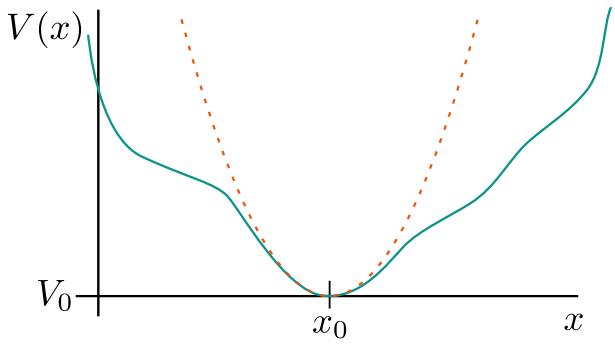
energy eigenfunctions of a free particle. Finally, the term $e^{-ip^2t/2M\hbar}$ is $e^{-iEt/\hbar}$, with $E = p^2/2M$ the energy eigenvalue of $v_p(x)$ for a free particle. Thus the form in both cases is in fact **identical**, up to the change from continuous to discrete energies. This is reassuring, as it shows that what we have done is the same, just in a different context here.

8. The quantum harmonic oscillator

We now turn our attention to arguably the most important system in all of quantum mechanics — the quantum harmonic oscillator. Here we will study the quantum mechanics of a particle subject to a force of the form $F(x) = -kx$. That is, a restorative force proportional to the displacement from the origin. We know that in classical mechanics systems with such forces lead to simple harmonic motion. Our goal now is to understand the quantum mechanics of such systems.

Why is this one of the most important systems to study? The first reason is that it provides an approximation to the behaviour of **almost all** potential wells, as we outline briefly as an aside. Second, you will see later in your degree that the quantum harmonic oscillator reappears in a huge range of situations, from describing light in quantum field theory, to describing phonons in condensed matter physics and far beyond.

The method we will use to find the energy eigenstates of the quantum harmonic oscillator is also a very deep and powerful method. It is often called the **algebraic** approach, and provides a fascinating way of **turning one energy eigenstate into another**. The alternative approach is called the **analytical** approach, and relies on solving the TISE directly. Most textbooks choose one approach or the other (or present them both). Here, we will focus on the former, as it is also the approach which is widely used in other contexts.

> ***Approximation of an arbitrary potential well.*** *Imagine a situation where a particle is acted on by a conservative force $F(x)$, with associated potential energy $V(x)$. Assume that this force pushes the particle back to a position x_0. An example is depicted below:*
>
> 
>
>
> *What is the form of the potential when the particle is close to the bottom of the well? We can **expand** the potential energy around the bottom, to obtain*
>
> $$V(x) \approx V_0 + \left.\frac{dV}{dx}\right|_{x_0} (x - x_0) + \frac{1}{2}\left.\frac{d^2V}{dx^2}\right|_{x_0} (x - x_0)^2 \tag{8.1}$$
>
> *where $V_0 = V(x_0)$.*
>
> *At $x = x_0$, the bottom of the well, the first derivative of the potential vanishes, $\left.\frac{dV}{dx}\right|_{x_0} = 0$, since the bottom is, by definition, a **turning point**. Thus, to leading order, the potential will be approximately*
>
> $$V(x) \approx V_0 + \frac{1}{2}\left.\frac{d^2V}{dx^2}\right|_{x_0} (x - x_0)^2. \tag{8.2}$$
>
> *We can finally make two simplifications without loss of generality: First, we can **redefine the zero of the energy scale**, so that $V_0 = V(x_0) = 0$. Second, we can change our co-ordinate so that the bottom of the well corresponds to $x_0 = 0$. With these two simplifications, we arrive at*
>
> $$V(x) \approx \frac{1}{2}\left.\frac{d^2V}{dx^2}\right|_{0} x^2 \tag{8.3}$$

If we call $k = \left.\frac{d^2V(x)}{dx^2}\right|_0$*, then we arrive precisely at the potential energy* $V(x) = \frac{1}{2}kx^2$ *associated to the force* $F(x) = -kx$*, through* $F(x) = -dV/dx$*.*

This shows that the harmonic oscillator is the leading-order approximation to any potential well in the vicinity of the bottom of the well. It means that if the particle is not too far from $x_0 = 0$*, then the force it experiences (and the potential energy it will have) will be essentially indistinguishable from the linear force of simple harmonic motion.*

This region of small x *is precisely the region that is explored classically by a particle with low energy. It is thus reasonable to assume that for low energies, all systems behave roughly like the harmonic oscillator, and that this is also true in quantum mechanics. Hence, by solving the quantum mechanics of a harmonic oscillator, we obtain an approximation to the quantum mechanics of any particle trapped in a potential well.*

The only caveat to the above is when d^2V/dx^2 *vanishes. In this case, we need to go to higher-order derivatives in order to find the leading-order approximation to the potential well, and the behaviour will not be well approximated by the harmonic oscillator. The infinite square well is precisely a potential of this kind — at the bottom of the well* ***all derivatives*** d^nV/dx^n ***vanish****. However, this indicates that there is no force on the particle whatsoever, which is a very special situation. In general, it will be the case that* d^2V/dx^2 *does not vanish, and that the leading-order behaviour is well approximated by the harmonic oscillator.*

8.1 The harmonic oscillator potential

For the classical harmonic oscillator, the relation between the **spring-constant** k, the **mass** M and **angular frequency** ω is $\omega = \sqrt{k/M}$. For this reason, it is customary to write $k = M\omega^2$ and take the definition of the harmonic oscillator potential to be

$$V(x) = \frac{1}{2}M\omega^2 x^2. \tag{8.4}$$

This is depicted below in Figure 8.1.

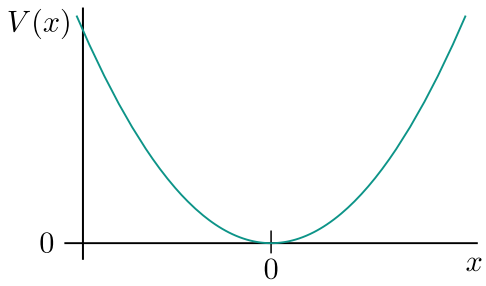


Figure 8.1: **Harmonic oscillator potential.** The potential energy of the harmonic oscillator.

8.2 Generating energy eigenstates using the creation and annihilation operators

Our first goal, as always, is to determine the **energy eigenstates** of the quantum harmonic oscillator. In this chapter instead of going about this directly — by solving the TISE with the potential (8.4) (which is in fact not very straightforward to do at all) — we are going to take a different approach. While at first sight this might seem strange, we will ultimately realise that this is a very powerful technique for finding the energy eigenstates.

The approach we will take is to find a special **pair of operators** called the **creation** and **annihilation operators**. These remarkable operators **transform one energy eigenstate into another**. In fact, we will see that the creation operator generates the next energy level up, while the annihilation operator generates the next one down. In this way, once we have found just **one** energy eigenstate (by whatever means!) we can then use the creation and annihilation operators to **find them all**!

What are these remarkable creation and annihilation operators? To define them, we first need to define a **characteristic length scale** and a **characteristic momentum scale** for the harmonic oscillator. As you will show in the online exercises, the **physical** quantities of the harmonic oscillator — the **mass** M, the **angular frequency** ω (which specifies the scale of the **force** through the spring constant), along with $\hbar$ (the key quantum constant which has dimensions) — can be uniquely combined to specify a length ℓ and a momentum $q = \hbar/\ell$, given by

$$\ell = \sqrt{\frac{\hbar}{M\omega}}, \quad q = \frac{\hbar}{\ell} = \sqrt{M\hbar\omega}. \tag{8.5}$$

With these defined, we can now define the annihilation and creation operators, which we denote by $\hat{a}$ and $\hat{a}^\dagger$ respectively, as

$$\begin{aligned} \hat{a} &= \frac{1}{\sqrt{2}}\left(\frac{\hat{X}}{\ell} + i\frac{\hat{P}}{q}\right) = \frac{1}{\sqrt{2}}\left(\frac{\hat{X}}{\ell} + i\frac{\ell\hat{P}}{\hbar}\right) \quad \text{annihilation op.} \\ \hat{a}^\dagger &= \frac{1}{\sqrt{2}}\left(\frac{\hat{X}}{\ell} - i\frac{\hat{P}}{q}\right) = \frac{1}{\sqrt{2}}\left(\frac{\hat{X}}{\ell} - i\frac{\ell\hat{P}}{\hbar}\right) \quad \text{creation op.} \end{aligned} \tag{8.6}$$

That is, the creation operator is a **dimensionless** combination of the position and momentum operators, with the momentum operator multiplied by the imaginary number i (and an overall normalisation factor of $1/\sqrt{2}$).

Why did we call the creation operator $\hat{a}^\dagger$? Because it is indeed the **Hermitian conjugate operator to $\hat{a}$**: Recall that since position and momentum are **observable quantities** the operators $\hat{X}$ and $\hat{P}$ are **Hermitian**, satisfying $\hat{X}^\dagger = \hat{X}$ and $\hat{P}^\dagger = \hat{P}$. Thus, when we take the Hermitian conjugate of $\hat{a}$, the **only** change is to the sign of the imaginary number i, which is precisely the definition of $\hat{a}^\dagger$.

So, how do we see that these operators do as stated? The claim is that if we act with the creation operator on an energy eigenstate $|E_n\rangle$ (of the quantum harmonic oscillator), then we will obtain (up to normalisation) **the energy eigenstate $|E_{n+1}\rangle$.** To be an energy eigenstate means that $\hat{H}|E_n\rangle = E_n|E_n\rangle$. Thus, the claim above, expressed mathematically, is that

$$\hat{H}\left(\hat{a}^\dagger|E_n\rangle\right) = E_{n+1}\left(\hat{a}^\dagger|E_n\rangle\right), \tag{8.7}$$

since if this equation is satisfied, it means that $\left(\hat{a}^\dagger|E_n\rangle\right)$ is an energy eigenstate of $\hat{H}$, with eigenvalue E_{n+1}, i.e. it is the next energy level.

In order to show (8.7) holds, it is easiest to return to the notion of the **commutator** that we introduced previously in Commutation. In particular, if we calculate $[\hat{H}, \hat{a}^\dagger]$, and find that it is equal to the operator $\hat{b}$ (which we will come to below), then this would mean that $\hat{H}\hat{a}^\dagger - \hat{a}^\dagger\hat{H} = \hat{b}$, and so by re-arranging, $\hat{H}\hat{a}^\dagger = \hat{a}^\dagger\hat{H} + \hat{b}$. We can then substitute this into the left-hand side of (8.7) to obtain

$$\begin{aligned} \hat{H}\hat{a}^\dagger|E_n\rangle &= \left(\hat{a}^\dagger\hat{H} + \hat{b}\right)|E_n\rangle, \\ &= \left(E_n\hat{a}^\dagger + \hat{b}\right)|E_n\rangle. \end{aligned} \tag{8.8}$$

At first sight this might not look too helpful. However, at this stage it is useful to determine $\hat{b}$. In the online exercises below, by using the explicit form of $\hat{a}^\dagger$ from (8.6), as well as the explicit form of the Hamiltonian operator, with $\hat{V} = \frac{1}{2}M\omega^2\hat{X}^2$ from (8.4), and the two commutators

$$[\hat{P}^2, \hat{X}] = -2i\hbar\hat{P}, \quad [\hat{X}^2, \hat{P}] = 2i\hbar\hat{X}, \tag{8.9}$$

both of which you prove as problems in problem classes, you will show that

$$[\hat{H}, \hat{a}^\dagger] = \hbar\omega\hat{a}^\dagger. \tag{8.10}$$

(i.e. $\hat{b} = \hbar\omega\hat{a}^\dagger$). Substituting this into (8.8), we therefore see that

$$\hat{H}\hat{a}^\dagger|E_n\rangle = (E_n + \hbar\omega)\hat{a}^\dagger|E_n\rangle. \tag{8.11}$$

This means that $(\hat{a}^\dagger|E_n\rangle)$ is an **energy eigenstate** of energy $(E_n + \hbar\omega)$ (up to normalisation). This is in fact the energy of the **next** energy level E_{n+1}. **We will not prove this here, but state it as a fact** (which has been shown to be true). One crucial point to note is that the **spacing between the energy levels of the harmonic oscillator is constant, and equal to $\hbar\omega$.** This is a special feature of the harmonic oscillator and for other potential wells, the spacing is not constant (e.g. as we saw for the infinite square well in (7.13), the spacing **grows**).

In a similar fashion, in the online exercises you will show the commutation relation

$$[\hat{H}, \hat{a}] = -\hbar\omega\hat{a}, \tag{8.12}$$

between the Hamiltonian and the **annihilation operator**. From this, you will also show that

$$\hat{H}\hat{a}|E_n\rangle = (E_n - \hbar\omega)\hat{a}|E_n\rangle. \tag{8.13}$$

This shows that $(\hat{a}|E_n\rangle)$ is an energy eigenstate of $\hat{H}$ with energy $(E_n - \hbar\omega) = E_{n-1}$. That is, the annihilation operator transforms an energy level into the **next lowest energy level** (up to normalisation).

We thus have a procedure, starting from any given energy level $|E_n\rangle$, to create any other energy level, by either successively applying the creation operator (to **'climb up'** the energy levels) or by applying $\hat{a}$ (to **'climb down'** them). All we need is a way to find **one** eigenstate, and then we have everything we need. We will now see how we can obtain the **ground state** of the oscillator.

8.3 The ground state of the harmonic oscillator

You might have noticed something strange in the above — we appear to be able to apply the annihilation operator as many times as we like, and continually create new energy levels of progressively lower energy. However, we know that at some point we **must** hit the state of lowest energy — the ground state. We seem to thus have a problem with this approach.

A little thought shows that there is a way out of this problem: we pointed out a couple of times above that $\hat{a}|E_n\rangle$ isn't necessarily **normalised**, and in fact we will see below that it isn't, and this is the key to escaping our problem. In particular, let us imagine that we have indeed found the ground state, which we will denote by $|E_0\rangle$. According to the above, $\hat{a}|E_0\rangle$ will be **proportional** to a state of energy $E_0 - \hbar\omega$. However, if the proportionality constant **vanishes**, then our sequence **terminates**. That is, if $\hat{a}|E_0\rangle = 0$, then we **don't** obtain a new state, and hence we can no longer climb down!

Remarkably, this insight **also allows us to find the ground state** $|E_0\rangle$. More precisely, we can use it to find the ground state eigenfunction $u_0(x)$. *How do we do this?* Well, we know that eigenfunctions satisfy the same equation as eigenstates, except we have to use operators on wavefunctions instead. Therefore, $\hat{a}|E_0\rangle = 0$ is equivalent to

$$\hat{a}_\mathrm{w} u_0(x) = 0, \tag{8.14}$$

where $\hat{a}_\mathrm{w} = \frac{1}{\sqrt{2}}(\hat{X}_\mathrm{w}/\ell + i\ell\hat{P}_\mathrm{w}/\hbar)$, as in (8.6). Substituting in $\hat{P}_\mathrm{w} = -i\hbar\frac{\partial}{\partial x}$ and $\hat{X}_\mathrm{w} = x$, and re-arranging, we arrive at a simple differential equation that must be satisfied by $u_0(x)$:

$$\frac{\partial}{\partial x}u_0(x) = -\frac{x}{\ell^2}u_0(x). \tag{8.15}$$

It isn't too difficult to spot the solution to this: it is nothing but a Gaussian:

$$\frac{\partial}{\partial x}\left(N_0 e^{-x^2/2\ell^2}\right) = -\frac{x}{\ell^2}N_0 e^{-x^2/2\ell^2}, \tag{8.16}$$

where N_0 is a normalisation constant, that we will fix below. This shows that the **ground state of the harmonic oscillator is a Gaussian wavefunction**, with standard deviation $\Delta x = \ell/\sqrt{2}$ (i.e. confirming that, indeed, ℓ is really a characteristic length scale of the harmonic oscillator). We can thus use (3.1) to fix the normalisation, and so the ground state wavefunction is

$$u_0(x) = \sqrt{\frac{1}{\ell\sqrt{\pi}}} e^{-x^2/2\ell^2}. \tag{8.17}$$

As an exercise, by evaluating $\hat{H}_{\rm w} u_0(x)$, you will show that the energy of the ground state is

$$E_0 = \frac{1}{2}\hbar\omega. \tag{8.18}$$

Before determining the rest of the energy eigenstates and eigenvalues, we will first introduce the **number operator** as this will allow us to take care of the normalisation of energy eigenstates.

8.4 The number operator

Having found the energy of the ground state, and given that we saw above that the energy spacing is $\hbar\omega$ (as we saw in (8.11)), this means we now in fact know the energy of all of the energy levels, which is given by

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \ldots \tag{8.19}$$

Note, that whereas for the energy levels of the infinite square well we took n to start at $n = 1$, for the harmonic oscillator we take n to start at $n = 0$. This is **standard** and somewhat unfortunate, but there is a reason for it (which we will explain below).

We can use (8.19) to learn something very interesting about the product operator $\hat{a}^\dagger\hat{a}$. Using the definitions (8.6), we see that

$$\begin{aligned}\hat{a}^\dagger\hat{a} &= \frac{1}{2}\left(\frac{\hat{X}}{\ell} - i\frac{\ell\hat{P}}{\hbar}\right)\left(\frac{\hat{X}}{\ell} + i\frac{\ell\hat{P}}{\hbar}\right), \\ &= \frac{1}{2}\left(\frac{\hat{X}^2}{\ell^2} + \frac{i}{\hbar}\left(\hat{X}\hat{P} - \hat{P}\hat{X}\right) + \frac{\ell^2\hat{P}^2}{\hbar^2}\right).\end{aligned} \tag{8.20}$$

Using the definition of ℓ (8.5), and the canonical commutation relation (3.10), this can be further simplified to give

$$\begin{aligned}\hat{a}^\dagger\hat{a} &= \frac{1}{2}\left(\frac{M\omega\hat{X}^2}{\hbar} + \frac{i}{\hbar}i\hbar + \frac{\hat{P}^2}{\hbar M\omega}\right), \\ &= \frac{1}{\hbar\omega}\left(\frac{\hat{P}^2}{2M} + \frac{1}{2}M\omega^2\hat{X}^2 - \frac{1}{2}\hbar\omega\right).\end{aligned} \tag{8.21}$$

The first two terms in the bracket are nothing but the kinetic energy operator and potential energy operator of the harmonic oscillator, and so together give the Hamiltonian $\hat{H}$. Thus, by re-arranging, we arrive at a special equation,

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right), \tag{8.22}$$

which shows how we can **express the Hamiltonian in terms of the creation and annihilation operators**! Finally, comparing to (8.19), we see that we can give $\hat{a}^\dagger\hat{a}$ a **very nice interpretation**. Since $\hat{H}|E_n\rangle = E_n|E_n\rangle$ by definition, we must have

$$\hat{a}^\dagger\hat{a}|E_n\rangle = n|E_n\rangle. \tag{8.23}$$

We therefore call $\hat{a}^\dagger\hat{a}$ the **number operator**,

$$\hat{n} = \hat{a}^\dagger \hat{a}, \tag{8.24}$$

such that $|E_n\rangle$ are **eigenstates**, with eigenvalue n — the **quantum number** of the energy level.

8.5 Energy eigenstates of the harmonic oscillator

We can use the number operator to figure out how the creation and annihilation operators change the **norm** of a state. Consider that we have an energy eigenstate that is normalised, satisfying $\||E_n\rangle\| = \sqrt{\langle E_n|E_n\rangle} = 1$. Then, the norm of $\hat{a}|E_n\rangle$ will be

$$\begin{aligned} \|\hat{a}|E_n\rangle\| &= \sqrt{(\langle E_n|\hat{a}^\dagger)(\hat{a}|E_n\rangle)}, \\ &= \sqrt{\langle E_n|\hat{a}^\dagger\hat{a}|E_n\rangle}, \\ &= \sqrt{\langle E_n|\hat{n}|E_n\rangle}, \\ &= \sqrt{n\langle E_n|E_n\rangle}, \\ &= \sqrt{n} \end{aligned} \tag{8.25}$$

where to obtain the third line we used (8.24), and in the fourth we used this together with (8.23) to replace the operator by the number n, and pulled it out the front. Thus, combining this with (8.13), we can conclude that the action of $\hat{a}$ on an energy eigenstate is

$$\hat{a}|E_n\rangle = \sqrt{n}|E_{n-1}\rangle. \tag{8.26}$$

To find out how $\hat{a}^\dagger$ changes the norm of a state, we can apply $\hat{a}^\dagger$ to $(\hat{a}|E_{n+1}\rangle)$. We see on the one hand that

$$\hat{a}^\dagger\hat{a}|E_{n+1}\rangle = \hat{n}|E_{n+1}\rangle = (n+1)|E_{n+1}\rangle. \tag{8.27}$$

On the other hand, from (8.26), $\hat{a}|E_{n+1}\rangle = \sqrt{n+1}|E_n\rangle$, and so

$$\hat{a}^\dagger\hat{a}|E_{n+1}\rangle = (\sqrt{n+1})\hat{a}^\dagger|E_n\rangle. \tag{8.28}$$

Equating the right-hand side of (8.27) and (8.28), and dividing by $\sqrt{n+1}$, we therefore see that

$$\hat{a}^\dagger|E_n\rangle = \sqrt{n+1}|E_{n+1}\rangle. \tag{8.29}$$

We finally have everything we need in order to determine the energy eigenstates of the harmonic oscillator! We can generate the nth energy eigenstate by applying $\hat{a}^\dagger$ n times to the ground state $|E_0\rangle$ (which we have already found), and dividing by $\sqrt{n!}$ (where $n! = 1 \times 2 \times \cdots \times n$),

$$|E_n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|E_0\rangle. \tag{8.30}$$

This way, we 'climb up' the energy eigenstates, and remain normalised as we go, successively generating all of them.

What is more interesting and useful however is to obtain the **energy eigenfunctions** which we can also do in a similar manner: In fact, we have

$$u_n(x) = \frac{(\hat{a}_{\text{w}}^\dagger)^n}{\sqrt{n!}}u_0(x). \tag{8.31}$$

For example,

$$\begin{aligned} u_1(x) &= \frac{1}{\sqrt{2}}\left(\frac{\hat{X}_{\mathrm{w}}}{\ell} - i\frac{\ell\hat{P}_{\mathrm{w}}}{\hbar}\right)u_0(x), \\ &= \frac{1}{\sqrt{2}}\left(\frac{x}{\ell} - \ell\frac{\partial}{\partial x}\right)\sqrt{\frac{1}{\ell\sqrt{\pi}}}e^{-x^2/2\ell^2}, \\ &= \frac{1}{\sqrt{2}}\left(\frac{x}{\ell} - \ell\left(-\frac{x}{\ell^2}\right)\right)\sqrt{\frac{1}{\ell\sqrt{\pi}}}e^{-x^2/2\ell^2}, \\ &= \sqrt{\frac{2}{\ell\sqrt{\pi}}}\frac{x}{\ell}e^{-x^2/2\ell^2}. \end{aligned} \tag{8.32}$$

Repeating this procedure, what we find in general is that all of the energy eigenfunctions have the form

$$u_n(x) = N_n h_n(x/\ell)e^{-x^2/2\ell^2}, \tag{8.33}$$

where N_n is a **normalisation factor**, and $h_n(y)$ are functions known as **Hermite polynomials**. These are polynomials of degree n (and this is finally why we start counting at $n = 0$ and not $n = 1$), which alternate between being **even** and **odd** (with n). The first few Hermite polynomials (as well as the associated normalised energy eigenfunction) are given in the following table:

The energy eigenfunctions and corresponding probability densities of the first few energy eigenstates are plotted in Figure 8.2 and Figure 8.3 respectively.

It is insightful to compare to corresponding plots for the eigenfunctions and probability densities of the infinite square well potential from the previous chapter, as given in Figure 7.4 and Figure 7.5. Even though the **functional form** is very different, graphically they are **remarkably similar**! This is a type of **universality**: low energy states generically have similar energy eigenfunctions (even though the energies in general will be very different).

8.6 Tunnelling and barrier penetration

There is one new and very important aspect of quantum mechanics that arises for the harmonic oscillator, which isn't present in the infinite square well potential, known as **tunnelling** (first used by Gamow in 1928 to explain radioactive alpha decay [Gamow, 1928]) or alternatively **barrier penetration**.

Looking back at the potential energy $V(x)$ in Figure 8.1, it is evident that as the particle moves away from the origin, its potential energy increases quadratically. We have now found the energy eigenstates, and their corresponding energy eigenfunctions, as listed in Table 8.1 and depicted in Figure 8.2 and Figure 8.3.

Table 8.1: **Hermite polynomials and normalised energy eigenfunctions of the harmonic oscillator**. We give here the first 4 Hermite polynomials, and the corresponding normalised wavefunctions of the energy eigenstates of the quantum harmonic oscillator.

n	Hermite polynomial $h_n(y)$	Energy eigenfunction $u_n(x)$
0	1	$\left(\frac{1}{\ell\sqrt{\pi}}\right)^{1/2} e^{-x^2/2\ell^2}$
1	$2y$	$\left(\frac{2}{\ell\sqrt{\pi}}\right)^{1/2} \left(\frac{x}{\ell}\right) e^{-x^2/2\ell^2}$
2	$4y^2 - 2$	$\left(\frac{2}{\ell\sqrt{\pi}}\right)^{1/2} \left(\frac{x^2}{\ell^2} - \frac{1}{2}\right) e^{-x^2/2\ell^2}$
3	$8y^3 - 12y$	$2\left(\frac{1}{3\ell\sqrt{\pi}}\right)^{1/2} \frac{x}{\ell} \left(\frac{x^2}{\ell^2} - \frac{3}{2}\right) e^{-x^2/2\ell^2}$

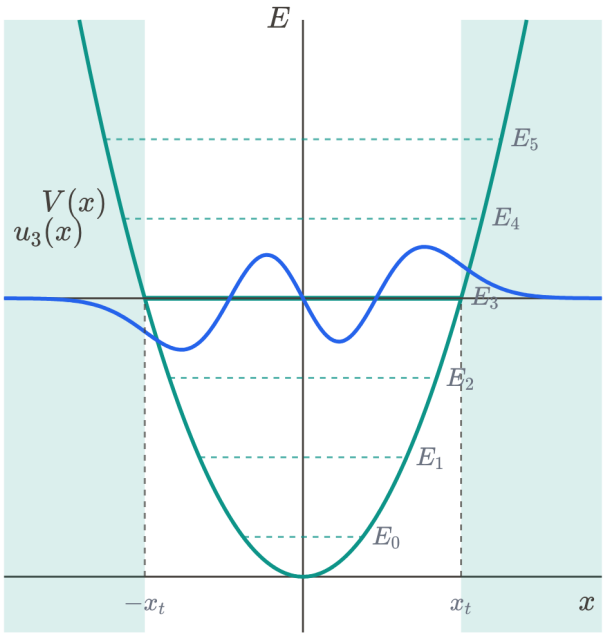


Figure 8.2: **Energy eigenfunctions of the harmonic oscillator.** The energy eigenfunctions $u_n(x)$ of the harmonic oscillator for $n = 0, \ldots, 5$, each drawn on its energy level E_n across the potential $V(x)$. The two markers on the x axis show where **tunnelling** starts: the classically-allowed region is between the markers, where the total energy of the particle is more than the potential energy. A still of the interactive figure in the online notes, showing $n = 3$.

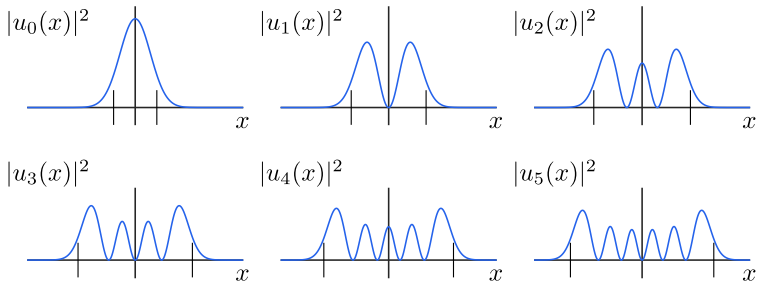


Figure 8.3: **Probability densities of the eigenstates of the harmonic oscillator.** The probability densities for finding the particle at x for the first six energy eigenstates of the harmonic oscillator. The markers again indicate where tunnelling starts.

Now, while the Gaussian function decays rather rapidly, it is in fact non-zero **for all values of** x, and hence when a particle has a definite energy in the harmonic oscillator, there is a non-zero probability to find it in any finite region, no matter how far from the origin.

This is however very strange, as above we just noted that the potential energy increases as we move away from the origin. This means that, no matter what finite energy the particle has (and all the energy levels have finite energy), it has a chance of being found far away **in a region where the potential energy is larger than the total energy** E_n **of the particle!**

This is known as **tunnelling** or **barrier penetration**: a quantum particle manages to **enter a region** in which classically it does not have enough energy to enter!

Remarkably, this isn't even a small effect! In Figure 8.2 and Figure 8.3 we have added vertical lines on the axis to denote the position where the potential energy $V(x) = E_n$. Beyond these points, classically the particle should not be found. As you can see, there is a significant probability to be found in the classically forbidden region, even in the ground state.

Although we won't study it in this course, tunnelling is interesting and important. Consider for example that we have **two potential wells**, separated by an energy barrier. Classically a particle can be trapped in one of them (for example, a planet can be bound in orbit around one star, even if there is a second star that it could equally be bound in orbit around). In quantum mechanics, **because a particle tunnels**, it is able to move between the two wells. This effect is extremely important, for example explaining α decay in particle physics, and being utilised in **semiconductor devices**.

In this unit we won't go further into this fascinating effect, but just highlight its existence, and leave it for future courses to elaborate on the physical consequences of it.

9. Scattering

In the previous two chapters we have focused our attention on situations where a particle is bound inside a potential well. This is a situation which is very prevalent and important. In these cases, we saw that a particle can only have special discrete energies.

Prior to this we studied a free particle — when there were no forces acting, and so far this has been the only instance of a **non-bound** state that we have considered. In this final chapter considering quantum mechanics in 1D, we will now return to a situation where a particle is non-bound, but one that is more interesting and physical than the free particle.

We will consider the situation where a particle **scatters** off of a target, which we model by a change in potential. This is a very important physical situation which arises in a wide range of situations, from neutrons scattering in condensed matter physics, to the scattering of particles off each other in particle detectors. Our goal here is to understand the basics of scattering in quantum mechanics.

We will consider just one simple potential — an abrupt change. The motivation for studying this is somewhat similar to our motivation for studying the infinite square well. While this is far from realistic, it is simple enough that we can solve exactly for the energy eigenstates (which are now no longer momentum eigenstates, like for the free particle). We can then use these to numerically study what happens when a wavepacket scatters off of the step.

We will uncover one final new aspect of quantum mechanics, which is almost the complete **opposite** of tunnelling. Whereas in tunnelling a quantum particle enters a region of space it should not have enough energy to enter, in this chapter we will see that quantum particles **reflect** off of abrupt changes in potential — **whether those changes are positive or negative**. That is, quantum particles reflect off of cliff edges, and therefore **fail to enter a region that they would definitely enter**.

9.1 The finite step potential

In this chapter we will consider the following potential function

$$V(x) = \begin{cases} 0 & \text{if } x < 0, \\ V_0 & \text{if } x \geq 0. \end{cases} \tag{9.1}$$

depicted in Figure 9.1.

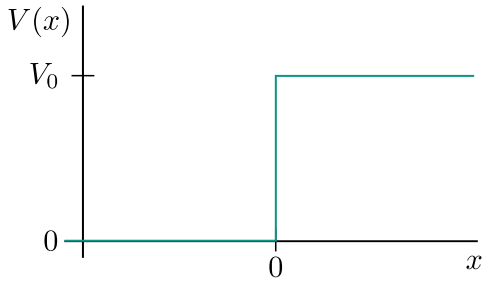


Figure 9.1: **The potential step.** The potential energy vanishes for $x < 0$, and is equal to V_0 for $x \geq 0$. In the figure we have taken $V_0 > 0$, although in the mathematics that follows, we may consider V_0 to be either positive or negative.

This corresponds to an abrupt change in potential from $V(x) = 0$ to $V(x) = V_0$ at the origin. It is important to note that V_0 could be either **positive** or **negative**, meaning that the potential energy either steps up or down. We will primarily keep in the back of our mind that $V_0 > 0$, but will also discuss what happens if $V_0 < 0$.

Our goal, as always, is to find the energy eigenstates of the potential step, which we will do by solving the TISE to find the energy eigenfunctions. Since the particle isn't bound, we expect that the energy should vary **continuously** and the eigenfunctions won't be physical or normalisable. We will indeed see that this is the case below.

9.2 Energy eigenfunctions of the potential step

The approach that we will take is going to be similar to the infinite square well — since the potential energy is specified in pieces, we will seek to solve for the energy eigenfunctions in pieces also. We will call the region to the left of the step **region I** ($x < 0$) and the region to the right of the step **region II** ($x \geq 0$). We will call the energy eigenfunctions in these regions $u_E^{\mathrm{I}}(x)$ and $u_E^{\mathrm{II}}(x)$ respectively, such that

$$u_E(x) = \begin{cases} u_E^{\mathrm{I}}(x) & \text{if } x < 0, \\ u_E^{\mathrm{II}}(x) & \text{if } x \geq 0. \end{cases} \tag{9.2}$$

After solving for the two pieces, we will join them together using the **continuity** of energy eigenfunctions, as we did in the case of the eigenfunctions of the infinite square well (with one additional new ingredient).

9.2.1 Energy eigenfunctions in Region I

We will start off considering Region I, to the left of the step. In this region $V(x) = 0$. Therefore, just as happened inside the infinite square well in Section 7.2.2,

$$\hat{H}_{\mathrm{w}} = \frac{\hat{P}_{\mathrm{w}}^2}{2M} \tag{9.3}$$

and so, just as we were able to do there, we can **immediately write down the energy eigenfunction**, as this is the same Hamiltonian (and the same as the free particle also!). As we discussed in Section 7.2.2, any function of the form

$$u_E^{\mathrm{I}}(x) = Ae^{ikx} + Be^{-ikx}, \tag{9.4}$$

will be an eigenfunction, where A and B are arbitrary complex numbers, and the corresponding energy eigenvalue is

$$E = \frac{\hbar^2 k^2}{2M}. \tag{9.5}$$

Just as in the case of the infinite square well, we **don't yet know which superposition of e^{ikx} and e^{-ikx} we should take** (i.e. how to choose A and B). For the moment we will leave them general, and see that we can determine them later, using continuity.

9.2.2 Energy eigenfunctions in Region II

We now turn our attention to Region II, to the right of the step. In this region $V(x) = V_0$. In this region $\hat{H}_{\mathrm{w}} = \frac{\hat{P}_{\mathrm{w}}^2}{2M} + V_0$, and the TISE we want to solve is

$$\left(\frac{\hat{P}_{\mathrm{w}}^2}{2M} + V_0 \right) u_E^{\mathrm{II}}(x) = E u_E^{\mathrm{II}}(x). \tag{9.6}$$

We can move the term involving V_0 from the left-hand to the right-hand side, to express this as

$$\frac{\hat{P}_{\mathrm{w}}^2}{2M} u_E^{\mathrm{II}}(x) = (E - V_0) u_E^{\mathrm{II}}(x). \tag{9.7}$$

Since V_0 is just a constant, this has again **exactly the same form as for a free particle** except we are looking for an eigenvalue of $E' = (E - V_0)$, instead of E. The only **subtlety** we need to be careful about is that previously we tacitly had $E \geq 0$. We will therefore **make the same assumption that $E' \geq 0$**, i.e.

that $E > V_0$. Physically this means we are going to consider situations where **the particle has enough energy to classically climb the potential step**. We will study the opposite situation as an exercise.

Given the above, we can once again **write down the solution**! In particular, the energy eigenfunction in region II will be identical in form to an energy eigenfunction **without a potential** and with energy E'. The solution is therefore

$$u_E^{\mathrm{II}}(x) = Ce^{ik'x} + De^{-ik'x}, \tag{9.8}$$

where C and D are two new constants (that we will also need to determine in due course), and

$$E - V_0 = \frac{\hbar^2 k'^2}{2M}. \tag{9.9}$$

Taking a step back (no pun intended), this makes **perfect sense**. Once the particle has passed the step, its potential energy has increased by V_0. The total energy is still E, and so the kinetic energy of the particle is now $E - V_0$. This is why we see that the energy eigenfunction has exactly the same form as an energy eigenfunction with $E' = E - V_0$.

9.2.3 Right-moving and Left-moving eigenstates

Putting everything together that we have learnt so far, we know that the energy eigenfunctions of the potential step have the form

$$u_E(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & \text{if } x < 0, \\ Ce^{ik'x} + De^{-ik'x} & \text{if } x \geq 0. \end{cases} \tag{9.10}$$

where A, B, C and D are complex constants, that we need to determine, and k and k' are functions of the energy and step height, specified via (9.5) and (9.9) respectively.

In order to make progress, we first need to cast our mind back to Chapter 6, and the fact that we found there **two degenerate eigenstates for each energy**. *What were the two degenerate eigenstates?* They were the two momentum eigenstates with equal and opposite momentum, $|p\rangle$ and $|-p\rangle$.

Here we are going to have something **very similar**. What we need to understand is **what is the analogue of these two eigenstates here?** The key insight is that a momentum eigenstate with **positive momentum** represents a particle moving to the right, while an eigenstate of **negative momentum** represents one moving to the left. We thus want to identify **right-moving** and **left-moving** eigenstates of the step, as the two basic states of interest.

We will be able to do this, but we need to use a bit of **foresight**. In particular, as stated in the introduction, **quantum particles reflect off of abrupt changes in potential**, irrespective of whether that is a positive or negative change. *What does this imply about left-moving and right-moving states?* Let us start off by considering a right-moving state in region I, which would correspond to e^{ikx} (since this has the same form as a positive momentum state in this region, if $k > 0$). When it reaches the step, it will reflect, creating a component e^{-ikx} (essentially a negative momentum state), and also pass through, creating a component $e^{ik'x}$ in region II (with positive momentum). In region II however, there will be **no negative momentum component** (i.e. component of the form $e^{-ik'x}$).

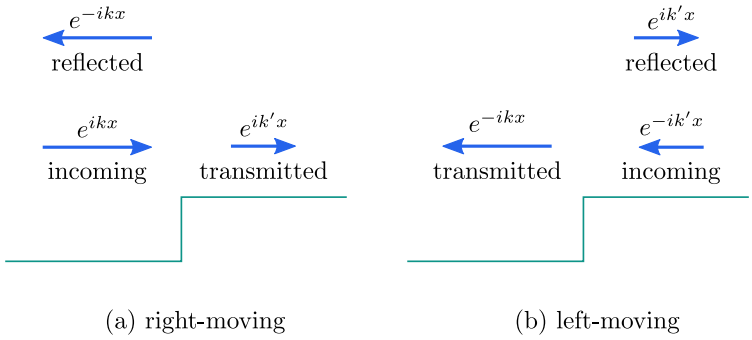
This is summarised graphically in Figure 9.2 (a).

This line of reasoning suggests that we should take $D = 0$ in (9.10) when considering right-moving eigenfunctions. We will denote this eigenfunction by $u_E^{\mathrm{R}}(x)$ ('R' for 'right-moving'), and its general form will be

$$u_E^{\mathrm{R}}(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & \text{if } x < 0, \\ Ce^{ik'x} & \text{if } x \geq 0. \end{cases} \tag{9.11}$$

A similar line of reasoning, which you are encouraged to think through, and which is summarised in Figure 9.2 (b), suggests that we should take $A = 0$ in order to define left-moving eigenfunctions, which we denote by $u_E^{\mathrm{L}}(x)$, and have the general form

$$u_E^{\mathrm{L}}(x) = \begin{cases} B'e^{-ikx} & \text{if } x < 0, \\ C'e^{ik'x} + D'e^{-ik'x} & \text{if } x \geq 0. \end{cases} \tag{9.12}$$



(a) right-moving (b) left-moving

Figure 9.2: **Right-moving and Left-moving situations**. (a) In a 'right-moving' situation, a particle approaches the step from the left, with positive momentum. It is then partially reflected by the step, and partially transmitted. (b) In a 'left-moving' situation, everything is reversed. The particle approaches the step from the right with negative momentum. It is then partially reflected, and partially transmitted.

(where we now use primes on the constants B', C' and D', to highlight the fact that this is a **distinct** eigenfunction with distinct unknown parameters at this stage).

In what follows, we will focus on **right-moving** particles, and leave the corresponding analysis of **left-moving** particles as an exercise.

9.2.4 Continuity of eigenfunctions and their first derivatives

We will now see how we can fix the unknown constants in the eigenfunctions. First, as we learnt in Section 7.2.3, **energy eigenfunctions must be continuous**. We must apply this fact at the potential step ($x = 0$), as for general choices of A, B and C this won't be the case. In particular, from (9.11) we see that at $x = 0$, $u_E^{\mathrm{R}}(0) = C$ (since we include $x = 0$ in region II). If we approach the origin from region I, i.e. take $x \to 0$ from below, we find $u_E^{\mathrm{R}}(0) = A + B$. In order for the eigenfunction to be continuous, we must therefore have

$$A + B = C. \tag{9.13}$$

We can use this equation to solve for one of the constants, e.g. to eliminate C directly. This still leaves us with two constants.

To fix another constant, we need to introduce one **further fact** about energy eigenfunctions:

> *The first spatial derivatives of energy eigenfunctions are **also continuous**.*

Like many good rules, this one **has an exception**! There is only one situation where the spatial derivative of an energy eigenfunction doesn't have to be continuous, and this is **precisely the situation we encountered when studying the infinite square well**: it is when the **potential is infinite**.

We therefore didn't introduce this fact previously, as we happened to be in the one exceptional case. We are introducing it **now**, as for the potential step the potential is **finite everywhere** and hence we do need to impose continuity of the spatial derivative.

What does it mean for a function to have a continuous derivative? This means that there will be no **kinks** in the graph — no points where the function changes slope abruptly. Thus energy eigenfunctions cannot have jumps or kinks (except where the potential is infinite).

Let us now see how to apply this. We can easily calculate the spatial derivative in the two regions, to find that

$$\frac{\partial}{\partial x} u_E^{\mathrm{R}}(x) = \begin{cases} ik(Ae^{ikx} - Be^{-ikx}) & \text{if } x < 0, \\ ik'(A+B)e^{ik'x} & \text{if } x \geq 0. \end{cases} \tag{9.14}$$

(where we have substituted $C = A + B$ already). We then demand continuity just as we did above. At $x = 0$, we have $\frac{\partial u_E^{\mathrm{R}}}{\partial x}\big|_{x=0} = ik'(A + B)$. On the other hand, if we approach 0 in region I from below,

we find $\frac{\partial u_E^{\mathrm{R}}}{\partial x}\big|_{x=0} = ik(A-B)$. Thus, in order to have a continuous first spatial derivative, we must have

$$ik'(A+B) = ik(A-B). \tag{9.15}$$

We can use this to solve for B, which after a small amount of re-arranging, leads to

$$B = \frac{k-k'}{k+k'}A. \tag{9.16}$$

What about the final constant A? As always, this is in fact just an overall **normalisation constant**. Since here we are considering a **non-bound** state, energy varies continuously, and **energy eigenfunctions are unnormalisable**. We can therefore take $A = 1$ **for simplicity**. There is no real natural choice for A, and this is as good as any. Altogether, we therefore find that the energy eigenfunction of energy E for a **right-moving** particle is

$$u_E^{\mathrm{R}}(x) = \begin{cases} e^{ikx} + \frac{k-k'}{k+k'}e^{-ikx} & \text{if } x < 0, \\ \frac{2k}{k+k'}e^{ik'x} & \text{if } x \geq 0. \end{cases} \tag{9.17}$$

This eigenfunction (for one choice of E) is depicted below in Figure 9.3.

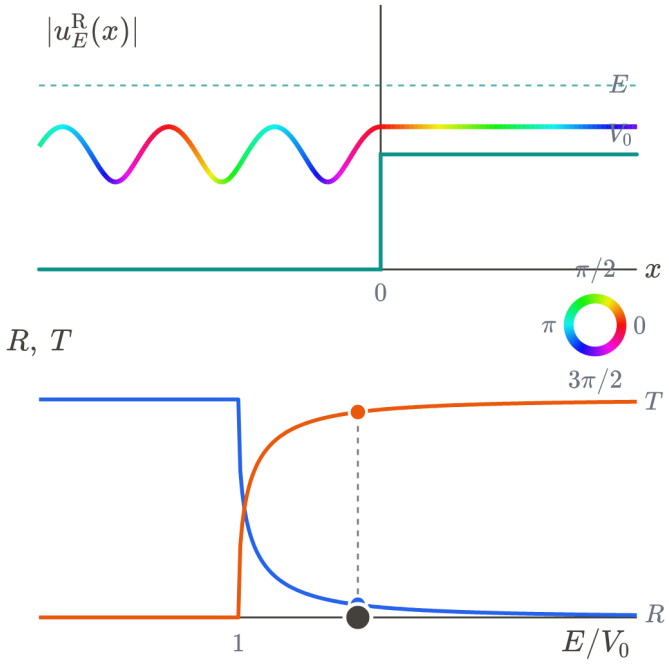


Figure 9.3: **Right-moving energy eigenfunction of the potential step**. Colour plot of a right-moving energy eigenfunction, given in (9.17), for an energy E that can be changed, together with the reflection and transmission coefficients R and T of the next section. For illustrative purposes, we also plot the potential step, to highlight where it is in relation to the wavefunction (the height of the step has no relation to the amplitude of the wavefunction). On the left of the step, where there is both the incoming and reflected component of the wavefunction, there is interference, leading to sinusoidal oscillations. On the right, where there is only the transmitted component, the modulus of the wavefunction is constant. A still of the interactive figure in the online notes, at $E = 1.6\,V_0$.

As a summary of what we have achieved: in (9.17) we have found the wavefunction of the quantum step Hamiltonian with constant energy E, that is 'right-moving', in the sense that when we look on the left

of the step, we only have a positive momentum component. We found this by using two key **mathematical properties** of energy eigenfunctions: that they are **continuous** (as previously seen), and that they have a **continuous derivative** (which is new here). More informally, this ensured that the function has no 'jumps' and no 'kinks'. Just as with the free particle, there is a **second degenerate energy eigenfunction of energy** E. This corresponds to a **left-moving** particle, and you will solve for it as an exercise below.

We will now analyse some of the interesting **physics** of these eigenfunctions, before briefly outlining how **they can be used** to study the dynamics of particles (although you won't be expected to do this yourself, only to appreciate qualitatively what is found).

9.3 Reflection and transmission coefficients

What we would like to understand now as quantitatively as possible, is the meaning of the superposition appearing in (9.17). We will see how to do this in terms of so-called **reflection** and **transmission** coefficients.

The main point we would like to understand is the significance of (9.16), namely how the **amplitude** of the **reflected** component e^{-ikx} relates to the **amplitude** of the **incoming** component e^{ikx} (which we have taken to be $A = 1$ for simplicity).

We can start by considering an extreme case: that of **no potential step**. That is, we can set $V_0 = 0$, and ask what happens to the energy eigenfunction. In this case, we know what we should hope to find, since this corresponds to a free particle, and hence the energy eigenfunctions should just be **momentum eigenfunctions**. This is precisely what we find, since from (9.9), in this case $k' = k$, and hence $B = 0$, leaving us with simply the component e^{ikx} for all x, corresponding to a momentum eigenfunction with momentum $p = \hbar k$.

This is reassuring, but what about for $V_0 \neq 0$? Here we see that B **depends upon the energy relative to the step height**, since E and V_0 specify k and k'. This shows that the **reflected component** is energy dependent.

The way this is usually studied is by introducing the so-called **reflection coefficient**: it is taken to be, by definition, the squared-modulus of B:

$$R = |B|^2 = \left(\frac{k - k'}{k + k'}\right)^2 \tag{9.18}$$

> ***Why is this a good definition?.*** *It is a little subtle, but **essentially** this captures roughly the **probability** that a particle of energy E will be reflected. This isn't strictly true, since energy eigenstates aren't physical states (being unnormalisable). More formally, one can define the notion of a **probability current** (which unfortunately we don't have time to do in this course). R can then be shown to be exactly the **size** of the reflected probability current relative to the incoming probability current.*

It is most insightful to re-express this not in terms of k and k', but instead in terms of the physical parameters: the **energy** E and the height of the potential step V_0. As you will show in an exercise,

$$R = \left(\frac{\sqrt{\frac{E}{V_0}} - \sqrt{\frac{E}{V_0} - 1}}{\sqrt{\frac{E}{V_0}} + \sqrt{\frac{E}{V_0} - 1}}\right)^2 \tag{9.19}$$

We already saw that $B = 0$ when $V_0 = 0$, hence $R = |B|^2 = 0$, so the reflection coefficient can be as small as 0. On the other hand, if E becomes **very close** to V_0 (so that classically, it would only just be able to pass the barrier, after which it would then come to rest, having essentially no kinetic energy), R will become very close to unity, since the second term in both the numerator and denominator will become very small. Thus, even though $E > V_0$, we can have **almost perfect reflection**! This is depicted in Figure 9.4. This is the **new quantum feature we discussed in the introduction**.

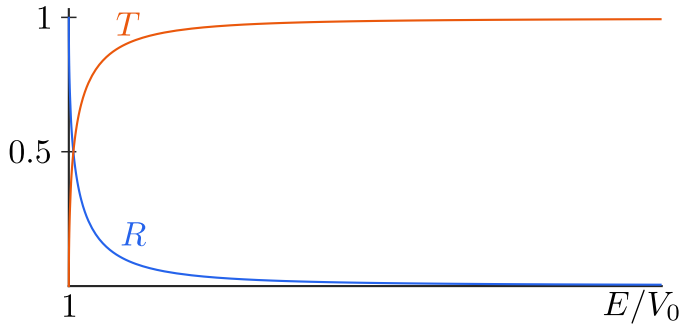


Figure 9.4: **Reflection and transmission coefficients**. Plot of reflection and transmission coefficients for the finite step, as given in (9.19) and (9.20). We see that the particle is perfectly reflected when $E = V_0$, and that the reflection coefficient is large only for E close to V_0. For large values of E/V_0, the particle is transmitted with almost unit probability.

In an exercise, you will furthermore see that we have reflection **even when V_0 is negative** — where naively we would not expect to see any reflection at all, since the particle always has enough energy to pass the step.

Finally, since we see in the graph that $0 \leq R \leq 1$, we can define the **transmission coefficient** in a very natural way, as $T = 1 - R$, so that, by construction $R + T = 1$. That is, if the particle isn't reflected, then it is transmitted. With this definition, we have

$$T = \frac{4\sqrt{\frac{E}{V_0}}\sqrt{\frac{E}{V_0} - 1}}{\left(\sqrt{\frac{E}{V_0}} + \sqrt{\frac{E}{V_0} - 1}\right)^2} \tag{9.20}$$

This is also plotted in Figure 9.4.

9.4 Evolution of wavefunctions

Finally, we will briefly discuss how we can **use the energy eigenfunctions** in order to study the behaviour of **normalised wavefunctions**. The idea is very similar in fact to how we used the energy eigenstates of a free particle in order to study their evolution: although an individual energy eigenstate in that case was a momentum state, and therefore an unphysical and unnormalisable state, we nevertheless **use them** as a convenient **basis** from which we can form well-behaved **superpositions**.

For the finite step potential, we can do **exactly the same**! In particular, we can form initial quantum states of the form

$$|\psi_{\text{in}}\rangle = \int_{V_0}^{\infty} \phi_{\text{in}}(E)|E^{\text{R}}\rangle dE, \tag{9.21}$$

where

$$|E^{\text{R}}\rangle = \int_{-\infty}^{\infty} u_E^{\text{R}}(x)|x\rangle dx, \tag{9.22}$$

is the **right-moving** energy eigenstate of energy E associated to the energy eigenfunction $u_E^{\text{R}}(x)$ in the usual way.

Why do we take only superpositions of right-moving eigenstates? This isn't necessary, but it is very **natural**: we want to consider sending in initial wavefunctions from the left, moving **towards the step** so that it **scatters off of it**. In order to study this, we just need to use the right-moving states. This is just like how we would only expect to use superpositions of **positive momentum** states if we were considering a free particle moving towards the right.

Now, just as in all previous cases, since in (9.21) the initial state is written as a **superposition** of energy eigenstates, we can use the **superposition** principle to **write down** the quantum state at time t, assuming that the state at $t = 0$ is $|\psi(0)\rangle = |\psi_{\text{in}}\rangle$. It will be

$$|\psi(t)\rangle = \int_{V_0}^{\infty} \phi_{\text{in}}(E) e^{-iEt/\hbar} |E^{\text{R}}\rangle dE, \tag{9.23}$$

and the corresponding wavefunction will be

$$\psi(x,t) = \int_{V_0}^{\infty} \phi_{\text{in}}(E) u_E^{\text{R}}(x) e^{-iEt/\hbar} dE. \tag{9.24}$$

In the interactive figure below, we show the evolution of a **Gaussian** wavefunction:

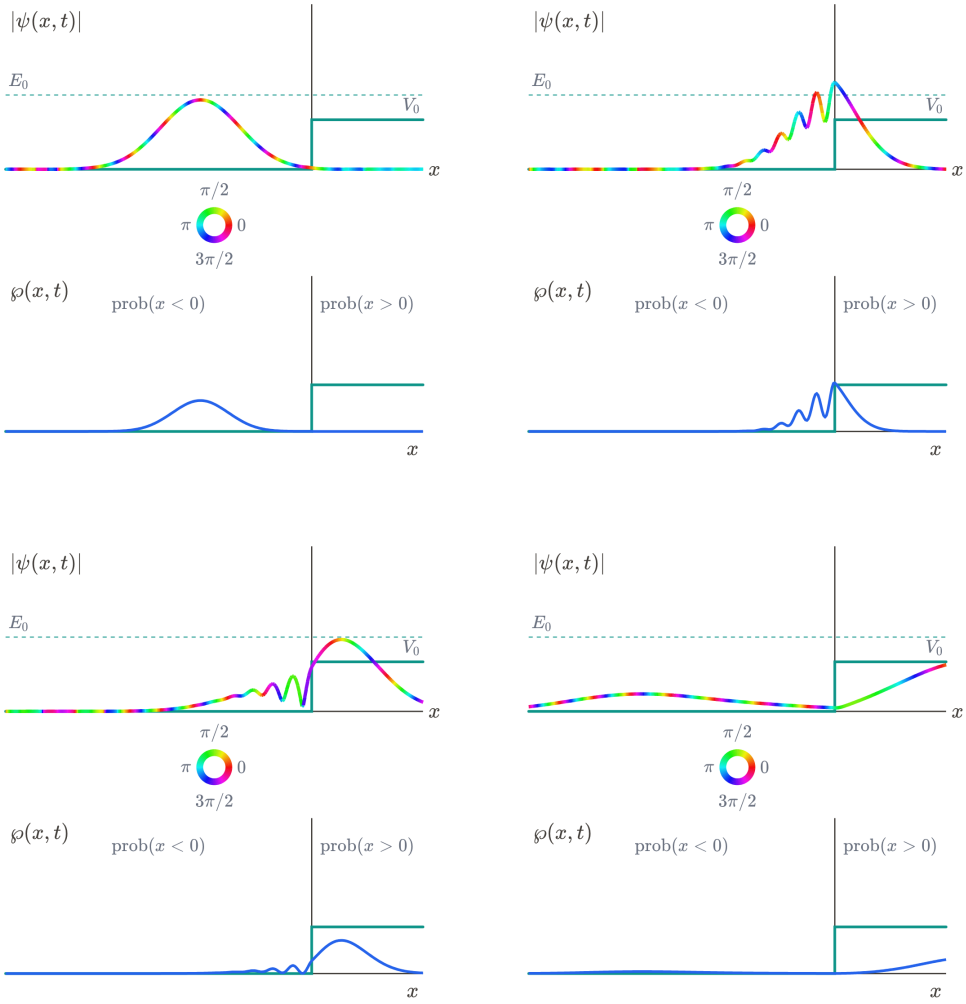


Figure 9.5: **Scattering of a Gaussian wavepacket by a potential step.** In the top plot, we show the wavefunction $|\psi(x,t)|$ using a colour plot. In the bottom plot, we show the associated probability density $\wp(x,t)$. The potential step is also plotted, for illustrative purposes only. The particle approaches the step from the left (with positive momentum), and mean energy E_0 larger than the potential height V_0. Nevertheless, the particle is partially reflected by the step, such that the final state of the particle is a superposition of a left-moving Gaussian after the step, and a right-moving Gaussian that has been reflected by the step. Four stills, at $t = 0$, $6.5\tau_0$, $10\tau_0$ and $20\tau_0$ with $\tau_0 = \hbar/V_0$, of the interactive figure in the online notes ($E_0 = 1.5\,V_0$, $a = 4\,\ell_0$).

We thus see that, as stated, the particle really does partially reflect and partially transmit. In the process, we also see some interesting dynamical features — in particular we see the presence of **interference fringes** during the period before the two distinct Gaussian components of the wavefunction emerge.

The above is just one example, but captures the most important features of scattering in quantum mechanics. We could of course also study more realistic changes in potential (i.e. different forces exerted by the scatterer), however then in general it isn't possible to solve everything analytically as we have done here.

This brings to a close our study of quantum mechanics in one dimension! In the next chapter, we will begin exploring the more physically relevant situation of the mechanics of a quantum particle in three dimensions.

10. Quantum mechanics in three dimensions

In this chapter we are now ready to move onto the study of quantum mechanics in three dimensions! Much of what we have learnt up until this stage will carry across in a rather straightforward manner, which is good news. Obviously the world around us is three dimensional, and so it is necessary to broaden our application of quantum mechanics to three dimensions, so that we can go on to study real-world situations, for example, laying the foundations for the study of the **Hydrogen atom** — arguably the most famous quantum system of all, and one of its biggest early successes. While we won't get all the way there in this unit, we will nevertheless introduce the basics of quantum mechanics in 3D, so that you will be able to go straight there next year.

10.1 The position of a particle in 3D

We will largely follow the same path in 3D that we followed in 1D, and start off with the most basic of all properties of a particle: its **position**. One of the biggest changes in 3D compared to 1D is that many physical quantities now become **vectors**, and the position is the first of these. In order to specify a point in space, we now need to use the **position vector** $\mathbf{r}$,

$$\boldsymbol{r} = x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}} = (x, y, z). \tag{10.1}$$

Just as before we associated a quantum state to each position x, we will now introduce **3D position states** $|\boldsymbol{r}\rangle$, and an associated **3D spatial wavefunction** $\psi(\boldsymbol{r})$, such that a general quantum state of a particle in three dimensions can be written as

$$|\psi\rangle = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \psi(\boldsymbol{r})|\boldsymbol{r}\rangle d^3\boldsymbol{r}, \tag{10.2}$$

which should be compared to (1.4).

> ***Notation for vectors and 3D integrals.*** *We will mostly use the notation $\boldsymbol{r} = (x, y, z)$ as convenient shorthand for 3D vectors, instead of $\boldsymbol{r} = x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$, which proves more cumbersome in general. In this way, we can write $\psi(\boldsymbol{r})$ and $\psi(x, y, z)$ interchangeably, as both should make sense. We will exclusively use **boldface** in the lecture notes for vectors, instead of $\vec{r}$.*
>
> *We will also use the notation $d^3\boldsymbol{r}$ for the **volume element** $dxdydz$. Previously you have seen this as $d\tau$. Both are common (Griffiths for example uses the former, and we will follow their notation here).*
>
> *Finally, we will often simply write $\int_V f(\boldsymbol{r})d^3\boldsymbol{r}$ in place of $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\boldsymbol{r})d^3\boldsymbol{r}$, to indicate an integration over **all space** when this will not cause any confusion.*

This state corresponds to a **superposition** of the particle being at position $\boldsymbol{r}$. The 3D position states $|\boldsymbol{r}\rangle$ are **unphysical states**, which satisfy the orthogonality and normalisation condition

$$\langle \boldsymbol{r}'|\boldsymbol{r}\rangle = \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}'), \tag{10.3}$$

where $\delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}')$ is the **3D Dirac delta function**, given by

$$\delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}') = \delta(x - x')\delta(y - y')\delta(z - z'), \tag{10.4}$$

i.e. it is the **product** of three delta functions, one for each coordinate. Using the 3D delta function, as an exercise you will show that the normalisation condition $\||\psi\rangle\|^2 = 1$, in terms of the wavefunction $\psi(\boldsymbol{r})$ becomes

$$\int_V |\psi(\boldsymbol{r})|^2 d^3\boldsymbol{r} = 1, \tag{10.5}$$

which should be compared to (1.7). The **interpretation** of this equation is in fact identical to the interpretation we gave to (1.7). However, to arrive at this, we first need to introduce **operators** corresponding to the **coordinates** of the particle.

Finally, using the 3D Dirac delta function, as an exercise you will show that we have the direct analogue of (2.17) in 3D, namely

$$\langle \boldsymbol{r}|\psi\rangle = \psi(\boldsymbol{r}), \tag{10.6}$$

which shows that the wavefunction is nothing but the scalar product between a 3D position state and the quantum state itself.

10.1.1 Position and coordinate operators

One new question that arises in 3D with the introduction of **vector quantities** is *how to associate operators to them?* It turns out that the correct way to do this is to **associate a separate operator to each component of the vector**. In the context of the **position**, this means we will associate an operator to **each coordinate** of the particle. We thus introduce 3 operators, $\hat{X}$, $\hat{Y}$ and $\hat{Z}$. The position states $|\boldsymbol{r}\rangle$ are **joint eigenstates of all 3 operators**, satisfying the eigenvalue equations

$$\hat{X}|\boldsymbol{r}\rangle = x|\boldsymbol{r}\rangle, \quad \hat{Y}|\boldsymbol{r}\rangle = y|\boldsymbol{r}\rangle, \quad \hat{Z}|\boldsymbol{r}\rangle = z|\boldsymbol{r}\rangle \tag{10.7}$$

As you will confirm as an exercise, we now have a **vast amount of degeneracy** in these operators. Recall that **degeneracy** refers to the fact that there can be **multiple eigenvectors with the same eigenvalue**. We see that any two position states $|\boldsymbol{r}\rangle$ and $|\boldsymbol{r}'\rangle$ that have the same x coordinate, but arbitrary y and z coordinates, will be degenerate as far as $\hat{X}$ is concerned (and similarly for $\hat{Y}$ and $\hat{Z}$). There are **infinitely many such states**.

Just as it is convenient to combine coordinates together into a vector in classical mechanics, it is also convenient and useful to combine **operators together into a vector of operators**. In particular, we can combine the coordinate operators to form a **3D position operator** $\hat{\mathbf{R}}$, given by

$$\hat{\mathbf{R}} = (\hat{X}, \hat{Y}, \hat{Z}). \tag{10.8}$$

With this notation, we therefore see that

$$\begin{aligned} \hat{\mathbf{R}}|\boldsymbol{r}\rangle &= (\hat{X}, \hat{Y}, \hat{Z})|\boldsymbol{r}\rangle, \\ &= (x, y, z)|\boldsymbol{r}\rangle, \\ &= \boldsymbol{r}|\boldsymbol{r}\rangle, \end{aligned} \tag{10.9}$$

where to obtain the second line we have used (10.7). Apart from the fact that $\hat{\mathbf{R}}$ and $\boldsymbol{r}$ are **vectors**, this is exactly in the form of an **eigenvalue equation**, and this is one reason why it is useful: we see that $|\boldsymbol{r}\rangle$ can be viewed as the eigenvector of the 3D position operator $\hat{\mathbf{R}}$.

> ***Vector eigenvalues?.*** *You might be concerned by this, as it appears that we now have eigenvalues which are themselves vectors? Although this isn't a common way to view things, it does nevertheless make sense, and simply tells us in a succinct way that $|\boldsymbol{r}\rangle$ is a **simultaneous eigenstate** of $\hat{X}$, $\hat{Y}$ and $\hat{Z}$. So just as with many other uses of vectors, here we condense three separate equations into a **single** vector equation.*

We can also introduce the **coordinate operators acting on wavefunctions** using the same prescription as

before, (1.21). Using (10.7), we see for example that

$$\hat{X}|\psi\rangle = \int_V \psi(\boldsymbol{r}) x |\boldsymbol{r}\rangle d^3\boldsymbol{r}, \tag{10.10}$$

i.e. the action of the operator is to transform $\psi(\boldsymbol{r})$ into $\psi'(\boldsymbol{r}) = x\psi(\boldsymbol{r})$, and so we have $\hat{X}_{\rm w} = x$, just as before. That is, the X coordinate operator multiplies a wavefunction by the coordinate x, as we should expect. We similarly have $\hat{Y}_{\rm w} = y$ and $\hat{Z}_{\rm w} = z$. Interestingly, we can **also** combine these into a single **vector operator**. In particular, we see that

$$\begin{aligned} \hat{\mathbf{R}}|\psi\rangle &= \hat{\mathbf{R}} \int_V \psi(\boldsymbol{r})|\boldsymbol{r}\rangle d^3\boldsymbol{r}, \\ &= \int_V \psi(\boldsymbol{r})\hat{\mathbf{R}}|\boldsymbol{r}\rangle d^3\boldsymbol{r}, \\ &= \int_V \psi(\boldsymbol{r})\boldsymbol{r}|\boldsymbol{r}\rangle d^3\boldsymbol{r}. \end{aligned} \tag{10.11}$$

Thus the action of the operator is to turn the wavefunction into the **vector**

$$\psi(\boldsymbol{r})\boldsymbol{r} = (x\psi(\boldsymbol{r}), y\psi(\boldsymbol{r}), z\psi(\boldsymbol{r})), \tag{10.12}$$

i.e. **each component** of $\boldsymbol{r}$ is multiplied by the wavefunction $\psi(\boldsymbol{r})$ (which is itself just a **scalar** quantity, being a single complex number for each position $\boldsymbol{r}$). Given (10.12), we therefore write

$$\hat{\mathbf{R}}_{\rm w} = \boldsymbol{r} \tag{10.13}$$

Finally, using the definitions of $\hat{X}_{\rm w}$, $\hat{Y}_{\rm w}$ and $\hat{Z}_{\rm w}$, it is also straightforward to see that the three coordinates are **compatible observables** that can therefore be **simultaneously measured** (in principle at least). Recall that the **mathematical** property behind compatibility is **commutativity**. We can easily check that the commutator of any two coordinate wavefunction operators vanishes, for example,

$$\begin{aligned} [\hat{X}_{\rm w}, \hat{Y}_{\rm w}]\psi(\boldsymbol{r}) &= (\hat{X}_{\rm w}\hat{Y}_{\rm w} - \hat{Y}_{\rm w}\hat{X}_{\rm w})\psi(\boldsymbol{r}), \\ &= \hat{X}_{\rm w}(y\psi(\boldsymbol{r})) - \hat{Y}_{\rm w}(x\psi(\boldsymbol{r})), \\ &= (xy - yx)\psi(\boldsymbol{r}), \\ &= 0, \end{aligned} \tag{10.14}$$

from which it follows that $[\hat{X}, \hat{Y}] = 0$, according to (3.11).

10.1.2 Probability to find a particle somewhere

We can now think about measuring the **position of the particle** in 3D, and ask *what is the probability to find it in an (arbitrarily small) region around* $\boldsymbol{r}$*?* That is, if we performed an arbitrarily good simultaneous measurement of the 3 coordinates of the particle. The answer is the direct generalisation of what we saw for a particle in one dimension: the **probability density** at position $\boldsymbol{r}$ is precisely

$$\wp(\boldsymbol{r}) = |\psi(\boldsymbol{r})|^2, \tag{10.15}$$

so that the probability to find the particle in a small region of volume dV around the point $\boldsymbol{r}$ is $|\psi(\boldsymbol{r})|^2 dV$. We can then ask similar questions about the probability of a measurement finding the particle in a finite region, and the **total** probability in any region is obtained by **integrating** the probability density over the region.

This then allows us to interpret the normalisation condition (10.5) in a natural way: it says that the **total probability** to find the particle somewhere (in 3D now!) must be unity.

10.2 The momentum of a particle in 3D

Let us now turn our attention to the **momentum** of a particle in 3D. As with the position, the momentum is now a **vector** quantity, $\boldsymbol{p} = (p_x, p_y, p_z)$. In Chapter 2 we saw that if a particle had a definite momentum p_x in the x-direction, then the spatial wavefunction was the momentum eigenfunction $v_{p_x}(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{ip_x x/\hbar}$.

Even though we are now in 3D, we still want this to be the **behaviour of the wavefunction as x varies if the particle has a definite x component of momentum**. We can apply the same logic along the y and z directions, and this suggests that **momentum eigenfunctions** of momentum $\boldsymbol{p}$ should be given by

$$\begin{aligned} v_{\boldsymbol{p}}(\boldsymbol{r}) &= \frac{1}{(\sqrt{2\pi\hbar})^3} e^{ip_x x/\hbar} e^{ip_y y/\hbar} e^{ip_z z/\hbar}, \\ &= \frac{1}{(2\pi\hbar)^{3/2}} e^{i\boldsymbol{p}\cdot\boldsymbol{r}/\hbar} \end{aligned} \tag{10.16}$$

where in the second line we have realised that $p_x x + p_y y + p_z z$ (that arises when combining the exponents of the three exponential functions) can be written as the **scalar product** between $\boldsymbol{r}$ and $\boldsymbol{p}$, i.e. as $\boldsymbol{p} \cdot \boldsymbol{r}$.

In what follows, we will see that **this is the correct definition of the 3D momentum eigenfunctions**. They correspond precisely to **3D complex plane waves** with wave vector $\mathbf{k} = \boldsymbol{p}/\hbar$, and **definite wavelength** $\lambda = 2\pi/|\mathbf{k}| = h/|\boldsymbol{p}|$. This is precisely the de Broglie relation, for a particle with momentum of magnitude $|\boldsymbol{p}|$.

Having defined the momentum eigenfunctions, we can now write down — using the standard prescription — the corresponding **3D momentum eigenstates**,

$$\begin{aligned} |\boldsymbol{p}\rangle &= \int_V v_{\boldsymbol{p}}(\boldsymbol{r}) |\boldsymbol{r}\rangle d^3\boldsymbol{r}, \\ &= \frac{1}{(2\pi\hbar)^{3/2}} \int_V e^{i\boldsymbol{p}\cdot\boldsymbol{r}/\hbar} |\boldsymbol{r}\rangle d^3\boldsymbol{r}. \end{aligned} \tag{10.17}$$

These states are also **unphysical** and satisfy the same orthogonality-normalisation condition as the 3D position states $|\boldsymbol{r}\rangle$, namely

$$\langle \boldsymbol{p}' | \boldsymbol{p} \rangle = \delta^{(3)}(\boldsymbol{p} - \boldsymbol{p}'). \tag{10.18}$$

We can use the momentum eigenstates in order to introduce the **3D momentum wavefunction**, in exact analogy to (2.12):

$$|\psi\rangle = \int_V \tilde{\psi}(\boldsymbol{p}) |\boldsymbol{p}\rangle d^3\boldsymbol{p}, \tag{10.19}$$

i.e. it is the way that we can express an **arbitrary state** as a **superposition** of momentum states. Just as we saw above for the spatial wavefunction, we can similarly directly show that the momentum wavefunction is the scalar product of the quantum state with a 3D momentum state, namely

$$\langle \boldsymbol{p} | \psi \rangle = \tilde{\psi}(\boldsymbol{p}). \tag{10.20}$$

10.2.1 Momentum operator

Just as for the position, we will need to introduce **one momentum operator for each component of momentum**, and we can **collect these together** into a single **vector momentum operator**,

$$\hat{\mathbf{P}} = (\hat{P}_x, \hat{P}_y, \hat{P}_z). \tag{10.21}$$

Momentum eigenstates will then be — by definition — joint eigenstates of each component of the momentum operator, i.e.

$$\hat{P}_x|\boldsymbol{p}\rangle = p_x|\boldsymbol{p}\rangle, \quad \hat{P}_y|\boldsymbol{p}\rangle = p_y|\boldsymbol{p}\rangle, \quad \hat{P}_z|\boldsymbol{p}\rangle = p_z|\boldsymbol{p}\rangle. \tag{10.22}$$

We can again write this in a succinct way as

$$\hat{\mathbf{P}}|\boldsymbol{p}\rangle = \boldsymbol{p}|\boldsymbol{p}\rangle, \tag{10.23}$$

just as we did in (10.9) for the 3D position operator $\hat{\mathbf{R}}$.

What about the wavefunction operator $\hat{\mathbf{P}}_{\mathrm{w}}$? Well, for the position operator, we saw in (10.13) that we in fact **kept the same wavefunction operator as we had in 1D**. Namely, we just multiplied by the coordinate. This suggests that for momentum we might have the same: that each component of the momentum operator should be specified the same way as in 1D, as in (2.27). We will thus take

$$\begin{aligned}\hat{\mathbf{P}}_{\mathrm{w}} &= \left(-i\hbar\frac{\partial}{\partial x}, -i\hbar\frac{\partial}{\partial y}, -i\hbar\frac{\partial}{\partial z}\right), \\ &= -i\hbar\boldsymbol{\nabla}.\end{aligned} \tag{10.24}$$

As an exercise, you will show that the 3D momentum eigenfunctions $v_{\boldsymbol{p}}(\boldsymbol{r})$ from (10.16) are indeed eigenfunctions of the 3D wavefunction momentum operator, satisfying

$$\hat{\mathbf{P}}_{\mathrm{w}} v_{\boldsymbol{p}}(\boldsymbol{r}) = \boldsymbol{p} v_{\boldsymbol{p}}(\boldsymbol{r}), \tag{10.25}$$

which is the **same** eigenvalue equation for the momentum operator as (10.23), just in terms of the wavefunction operator $\hat{\mathbf{P}}_{\mathrm{w}}$ instead of $\hat{\mathbf{P}}$.

Just as the three coordinate operators were **compatible** observables, so too are the three components of momentum. We can see that this is the case by using the wavefunction operators. For example, for the x-component and y-component of momentum, we have

$$\begin{aligned}[\hat{P}_{x,\mathrm{w}}, \hat{P}_{y,\mathrm{w}}]\psi(\boldsymbol{r}) &= (\hat{P}_{x,\mathrm{w}}\hat{P}_{y,\mathrm{w}} - \hat{P}_{y,\mathrm{w}}\hat{P}_{x,\mathrm{w}})\psi(\boldsymbol{r}), \\ &= \left(-i\hbar\frac{\partial}{\partial x}\right)\left(-i\hbar\frac{\partial\psi(\boldsymbol{r})}{\partial y}\right) - \left(-i\hbar\frac{\partial}{\partial y}\right)\left(-i\hbar\frac{\partial\psi(\boldsymbol{r})}{\partial x}\right), \\ &= -\hbar^2\frac{\partial^2\psi(\boldsymbol{r})}{\partial x\partial y} + \hbar^2\frac{\partial^2\psi(\boldsymbol{r})}{\partial x\partial y} \\ &= 0,\end{aligned} \tag{10.26}$$

where we used the fact in the third line that **we can take a partial derivative in either order** (i.e. partial differentiation commutes!). Similar calculations show that all components of momentum pairwise commute, and hence all 3 components are jointly compatible. This means it is (in principle) possible to measure the momentum $\boldsymbol{p}$ of a particle. Note that it isn't **a given** that the components of a vector quantity are necessarily compatible — and is in fact **not true** for the components of angular momentum, which is why the quantum mechanical treatment of angular momentum is in fact so rich and interesting (analogous somewhat to the incompatibility of kinetic and potential energy, and how this leads to interesting dynamics).

10.3 Canonical commutation relations

In Chapter 3 we saw that the position and momentum operators are **incompatible observables** and satisfy the so-called **canonical commutation relation** (3.10). In 3D we have already seen above that the coordinates of a particle are compatible, and that the components of momentum are also compatible. *What about the coordinates and components of momentum?* It turns out that it is only the **pairs** of coordinate and momentum component **in the same direction** that are **incompatible**, all of which obey the **canonical commutation relation**:

$$[\hat{X}, \hat{P}_x] = i\hbar, \quad [\hat{Y}, \hat{P}_y] = i\hbar, \quad [\hat{Z}, \hat{P}_z] = i\hbar, \tag{10.27}$$

while all other pairs are in fact **compatible**,

$$\begin{aligned} [\hat{Y}, \hat{P}_x] = 0, \quad [\hat{X}, \hat{P}_y] = 0, \quad [\hat{X}, \hat{P}_z] = 0, \\ [\hat{Z}, \hat{P}_x] = 0, \quad [\hat{Z}, \hat{P}_y] = 0, \quad [\hat{Y}, \hat{P}_z] = 0. \end{aligned} \tag{10.28}$$

You will prove one example of the above as an exercise, from which all of the others follow by direct analogy.

The physical significance of the above is that we have analogues of the HUP (3.5) for all of the coordinates, namely

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}, \quad \Delta y \Delta p_y \geq \frac{\hbar}{2}, \quad \Delta z \Delta p_z \geq \frac{\hbar}{2}, \tag{10.29}$$

and as such, a particle cannot have a very well-defined coordinate and momentum along the same axis at the same time. However, since for example the x-coordinate and the y-component of momentum commute, there is no uncertainty principle for them, and a particle can have a very well defined x-coordinate and a very well defined y-component of momentum at the same time!

10.4 Relationship between 3D spatial & momentum wavefunctions

In one dimension we saw in (2.19) and (2.20) that the relationship between the position and momentum wavefunctions was that they are a **Fourier transform pair**. This is very useful, as it provides a useful way of obtaining the momentum wavefunction from the spatial wavefunction and vice versa, **without having to go via the quantum state** $|\psi\rangle$.

In 3D (hopefully not unexpectedly at this stage of the chapter!) **the same relationship holds**: The spatial and momentum wavefunctions are **3D Fourier transform pairs**. On the one hand, starting from (10.19), taking the scalar product with a 3D position state $\langle \boldsymbol{r}|$, and using the fact that $\langle \boldsymbol{r}|\boldsymbol{p}\rangle = v_{\boldsymbol{p}}(\boldsymbol{r})$ (as given in (10.16)), we readily obtain

$$\psi(\boldsymbol{r}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_V \tilde{\psi}(\boldsymbol{p}) e^{i\boldsymbol{p}\cdot\boldsymbol{r}/\hbar} d^3\boldsymbol{p}. \tag{10.30}$$

Alternatively, starting from (10.2), taking the scalar product with a 3D momentum state $\langle \boldsymbol{p}|$, and using the fact that $\langle \boldsymbol{p}|\boldsymbol{r}\rangle = (\langle \boldsymbol{r}|\boldsymbol{p}\rangle)^* = v^*_{\boldsymbol{p}}(\boldsymbol{r})$, we also readily obtain

$$\tilde{\psi}(\boldsymbol{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_V \psi(\boldsymbol{r}) e^{-i\boldsymbol{p}\cdot\boldsymbol{r}/\hbar} d^3\boldsymbol{r}. \tag{10.31}$$

10.5 The energy of a particle in 3D

Having covered the position and momentum of a particle above, we are now ready to consider the **energy** of a particle in 3D. The Hamiltonian still contains two contributions to the energy — kinetic and potential. The latter is rather straightforward, as we simply now have $V(\boldsymbol{r})$, i.e. a potential which depends upon the (3D) position of the particle. We will study explicit examples later (e.g. a 3D 'box' and the 3D harmonic oscillator potential).

For the kinetic energy, the situation is slightly different, given that momentum is now a **vector** quantity in 3D. The kinetic energy depends only upon the **squared magnitude** of the momentum, that is (classically for the moment),

$$K = \frac{p_x^2 + p_y^2 + p_z^2}{2M} \tag{10.32}$$

We will thus form a kinetic energy operator by substituting each component of squared momentum by the corresponding **squared operator**, and thus arrive at the general **Hamiltonian operator**

$$\hat{H} = \frac{\hat{P}_x^2 + \hat{P}_y^2 + \hat{P}_z^2}{2M} + \hat{V}. \tag{10.33}$$

As always, a large part of our interest in 3D will be to find the eigenstates and eigenvalues of the Hamiltonian operator, in different physical situations, with different choices of $\hat{V}$ (specifying the forces acting on the particle).

The associated **operator acting on wavefunctions** is

$$\begin{aligned}\hat{H}_{\mathrm{w}} &= -\frac{\hbar^2}{2M}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) + V(\boldsymbol{r}), \\ &= -\frac{\hbar^2}{2M}\nabla^2 + V(\boldsymbol{r}).\end{aligned} \tag{10.34}$$

From this we arrive at the 3D TISE,

$$\hat{H}_{\mathrm{w}} u_E(\boldsymbol{r}) = E u_E(\boldsymbol{r}), \tag{10.35}$$

the **eigenvalue equation** for the Hamiltonian operator in terms of energy **eigenfunctions**. This is our **central tool** for finding the energy eigenstates in 3D, just as the 1D TISE was our central tool for finding energy eigenstates (via energy eigenfunctions) in 1D.

As in 1D, there are two **distinct** physical situations which we encounter, when a particle is either **bound** (remaining in a finite region of space), or **non-bound** (in which it can escape off to infinity). In 3D, there is a potential subtlety: what if the particle was **confined in one direction but not the others?** *What happens then?* In this case, the region that the particle can explore will **still be infinite in volume**, and as such we find that the energy of the particle varies **continuously**. Only if the particle is confined **in all three spatial dimensions** will we find **discrete energy levels**.

One update is that previously we only ever encountered a **single quantum number** n **per energy level**. That is, in both the infinite square well and the harmonic oscillator we were able to enumerate the energy levels using a **single** integer n. In 3D, in contrast, we will typically see that we have **multiple quantum numbers per energy level**.

A common case is to have **three quantum numbers**, one **per degree of freedom** (i.e. each **independent** direction of travel). We will label these by n_x, n_y and n_z respectively, and sometimes collect them into the vector $\mathbf{n} = (n_x, n_y, n_z)$.

10.6 Evolution in 3D

The equation of motion of a particle in 3D is no different from the equation of motion in 1D. It is given once again by the Schrödinger Equation, which takes an **identical form** to (5.1) when written out abstractly,

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle. \tag{10.36}$$

In 1D we saw that we **never actually solve the SE directly** — we instead used the **superposition principle** and the fact that **energy eigenstates evolve simply** to solve it **indirectly**. In 3D, nothing will change on this front either!

For example, assuming that the particle is **bound** and therefore that it has a discrete set of **energy levels** $|E_{\mathbf{n}}\rangle$ (specified by the vector of quantum numbers $\mathbf{n}$), if this is the initial state of a particle at time $t = 0$, i.e. $|\psi(t=0)\rangle = |E_{\mathbf{n}}\rangle$, then the state at time t will simply be

$$|\psi(t)\rangle = e^{-iE_{\mathbf{n}}t/\hbar}|E_{\mathbf{n}}\rangle, \tag{10.37}$$

as can be directly verified by substituting into the SE.

For an initial state $|\psi_{\text{in}}\rangle$, it is again crucial to write it as a superposition of energy eigenstates as

$$|\psi_{\text{in}}\rangle = \sum_{\mathbf{n}} \alpha_{\mathbf{n}} |E_{\mathbf{n}}\rangle, \tag{10.38}$$

where the summation runs over all the possible values of $\mathbf{n}$, e.g. this is really **3 summations**, over n_x, n_y and n_z, all of which are potentially infinite.

In this form, we can **directly apply the superposition principle** to write down the evolution of the particle, which is given by

$$|\psi(t)\rangle = \sum_{\mathbf{n}} \alpha_{\mathbf{n}} e^{-iE_{\mathbf{n}}t/\hbar} |E_{\mathbf{n}}\rangle. \tag{10.39}$$

Taking the scalar product of both sides with $\langle \boldsymbol{r}|$, we arrive at the evolution of the 3D wavefunction in time,

$$\psi(\boldsymbol{r}, t) = \sum_{\mathbf{n}} \alpha_{\mathbf{n}} e^{-iE_{\mathbf{n}}t/\hbar} u_{\mathbf{n}}(\boldsymbol{r}), \tag{10.40}$$

where we have written the energy eigenfunction $u_{\mathbf{n}}(\boldsymbol{r}) = \langle \boldsymbol{r}|E_{\mathbf{n}}\rangle$ of the energy level $E_{\mathbf{n}}$ in the same **shorthand notation** as previously (i.e. with a subscript $\mathbf{n}$ in place of $E_{\mathbf{n}}$, just as we did for the energy levels of the infinite square well and harmonic oscillator).

How do we write an arbitrary initial quantum state $|\psi_{\text{in}}\rangle$ in the form (10.38)? That is, how do we determine $\alpha_{\mathbf{n}}$? We use the direct analogy of (7.21):

$$\alpha_{\mathbf{n}} = \int_V \psi_{\text{in}}(\boldsymbol{r}) u^*_{\mathbf{n}}(\boldsymbol{r}) d^3\boldsymbol{r}, \tag{10.41}$$

where $\psi_{\text{in}}(\boldsymbol{r}) = \langle \boldsymbol{r}|\psi_{\text{in}}\rangle$ is the 3D wavefunction of the initial state $|\psi_{\text{in}}\rangle$. We use this form, as typically it is the **energy eigenfunctions that we know explicitly**, and so this is the most useful form to have.

If we wanted to, we could combine (10.40) and (10.41) to write the 3D wavefunction at time t **directly** in terms of the initial wavefunction as

$$\psi(\boldsymbol{r}, t) = \sum_{\mathbf{n}} \int_V \psi_{\text{in}}(\boldsymbol{r}') u^*_{\mathbf{n}}(\boldsymbol{r}') u_{\mathbf{n}}(\boldsymbol{r}) e^{-iE_{\mathbf{n}}t/\hbar} d^3\boldsymbol{r}', \tag{10.42}$$

where we have been **careful** to change our integration variable to $\boldsymbol{r}'$ (as always!), and where this should be compared to (7.22). This can be used to numerically solve for the quantum wavefunction at any time t, given any initial wavefunction.

11. The 3D infinite box

In this chapter we will apply what we learnt in the previous chapter to the mathematically simplest situation involving a particle confined in a three-dimensional potential well. The potential well we will consider is the direct generalisation of the **infinite square well** that we studied in Chapter 7. We will refer to this as the **3D infinite box** potential.

As we explained in Chapter 7, the idea that a particle could have an infinite potential energy in some region isn't particularly physical. Nevertheless, it provides us with a **clean mathematical** model which is easy to study, and it can be viewed as a fairly good approximation to a **very deep potential well**.

The main new ingredient we will encounter in this chapter is that in 3D **energy levels become degenerate**. This means that we will see that there are multiple different **distinct** energy eigenstates of the 3D infinite box, all with **the same energy**. While we have encountered energy degeneracy before in 1D, this only happened when the particle was **non-bound**. It is a new feature of 3D that we can have energy level degeneracy, and this is indeed a very important feature of nature (for example, the energy levels of Hydrogen are degenerate).

We will also see that even though we are now considering a 3D problem, **what we learnt in 1D is still highly relevant**. We will in particular see how the **1D energy eigenfunctions** in fact compose to give us the **3D eigenfunctions**. This is generically the case, and highlights the importance of studying 1D problems as a means for understanding the physics of more realistic 3D problems.

11.1 The 3D infinite box potential

We are going to consider a situation where a particle is **perfectly confined** inside a 3D 'box' — of volume L^3. That is, just like in 1D, that there are 'walls' at $x = 0$ and $x = L$, as well as at $y = 0$ and $y = L$ and $z = 0$ and $z = L$, leaving a perfectly **cubic** region in which the particle is confined. We can model this with the following potential energy function, the direct analogue of (7.1),

$$V(\boldsymbol{r}) = \begin{cases} 0 & \text{if } 0 \leq x, y, z \leq L, \\ \infty & \text{otherwise,} \end{cases} \tag{11.1}$$

where $0 \leq x, y, z \leq L$ means that **all three coordinates must individually be between 0 and** L. Unlike in 1D, it is no longer easy to depict this potential in a figure! Hopefully though it is straightforward to understand the basic geometry of this potential well.

Note that here we have chosen to take **all 3 side lengths to be equal**. This is a rather natural situation to consider, and will be the most interesting from the perspective of **degeneracy**. In the online exercises you will consider the most general situation, where all three side lengths differ.

11.2 Energy eigenstates of the 3D infinite box potential

As always, our first goal in studying the 3D infinite box potential is to determine the **energy eigenstates** and associated **energy eigenvalues**. Since we have a particle confined inside a well, these will be **bound states** and so we expect that we will find **discrete energy levels**, just as we found in 1D.

As we spent some time arguing in 1D, due to the fact that the potential energy is **infinite** outside of the box, and given that physically we only want to consider that the particle has **finite energy**, it is natural to assume that the energy eigenfunctions will **vanish** outside of the box. That is, if we refer to the inside of the box as **region I**, and outside the box as **region II**, then we will look for energy eigenfunctions of the form

$$u_E(\boldsymbol{r}) = \begin{cases} u_E^{\mathrm{I}}(\boldsymbol{r}) & \text{if } 0 \leq x, y, z \leq L, \\ 0 & \text{otherwise,} \end{cases} \tag{11.2}$$

where $u_E^{\mathrm{I}}(\boldsymbol{r})$ is the function we would like to find. Since studying the infinite square well in Chapter 7, we subsequently saw in Chapter 8 that particles can **tunnel** into regions where they don't have enough energy to be classically. We might therefore now be concerned about our assumption that the wavefunction vanishes outside the box. Although we don't go further into this here, it can be shown that **due to the infinite nature of the potential** the wavefunction really does vanish outside of the box, and we are again in the exceptional circumstance where there is **no tunnelling or barrier penetration**.

We can therefore restrict our attention to the **inside** of the well. In this region, the Hamiltonian of the particle takes the simple form

$$\hat{H} = \frac{\hat{P}_x^2 + \hat{P}_y^2 + \hat{P}_z^2}{2M}, \tag{11.3}$$

since inside the well the potential vanishes. The 3D TISE (10.35) therefore reads

$$-\frac{\hbar^2}{2M}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) u_E^{\mathrm{I}}(\boldsymbol{r}) = E u_E^{\mathrm{I}}(\boldsymbol{r}), \tag{11.4}$$

where the left-hand side is $\hat{H}_{\mathrm{w}} u_E^{\mathrm{I}}(\boldsymbol{r})$.

Up until now we have only solved the TISE for a particle in 1D, which was then a partial differential equation in one variable. Now, for the first time, we need to study a partial differential equation in three variables, and understand how we can find its solutions.

Instead of approaching this head-on, we will try to analyse the structure a little, and in so doing, realise that we can **spot** the solutions.

In order to do so, let us first recall our solution to the 1D infinite square well. This was given in (7.14), namely,

$$u_n(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} & \text{if } 0 \leq x \leq L, \\ 0 & \text{if } x < 0 \text{ or } x > L. \end{cases} \tag{11.5}$$

The fact that these are energy eigenfunctions means that **inside** the infinite square well ($0 \leq x \leq L$) they satisfy the relevant TISE, namely

$$-\frac{\hbar^2}{2M}\frac{\partial^2}{\partial x^2}\left(\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}\right) = \frac{\hbar^2 n^2 \pi^2}{2ML^2}\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \tag{11.6}$$

from which we confirm that the corresponding energy eigenvalue is $E_n = \frac{\hbar^2 n^2 \pi^2}{2ML^2}$.

Why is this relevant? Well, if we consider just the **first** term on the left-hand side of (11.4) — i.e. $\hat{P}_{x,\mathrm{w}}^2/2M$ — it is **identical** to the 1D TISE, except for the fact that in 1D the eigenfunctions were **only** a function of x, while now we have a 3D function, of $\boldsymbol{r}$. However, this isn't a big problem, since if we consider **any** function of the form

$$u(\boldsymbol{r}) = u_n(x) f(y, z), \tag{11.7}$$

where $f(y, z)$ is some arbitrary function of y and z alone, then we see that we have essentially **identical behaviour**, and that, in fact, **any such function will be an eigenfunction of** $\hat{P}_{x,\mathrm{w}}^2/2M$. Namely,

$$\begin{aligned} -\frac{\hbar^2}{2M}\frac{\partial^2}{\partial x^2} u(\boldsymbol{r}) &= -\frac{\hbar^2}{2M}\frac{\partial^2}{\partial x^2}\left(\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} f(y, z)\right) \\ &= \frac{\hbar^2 n^2 \pi^2}{2ML^2}\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} f(y, z), \\ &= E_n u(\boldsymbol{r}), \end{aligned} \tag{11.8}$$

where we have used the fact that the **partial derivative** treats $f(y, z)$ essentially as a **constant**. The above confirms that $u(\boldsymbol{r}) = u_n(x) f(y, z)$ is an eigenfunction, with the same eigenvalue E_n as $u_n(x)$.

Why has this helped us? Well, the next key step is to realise that we could make the **same argument**, but now focusing only on the y component of the kinetic energy, $\hat{P}^2_{y,\mathrm{w}}/2M$, or the z component of the kinetic energy, $\hat{P}^2_{z,\mathrm{w}}/2M$. In these two cases respectively, we would see that any functions of the form

$$u'(\boldsymbol{r}) = u_n(y)f'(x,z) \quad \text{and} \quad u''(\boldsymbol{r}) = f''(x,y)u_n(z), \tag{11.9}$$

would be eigenfunctions of $\hat{P}^2_{y,\mathrm{w}}/2M$ and $\hat{P}^2_{z,\mathrm{w}}/2M$ respectively, all with the same eigenvalue E_n, for arbitrary functions $f'(x,z)$ and $f''(x,y)$.

Now, we **aren't looking for eigenfunctions of an individual component of the kinetic energy**. However, what we have seen above shows us that there is a way to find a **joint eigenfunction** of all 3 components! In particular, we see that if we find a function such that the behaviour in x is $u_n(x)$, the behaviour in y is $u_n(y)$ and the behaviour in z is $u_n(z)$, then **this will do the job**! It is easy to write down such a function — it is **just the product of the three individual functions**! We only need to be **careful** and realise that we **shouldn't repeat n 3 times, but consider 3 different values of n**, which we will call n_x, n_y and n_z respectively. Thus, we arrive at the **educated guess** (also called **ansatz** frequently), that the energy eigenfunctions of the 3D box are of the form

$$\begin{aligned} u_{\mathbf{n}}(\boldsymbol{r}) &= u_{n_x}(x)u_{n_y}(y)u_{n_z}(z), \\ &= \begin{cases} \sqrt{\frac{8}{L^3}} \sin\frac{n_x\pi x}{L} \sin\frac{n_y\pi y}{L} \sin\frac{n_z\pi z}{L} & \text{if } 0 \le x,y,z \le L, \\ 0 & \text{otherwise,} \end{cases} \end{aligned} \tag{11.10}$$

where, just as we did in 1D, we have **changed notation** from $u_E(\boldsymbol{r})$ to $u_{\mathbf{n}}(\boldsymbol{r})$, labelling the energy eigenfunction by the **vector of quantum numbers**

$$\mathbf{n} = (n_x, n_y, n_z). \tag{11.11}$$

> ***A note on notation.*** *Please note that there is a slightly confusing notation going on here: $u_{\mathbf{n}}(\boldsymbol{r})$ are the **energy eigenfunctions of the 3D box**. On the other hand, $u_{n_x}(x)$, $u_{n_y}(y)$ and $u_{n_z}(z)$ are the eigenfunctions of the **1D infinite square well**. Since in 3D the eigenfunctions are a function of $\boldsymbol{r}$, and are labelled by the vector $\mathbf{n}$, while in 1D they are functions of a single variable (x or y or z), and labelled by a single quantum number (n_x or n_y or n_z) there shouldn't be any ambiguity here. Nevertheless, I want to flag this up, in case the distinction gets missed and causes confusion.*

As an exercise you will confirm explicitly that these functions **satisfy the TISE** (11.4) with energy eigenvalue

$$E_{\mathbf{n}} = \frac{\hbar^2\pi^2}{2ML^2}\left(n_x^2 + n_y^2 + n_z^2\right). \tag{11.12}$$

In what follows, it will be convenient to give a name to the combination $\frac{\hbar^2\pi^2}{2ML^2}$, which is a **common multiple** among all energy levels. We will denote this by

$$E_{\mathrm{box}} = \frac{\hbar^2\pi^2}{2ML^2}. \tag{11.13}$$

Although we won't prove it here, the energy eigenstates we have found in the above are in fact **all energy eigenstates**. That is, our educated guess in fact encompasses every energy level of the 3D infinite box, and doesn't miss anything out.

11.3 Degeneracy of the energy levels

Having obtained all of the energy levels of the 3D infinite box, we will now turn our attention to the one new property that arises in 3D compared to 1D — degeneracy. It will be most instructive to start **enumerating** all of the energy levels and their corresponding energies.

We do this in the following table:

The reason for including so many levels in this table is so that we can get a feeling for the **complexity** that arises. In particular, what we see is that the ground state is still a so-called **non-degenerate** energy level: there is a unique state of lowest energy (equal to $3E_{\text{box}}$), just like in 1D.

In contrast, the next two energies are both **triply degenerate**, at energy $6E_{\text{box}}$ and $9E_{\text{box}}$. *Where did this 3-fold degeneracy come from?* If we look at the structure of $\mathbf{n}$, we see it arises due to **combinatorics**: there are 3 vectors which have a single 2 and two 1s. With this insight, we see that **any level** of the form $\mathbf{n} = (n, n', n')$ (i.e. with a repetition in one of the quantum numbers) will be 3-fold degenerate, just as is the case for levels of energy $9E_{\text{box}}$.

It is important to note that even though these states have the same energy, as you'll confirm in an exercise, they are still nevertheless **orthogonal states**, e.g. $\langle E_{(1,1,2)}|E_{(1,2,1)}\rangle = 0$, because in the two cases the corresponding vector of quantum numbers is **different**.

The only other situation we can encounter is when $n_x \neq n_y \neq n_z$. This happens for the first time when the energy is $14E_{\text{box}}$. In this case, the combinatorics gives us **6 energy levels with the same energy**, as there are **6 permutations** of 3 numbers.

Finally, we also see that the level at energy $12E_{\text{box}}$ with $\mathbf{n} = (2, 2, 2)$ is non-degenerate, just like the ground state. This also follows from combinatorics: there is a single vector $\mathbf{n} = (n, n, n)$ for any integer n (and no permutations).

All of the above degeneracies can be seen to arise due to **symmetries** — our ability to swap say x and y when n_x and n_y were equal. This didn't actually change the energy eigenfunction, and this is why the energy has to be the same.

There is in fact another type of degeneracy that can arise, which **isn't due to symmetry**, but is in fact just **accidental**. For example, the two levels $E_{(7,1,1)}$ and $E_{(5,5,1)}$ are degenerate, with energy $51E_{\text{box}}$, because $5^2 + 5^2 + 1^2$ happens to be equal to $7^2 + 1^2 + 1^2$. This has **nothing** to do with symmetry, and is just a particularity of the specific dependence of $E_{\mathbf{n}}$ on $\mathbf{n}$ (i.e. that it is the sum of the components squared).

In other potentials — such as Hydrogen — we will have a different degeneracy structure. It is important to realise however that it is precisely degeneracy that is the origin of the famous s, p and d **shells** that you may have already encountered, each being a different degenerate collection of energy levels.

11.4 Evolution of particles inside a 3D box

As a final topic, we will now look at the evolution of a particle confined inside a 3D box. While on the one hand, the situation is just another example of what we have already encountered (i.e. an application of the superposition principle!), we will see that studying motion in 3D allows us to see more **complex** features, such as **correlation** between the different coordinates of the particle in time.

In the previous chapter, we already gave the **general formulas** for time evolution in 3D. Specifically, we saw that an initial state of the form

$$|\psi_{\text{in}}\rangle = \sum_{\mathbf{n}} \alpha_{\mathbf{n}} |E_{\mathbf{n}}\rangle, \tag{11.14}$$

evolves into

$$|\psi(t)\rangle = \sum_{\mathbf{n}} \alpha_{\mathbf{n}} e^{-iE_{\mathbf{n}}t/\hbar} |E_{\mathbf{n}}\rangle. \tag{11.15}$$

with corresponding wavefunction $\langle \boldsymbol{r}|\psi(t)\rangle$,

$$\psi(\boldsymbol{r}, t) = \sum_{\mathbf{n}} \alpha_{\mathbf{n}} e^{-iE_{\mathbf{n}}t/\hbar} u_{\mathbf{n}}(\boldsymbol{r}), \tag{11.16}$$

Table 11.1: **Energy levels of the 3D infinite box**. We enumerate here the energy levels of the box, in increasing energy. As we can see, we have **degeneracy**: multiple distinct energy levels all with the same energy eigenvalue. **Note**: In the final column, we only write the form of the energy eigenfunction **inside** the box (and leave it **implicit** that outside the box the wavefunction vanishes).

$\mathbf{n} = (n_x, n_y, n_z)$	Energy $E_{\mathbf{n}}$	Energy eigenfunction $u_{\mathbf{n}}(\boldsymbol{r})$
(1,1,1)	$3E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{\pi x}{L} \sin\frac{\pi y}{L} \sin\frac{\pi z}{L}$
(1,1,2)	$6E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{\pi x}{L} \sin\frac{\pi y}{L} \sin\frac{2\pi z}{L}$
(1,2,1)	$6E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{\pi x}{L} \sin\frac{2\pi y}{L} \sin\frac{\pi z}{L}$
(2,1,1)	$6E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{2\pi x}{L} \sin\frac{\pi y}{L} \sin\frac{\pi z}{L}$
(1,2,2)	$9E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{\pi x}{L} \sin\frac{2\pi y}{L} \sin\frac{2\pi z}{L}$
(2,1,2)	$9E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{2\pi x}{L} \sin\frac{\pi y}{L} \sin\frac{2\pi z}{L}$
(2,2,1)	$9E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{2\pi x}{L} \sin\frac{2\pi y}{L} \sin\frac{\pi z}{L}$
(1,1,3)	$11E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{\pi x}{L} \sin\frac{\pi y}{L} \sin\frac{3\pi z}{L}$
(1,3,1)	$11E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{\pi x}{L} \sin\frac{3\pi y}{L} \sin\frac{\pi z}{L}$
(3,1,1)	$11E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{3\pi x}{L} \sin\frac{\pi y}{L} \sin\frac{\pi z}{L}$
(2,2,2)	$12E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{2\pi x}{L} \sin\frac{2\pi y}{L} \sin\frac{2\pi z}{L}$
(1,2,3)	$14E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{\pi x}{L} \sin\frac{2\pi y}{L} \sin\frac{3\pi z}{L}$
(1,3,2)	$14E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{\pi x}{L} \sin\frac{3\pi y}{L} \sin\frac{2\pi z}{L}$
(2,1,3)	$14E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{2\pi x}{L} \sin\frac{\pi y}{L} \sin\frac{3\pi z}{L}$
(2,3,1)	$14E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{2\pi x}{L} \sin\frac{3\pi y}{L} \sin\frac{\pi z}{L}$
(3,1,2)	$14E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{3\pi x}{L} \sin\frac{\pi y}{L} \sin\frac{2\pi z}{L}$
(3,2,1)	$14E_{\text{box}}$	$\sqrt{\frac{8}{L^3}} \sin\frac{3\pi x}{L} \sin\frac{2\pi y}{L} \sin\frac{\pi z}{L}$

Therefore, here we will consider a simple example, and see how we can analyse the behaviour. To that end, consider the specific initial wavefunction

$$\psi_{\text{in}}(\boldsymbol{r}) = \frac{1}{\sqrt{2}}\left(u_{(1,1,1)}(\boldsymbol{r}) + u_{(2,2,1)}(\boldsymbol{r})\right), \tag{11.17}$$

the wavefunction associated to an **equal superposition** of the ground state $|E_{(1,1,1)}\rangle$ of energy $E_{(1,1,1)} = 3E_{\text{box}}$ and the state $|E_{(2,2,1)}\rangle$ of energy $E_{(2,2,1)} = 9E_{\text{box}}$, from the 3rd energy level. This state has $\alpha_{(1,1,1)} = \frac{1}{\sqrt{2}}$, $\alpha_{(2,2,1)} = \frac{1}{\sqrt{2}}$, and all other $\alpha_{\mathbf{n}} = 0$.

Using the superposition principle, we can immediately write down the wavefunction of the particle at time t, which is given by

$$\psi(\boldsymbol{r}, t) = \frac{1}{\sqrt{2}}\left(e^{-3iE_{\text{box}}t/\hbar}u_{(1,1,1)}(\boldsymbol{r}) + e^{-9iE_{\text{box}}t/\hbar}u_{(2,2,1)}(\boldsymbol{r})\right). \tag{11.18}$$

We could stop here, but what is interesting is to analyse the properties of the particle. First, we can calculate the **probability density** $\wp(\boldsymbol{r}, t) = |\psi(\boldsymbol{r}, t)|^2$. As you will confirm in an exercise, this is given by

$$\wp(\boldsymbol{r}, t) = \frac{1}{2}\left(|u_{(1,1,1)}(\boldsymbol{r})|^2 + |u_{(2,2,1)}(\boldsymbol{r})|^2 + 2\cos\frac{6E_{\text{box}}t}{\hbar}u_{(1,1,1)}(\boldsymbol{r})u_{(2,2,1)}(\boldsymbol{r})\right) \tag{11.19}$$

This is the probability density in 3D, but there is nothing to stop us asking questions about the probability density of any given coordinate. This is formally a **marginal probability density**, and we can obtain it by **integrating out** the coordinates we aren't interested in (so that we get the **total probability density** for the remaining coordinate). For example, for the x coordinate, we have

$$\wp(x, t) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\wp(\boldsymbol{r}, t)dydz. \tag{11.20}$$

In order to evaluate this, it will be advantageous to use the fact that the energy eigenfunctions of the 3D box are **products** of the eigenfunctions of the **1D infinite square well** (as given in (11.10)). Focusing first on the first term on the right-hand side of (11.19), we see that

$$\begin{aligned}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}|u_{(1,1,1)}(\boldsymbol{r})|^2dydz &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}|u_1(x)|^2|u_1(y)|^2|u_1(z)|^2dydz,\\ &= |u_1(x)|^2\int_{-\infty}^{\infty}|u_1(y)|^2dy\int_{-\infty}^{\infty}|u_1(z)|^2dz,\\ &= |u_1(x)|^2,\end{aligned} \tag{11.21}$$

where in the first line we write the eigenfunction in terms of the eigenfunctions of the 1D infinite well, in the second line we have reorganised the integrals, and in the third line we have used the fact that the **energy eigenfunctions of the 1D infinite square well are normalised**, hence the integrals evaluate to unity (without having to do any work!).

We arrive at a similar conclusion for the second term on the right-hand side of (11.19), and see that after integrating over y and z, we are left with $|u_2(x)|^2$. For the last term, the situation is interesting. The relevant integral is

$$\begin{aligned}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}u_{(1,1,1)}(\boldsymbol{r})u_{(2,2,1)}(\boldsymbol{r})dydz &= u_1(x)u_2(x)\int_{-\infty}^{\infty}u_1(y)u_2(y)dy\int_{-\infty}^{\infty}u_1(z)u_1(z)dz,\\ &= 0.\end{aligned} \tag{11.22}$$

Why is this zero? It is because of the integral over y. We should recall that **the energy eigenfunctions of the infinite square well are real**. This means we could equally write $u_1(y)$ as $u_1^*(y)$. The integral over y is thus nothing but $\langle E_1|E_2\rangle$ in disguise (as you have shown as an exercise in class, and where $|E_1\rangle$ and

$|E_2\rangle$ refer to energy eigenstates of the 1D infinite square well). Since the energy levels are **orthogonal** (see (4.13)), the integral over y vanishes.

Putting everything together, we therefore see that

$$\wp(x,t) = \frac{1}{2}\left(|u_1(x)|^2 + |u_2(x)|^2\right), \tag{11.23}$$

which is **independent of time**! That is, if we only keep track of the x coordinate of the particle, this doesn't change in time. Similar calculations, which you will do as an exercise show that similarly

$$\begin{aligned}\wp(y,t) &= \frac{1}{2}\left(|u_1(y)|^2 + |u_2(y)|^2\right),\\ \wp(z,t) &= |u_1(z)|^2,\end{aligned} \tag{11.24}$$

and are thus **also independent of time**.

What is this telling us? It actually shows that the **motion in the different coordinates must be correlated**, as the probability density $\wp(\boldsymbol{r},t)$ is definitely **not** independent of time. It is actually the x and y coordinates that are correlated. We can study these by integrating over z. As part of the online exercises, you will show that

$$\begin{aligned}\wp(x,y,t) &= \int_{-\infty}^{\infty} \wp(\boldsymbol{r},t)dz,\\ &= \frac{1}{2}\left(|u_1(x)|^2|u_1(y)|^2 + |u_2(x)|^2|u_2(y)|^2 + 2\cos\frac{6E_{\text{box}}t}{\hbar}u_1(x)u_2(x)u_1(y)u_2(y)\right)\end{aligned} \tag{11.25}$$

In the following animation, you can see what this marginal probability density looks like:

As you can see, the particle **oscillates** back and forth, in a **correlated fashion**. In an exercise, you will study this correlation using the so-called **covariance**, a measure for how **correlated** the two coordinates of the particle are, on average at any given time. You will see that the coordinates oscillate between being **correlated** (both being large and positive / negative at the same time) to being **anti-correlated** (having opposite signs at a given time).

The main message here is that even **simple superpositions** of energy eigenstates can exhibit **interesting correlated dynamics**. This example is artificial, but hopefully from it you can see a glimpse of the **rich dynamics** that quantum particles exhibit in 3D.

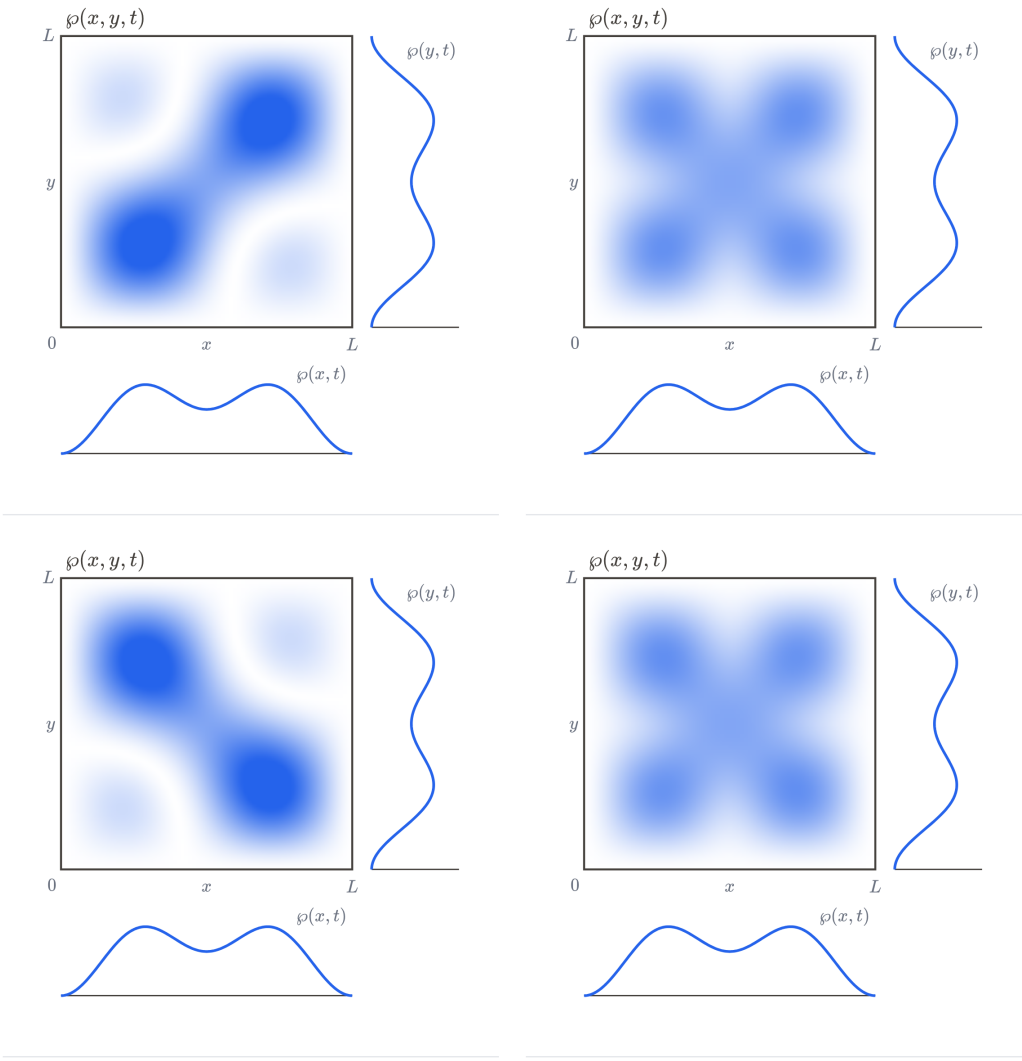


Figure 11.1: **Probability density in the x and y direction.** The probability density $\wp(x, y, t)$ as given in (11.25), shown as a heat-map over the x–y face of the box, with its marginals $\wp(x, t)$ and $\wp(y, t)$ beside it. The z axis in this plot is $\wp(x, y, t)$ (and **not** the box!). That is, we are ignoring motion along the z direction, just studying the motion of the particle in the x and y directions only. We see that the particle oscillates, between two distributions, each of which has two large peaks. If we look along only **one** direction, we can see how the individual probability densities $\wp(x, t)$ and $\wp(y, t)$ are time independent. Four stills over one period of the interactive figure in the online notes.

A. Gaussian integrals

$$\int_{-\infty}^{\infty} e^{-y^2}\,dy = \sqrt{\pi}. \tag{A.1}$$

$$\int_{-\infty}^{\infty} e^{-by^2}\,dy = \sqrt{\frac{\pi}{b}} \tag{A.2}$$

where $b > 0$ is a real and positive constant.

$$\int_{-\infty}^{\infty} y e^{-by^2}\,dy = 0, \tag{A.3}$$

where $b > 0$ is a real and positive constant.

$$\int_{-\infty}^{\infty} y^2 e^{-by^2}\,dy = \frac{1}{2}\sqrt{\frac{\pi}{b^3}}, \tag{A.4}$$

where $b > 0$ is a real and positive constant.

$$\int_{-\infty}^{\infty} e^{-\alpha y^2+\beta y}dy = \sqrt{\frac{\pi}{\alpha}}e^{\beta^2/4\alpha} \tag{A.5}$$

where α and β are complex numbers, and the real part of α is positive, $\mathrm{Re}(\alpha) > 0$.

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